

Space-Time Algebra (STA)

Dr. Luke A. Wendt

Abstract—This paper presents a compact paravector form of space-time algebra that turns geometry, electromagnetism, and relativistic wave mechanics into consequences of one common algebraic language. The treatment is designed to stay explicit and usable: scalar and vector parts remain visible, real and imaginary structure is kept in view, and familiar results emerge from a small set of consistent product rules. The payoff is a concise route from boosts and intervals to Maxwell, Klein–Gordon, Dirac, and Pauli equations without switching formalisms.

Index Terms—space-time algebra, geometric algebra, Lorentz transformations, Maxwell electrodynamics, Dirac equation, Pauli equation.

Introduction

The main structural emphasis of this manuscript is a paravector form of space-time algebra organized by three operations: the conjugate $(\cdot)^*$, the reverse $(\cdot)^R$, and the scalar paravector metric $\langle\langle Z \rangle\rangle^2$. Writing $Z \in \mathbb{C} \oplus \mathbb{C}^3$ as a complex scalar plus complex vector, or equivalently as real scalar, imaginary scalar, real vector, and imaginary vector parts, makes several layers of structure visible at once. The same product then encodes scalar-vector couplings, real-imaginary couplings, norms, interval formulas, field terms, and wave operators in one algebraic framework.

The document proceeds from algebraic construction to physical interpretation. It begins with the sigma basis, the conjugate and reverse operators, the decomposition of complex paravectors, dot and cross products, general paravector product laws, and the resulting norm and scalar-paravector-metric identities. It then develops projector algebra, exponential and logarithmic formulas, generalized transform identities, and the matrix and auxiliary 2×2 realizations. Only after that algebraic setup is fixed does the presentation turn to space-time kinematics, differential-operator notation, electromagnetism, and relativistic wave equations.

This perspective is consistent with the modern STA and paravector literature of Hestenes, Baylis, and Doran–Lasenby [12, 13, 15], while its algebraic lineage runs back through Hamilton’s quaternion construction and Clifford’s geometric extension [1, 3]. The present treatment emphasizes keeping the scalar/vector and real/imaginary decompositions explicit so that familiar physical relations appear as compact consequences of the same algebraic products. A one-page Equation Reference Sheet follows this introduction and collects the main numbered formulas.

Bold symbols denote three-vectors, and the remaining symbols are defined where they first appear. Whenever a new symbol is introduced, the text states whether it is a scalar, vector, or paravector and whether it is real or complex. Complexification is carried by the pseudoscalar i . Natural units $\hbar = c = 1$ and metric signature $(+, -, -, -)$ are used unless stated otherwise.

Reading Guide

1. The Equation Reference Sheet on the next page collects the main numbered equations in the order they are developed.
2. Section 1 develops the algebraic foundations: the sigma basis, the conjugate and reverse operators, complex paravectors, dot and cross products, paravector products, norms and scalar paravector metrics, projector algebra, exponential and logarithmic formulas, generalized transform identities, and the matrix and auxiliary 2×2 realizations.
3. Section 2 interprets the earlier scalar paravector metric as the Minkowski metric of interval geometry and develops proper velocity, proper momentum, acceleration, and free relativistic motion.
4. Section 4 collects the spatial and paravector differential operators in one place, fixing the notation for ∂_t , ∇ , ∂ , ∂^* , and $\square$ before they are used in the physical sections.
5. Section 5 introduces the field and potential paravectors, Maxwell theory, gauge structure, wave equations for the field, and the Lorentz force.
6. Section 6 develops the relativistic wave-equation material, from frequency splitting and Klein–Gordon factorization to the Dirac two-component field of complex paravectors, the Pauli limit, and the energy/projector split.
7. Section 7 summarizes the main structural consequences of the paravector split and records the principal directions that remain to be developed.
8. After the references, Section 8 records scope notes and cautions, and Section 9 records verification and formula status.

Equation Reference Sheet

Natural units: $\hbar = c = 1$. Metric signature: $(+, -, -, -)$.

		Ref.
Pseudoscalar	$i = \sigma_1 \sigma_2 \sigma_3$	(1)
Sigma product	$\sigma_n \sigma_m = \delta_{nm} + \sum_{k=1}^3 \epsilon_{nmk} i \sigma_k$	(2)
Conjugate operator	$(\cdot)^* = \text{conjugate that flips signs of } \sigma_n$	(3)
Reverse operator	$(\cdot)^R = \text{reverse order of multiplied } \sigma_n$	(4)
Complex paravector	$Z = (a + ib) + (\mathbf{A} + i\mathbf{B}) \cdot \boldsymbol{\sigma} = S + \mathbf{V} \cdot \boldsymbol{\sigma} = X + iY$	(11), (17)
Real function	$\text{real}(Z) = X$	(18)
Img function	$\text{img}(Z) = Y$	(19)
Scalar function	$\text{scalar}(Z) = S$	(20)
Vector function	$\text{vector}(Z) = \mathbf{V}$	(21)
Conjugate paravector	$Z^* = (a - ib) + (-\mathbf{A} + i\mathbf{B}) \cdot \boldsymbol{\sigma}$	(22)
Real conjugate	$X^* = a - \mathbf{A} \cdot \boldsymbol{\sigma}$	(25)
Reverse paravector	$Z^R = X - iY = (a - ib) + (\mathbf{A} - i\mathbf{B}) \cdot \boldsymbol{\sigma}$	(23)
Real reverse	$X^R = X = a + \mathbf{A} \cdot \boldsymbol{\sigma}$	(26)
Reverse conjugate	$Z^{*R} = (a + ib) - (\mathbf{A} + i\mathbf{B}) \cdot \boldsymbol{\sigma}$	(24)
Real reverse conjugate	$X^{*R} = a - \mathbf{A} \cdot \boldsymbol{\sigma} = X^*$	(27)
Vector product	$(\mathbf{A} \cdot \boldsymbol{\sigma})(\mathbf{B} \cdot \boldsymbol{\sigma}) = \mathbf{A} \cdot \mathbf{B} + i(\mathbf{A} \times \mathbf{B}) \cdot \boldsymbol{\sigma}$	(31)
Paravector product	$ZZ' = SS' + \mathbf{V} \cdot \mathbf{V}' + (S\mathbf{V}' + S'\mathbf{V} + i(\mathbf{V} \times \mathbf{V}') \cdot \boldsymbol{\sigma}) \cdot \boldsymbol{\sigma}$	(32)
Paravector commutator	$[Z, Z'] = 2i(\mathbf{V} \times \mathbf{V}') \cdot \boldsymbol{\sigma}$	(33)
Scalar norm	$ S = \sqrt{SS^*}$	(34)
Vector norm	$\ \mathbf{V}\ = \sqrt{\mathbf{V} \cdot \mathbf{V}^*}$	(35)
Paravector norm	$\ Z\ = \sqrt{ S ^2 + \ \mathbf{V}\ ^2}$	(36)
Real scalar paravector metric	$\langle\langle X \rangle\rangle^2 = XX^* = a^2 - \mathbf{A}^2$	(38)
Scalar paravector metric	$\langle\langle Z \rangle\rangle = \sqrt{S^2 - \mathbf{V}^2}$	(37)
Paravector inverse	$Z^{-1} = Z^{*R} / \langle\langle Z \rangle\rangle^2 = (S - \mathbf{V} \cdot \boldsymbol{\sigma}) / (S^2 - \mathbf{V}^2)$	(42)
Projector	$\Pi(\pm) = \frac{1}{2}(1 \pm \mathbf{u} \cdot \boldsymbol{\sigma})$	(45)
Functional calculus	$f(\mathbf{u} \cdot \boldsymbol{\sigma}) = f(+)\Pi(+) + f(-)\Pi(-)$	(46)
Vector exponential	$\exp(\theta \mathbf{u} \cdot \boldsymbol{\sigma}) = \cosh \theta + \sinh \theta \mathbf{u} \cdot \boldsymbol{\sigma}$	(48)
Paravector logarithm	$\log(Y) = \frac{1}{2} \log(\lambda(+)\lambda(-)) + \frac{1}{2} \log(\lambda(+)/\lambda(-)) \mathbf{u} \cdot \boldsymbol{\sigma}$	(50)
General transform	$X' = T^{-1} X T$	(53)
Rotor	$R(\theta, \mathbf{u}) = \exp(\frac{\theta}{2} \mathbf{u} \cdot i\boldsymbol{\sigma}), X' = R^R X R$	(52)
Boost	$L(\theta, \mathbf{u}) = \exp(\frac{\theta}{2} \mathbf{u} \cdot \boldsymbol{\sigma}), X' = L^* X L$	(49)
Space-time interval	$ds^2 = \langle\langle dX \rangle\rangle^2 = \langle\langle dt + \boldsymbol{\sigma} \cdot d\mathbf{r} \rangle\rangle^2 = dt^2 - d\mathbf{r}^2$	(64)
Proper velocity	$U = \gamma(1 + \mathbf{v} \cdot \boldsymbol{\sigma})$	(65)
Proper momentum	$mU = E + \mathbf{P} \cdot \boldsymbol{\sigma}$	(66)
Frequency operator	$\Omega = \langle\langle m + \boldsymbol{\sigma} \cdot \nabla \rangle\rangle = \sqrt{m^2 - \nabla^2}$	(91)
Frequency split	$(\partial_t \pm i\Omega)\Psi(\pm) = 0$	(91)
Spatial derivative	$\partial_{\mathbf{r}} = \{\partial/\partial r_1, \partial/\partial r_2, \partial/\partial r_3\}$	(70)
Gradient notation	$\nabla = \partial_{\mathbf{r}} = \{\partial_1, \partial_2, \partial_3\}$	(71)
Spatial Laplacian	$\nabla^2 = \partial_1^2 + \partial_2^2 + \partial_3^2$	(72)
Paravector derivative	$\partial = \partial_t + \boldsymbol{\sigma} \cdot \nabla = \partial_t + \sigma_1 \partial_1 + \sigma_2 \partial_2 + \sigma_3 \partial_3$	(73)
Conjugate derivative	$\partial^* = \partial_t - \boldsymbol{\sigma} \cdot \nabla = \partial_t - \sigma_1 \partial_1 - \sigma_2 \partial_2 - \sigma_3 \partial_3$	(74)
d'Alembertian	$\square = \langle\langle \partial \rangle\rangle^2 = \partial \partial^* = \partial^* \partial = \partial_t^2 - \nabla^2$	(75), (76)
Field paravector	$F = (\mathbf{E} + i\mathbf{B}) \cdot \boldsymbol{\sigma}$	(77)
Maxwell equation	$\partial F = \rho - \mathbf{J} \cdot \boldsymbol{\sigma}$	(81)
Field wave equation	$\square F = \partial^*(\rho - \mathbf{J} \cdot \boldsymbol{\sigma})$	(84)
Charge conservation	$\partial_t \rho + \nabla \cdot \mathbf{J} = 0$	(85)
Lorentz force	$m dU/ds = q \text{real}(FU)$	(87)
Potential paravector	$\Phi = V + \mathbf{A} \cdot \boldsymbol{\sigma}$	(77)
Potential gradient	$\partial \Phi = S + F^*$	(79)
Gauge law	$\Phi' = \Phi + \partial^* \lambda$	(86)
Klein-Gordon	$(-E^2 + \mathbf{P}^2 + m^2)\Psi = 0$	(93)
Covariant energy	$E = i\partial_t - qV$	(92)
Covariant momentum	$\mathbf{P} = -i\nabla - q\mathbf{A}$	(92)
Dirac factorization	$(E + \boldsymbol{\sigma} \cdot \mathbf{P})(E - \boldsymbol{\sigma} \cdot \mathbf{P}) = E^2 - \mathbf{P}^2 - iqF$	(96)
Dirac system	$(W(i\partial - q\Phi^*) - mI)\Psi = 0$	(98)
W map	$W(Z) = \{0, Z; Z^{*R}, 0\}$	(61)
Dirac square	$\{(-E^2 + \mathbf{P}^2 + m^2 + iqF^*)\Psi_1; (-E^2 + \mathbf{P}^2 + m^2 + iqF)\Psi_2\} = 0$	(99)
Pauli spin split	$i\partial_t \Psi(\pm) = ((\mathbf{P}^2 \mp q\ \mathbf{B}\)/(2m) + qV) \Psi(\pm)$	(104)
Pauli limit	$2mE\Psi = (\mathbf{P}^2 - q\mathbf{B} \cdot \boldsymbol{\sigma})\Psi$	(101)

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1 Algebraic Foundations

This section fixes the basis, the conjugate and reverse operators, complex paravectors, dot and cross products, norms and metrics, paravector products, projector formulas, exponential and logarithmic formulas, generalized transforms, and the standard matrix realizations.

1.1 Basis and pseudoscalar

For $n \in \{1, 2, 3\}$, the algebra is generated by the three basis elements σ_n . Their products generate scalars, vectors, bivectors, and pseudoscalars. The basis relations are

$$\begin{aligned} \sigma_n^2 &= 1, & \sigma_n \sigma_m &= -\sigma_m \sigma_n \quad (n \neq m), \\ i &:= \sigma_1 \sigma_2 \sigma_3, & i^2 &= -1. \end{aligned} \quad (1)$$

The element i is a pseudoscalar because $i\sigma_n = \sigma_n i$, and squaring i returns the scalar -1 :

$$\begin{aligned} (\sigma_1 \sigma_2 \sigma_3)^2 &= \sigma_1 \sigma_2 (\sigma_3 \sigma_1) \sigma_2 \sigma_3, \\ &= -\sigma_1 (\sigma_2 \sigma_1) (\sigma_3 \sigma_2) \sigma_3, \\ &= -\sigma_1^2 \sigma_2^2 \sigma_3^2, \\ &= -1. \end{aligned}$$

With the Kronecker delta and Levi-Civita symbol,

$$\sigma_n \sigma_m = \delta_{nm} + \sum_{k=1}^3 \epsilon_{nmk} i \sigma_k. \quad (2)$$

where

$$\begin{aligned} \delta_{nm} &:= \begin{cases} 1, & n = m, \\ 0, & n \neq m, \end{cases} \\ \epsilon_{nmk} &:= \begin{cases} +1, & (n, m, k) \text{ even permutation of } (1, 2, 3), \\ -1, & (n, m, k) \text{ odd permutation of } (1, 2, 3), \\ 0, & \text{otherwise.} \end{cases} \end{aligned}$$

1.2 The Conjugate and Reverse Operators

The two basic operators are

$$(\cdot)^* = \text{conjugate that flips signs of } \sigma_n, \quad (3)$$

$$(\cdot)^R = \text{reverse order of multiplied } \sigma_n, \quad (4)$$

In operator form,

$$\begin{aligned} (XY)^* &= X^* Y^*, & (XY)^R &= Y^R X^R, \\ X^{*R} &= (X^R)^* = (X^*)^R. \end{aligned} \quad (5)$$

Applied to the generators and the pseudoscalar, these rules reduce to

$$\sigma_n^* = -\sigma_n, \quad \sigma_n^R = \sigma_n, \quad \sigma_n^{*R} = -\sigma_n, \quad (6)$$

$$i^* = -i, \quad i^R = -i, \quad i^{*R} = i. \quad (7)$$

For a two-generator bivector pair with $n \neq m$,

$$\begin{aligned} (\sigma_n \sigma_m)^* &= \sigma_n^* \sigma_m^* = (-\sigma_n)(-\sigma_m) = \sigma_n \sigma_m, \\ (\sigma_n \sigma_m)^R &= \sigma_m^R \sigma_n^R = \sigma_m \sigma_n = -\sigma_n \sigma_m, \\ (\sigma_n \sigma_m)^{*R} &= (\sigma_m \sigma_n)^* = \sigma_m^* \sigma_n^* \\ &= (-\sigma_m)(-\sigma_n) = \sigma_m \sigma_n = -\sigma_n \sigma_m. \end{aligned} \quad (8)$$

Equation (8) records the same sign pattern on the sigma-number-two elements. These rules agree with standard STA conventions [12, 15].

The sigma number of a reduced basis product is the number of sigma generators that remain after simplification. Thus 1, σ_n , $\sigma_n \sigma_m$ with $n \neq m$, and i have sigma numbers 0, 1, 2, and 3. Any unreduced product with sigma number larger than 3 simplifies to one of these four cases. Tables 1 and 2 summarize the basis elements and operator signs by sigma number.

Table 1. Sigma basis elements by sigma number.

sigma number	representative basis elements
0 (scalar)	1
1 (vector)	$\sigma_1, \sigma_2, \sigma_3$
2 (bivector)	$\sigma_2 \sigma_3, \sigma_3 \sigma_1, \sigma_1 \sigma_2$
3 (pseudoscalar)	$\sigma_1 \sigma_2 \sigma_3$

Table 2. Operator signs by sigma number.

sigma number	$(\cdot)^*$	$(\cdot)^R$	$(\cdot)^{*R}$
0 (scalar)	+	+	+
1 (vector)	−	+	−
2 (bivector)	+	−	−
3 (pseudoscalar)	−	−	+

1.3 Generic Complex Paravector

Collect the three generators into the sigma-vector

$$\sigma := \{\sigma_1, \sigma_2, \sigma_3\}, \quad (9)$$

which is a vector of algebraic objects rather than a vector of scalar components. Multiplication by the pseudoscalar gives the associated bivector triple

$$i\sigma := \{i\sigma_1, i\sigma_2, i\sigma_3\} = \{\sigma_2 \sigma_3, \sigma_3 \sigma_1, \sigma_1 \sigma_2\}, \quad (10)$$

because $i = \sigma_1 \sigma_2 \sigma_3$ and each product $i\sigma_n$ reduces to the complementary ordered two-generator product. Equation (10) therefore records the canonical sigma-number-two basis built from the generators. With this basis fixed, for real scalars $a, b \in \mathbb{R}$ and real vectors $\mathbf{A}, \mathbf{B} \in \mathbb{R}^3$, with $\mathbf{A} \cdot \sigma = A_1 \sigma_1 + A_2 \sigma_2 + A_3 \sigma_3$, define the real paravectors $X, Y \in \mathbb{R} \oplus \mathbb{R}^3$ by

$$X = a + \mathbf{A} \cdot \sigma. \quad (11)$$

$$Y = b + \mathbf{B} \cdot \sigma. \quad (12)$$

We can define a complex paravector $Z \in \mathbb{C} \oplus \mathbb{C}^3$ either as a real paravector plus an imaginary paravector with

$$Z = X + iY. \quad (13)$$

or as a scalar plus a vector component with

$$Z = (a + ib) + (\mathbf{A} + i\mathbf{B}) \cdot \boldsymbol{\sigma} \quad (14)$$

$$= S + \mathbf{V} \cdot \boldsymbol{\sigma}. \quad (15)$$

where the complex scalar and vector components are given by

$$S = a + ib \in \mathbb{C}. \quad (16)$$

$$\mathbf{V} = \mathbf{A} + i\mathbf{B} \in \mathbb{C}^3. \quad (17)$$

Define the real, imaginary, scalar, and vector functions by

$$\text{real}(Z) := X, \quad (18)$$

$$\text{img}(Z) := Y. \quad (19)$$

$$\text{scalar}(Z) := S, \quad (20)$$

$$\text{vector}(Z) := \mathbf{V}. \quad (21)$$

For the scalar part and vector part of the paravector Z , we also write $Z_S := \text{scalar}(Z)$ and $Z_V := \text{vector}(Z)$.

The conjugate and reverse operators act on the generic complex paravector by

$$Z^* = (a - ib) + (-\mathbf{A} + i\mathbf{B}) \cdot \boldsymbol{\sigma}, \quad (22)$$

$$Z^R = X - iY = (a - ib) + (\mathbf{A} - i\mathbf{B}) \cdot \boldsymbol{\sigma}, \quad (23)$$

$$Z^{*R} = (a + ib) - (\mathbf{A} + i\mathbf{B}) \cdot \boldsymbol{\sigma}. \quad (24)$$

For the real paravector X from Eq. (11), these reduce to

$$X^* = a - \mathbf{A} \cdot \boldsymbol{\sigma}, \quad (25)$$

$$X^R = X = a + \mathbf{A} \cdot \boldsymbol{\sigma}, \quad (26)$$

$$X^{*R} = a - \mathbf{A} \cdot \boldsymbol{\sigma} = X^*. \quad (27)$$

Equations (22)–(24) are the generic operator formulas used throughout the later algebraic constructions, while Eqs. (25)–(27) record the purely real specialization used whenever the imaginary scalar and imaginary vector parts vanish. Equations (11)–(21) fix the real and complex paravector pieces and extractors used throughout the later algebra. Table 3 records the type assignments, Table 4 records the recovery formulas for the extractor definitions, and Table 5 records the same split in components.

Table 3. Type assignments for the generic complex paravector notation.

Type name	Underlying set	Example
real scalar	$\mathbb{R}$	a, b
real vector	$\mathbb{R}^3$	$\mathbf{A}, \mathbf{B}$
complex scalar	$\mathbb{C}$	S
complex vector	$\mathbb{C}^3$	$\mathbf{V}$
real paravector	$\mathbb{R} \oplus \mathbb{R}^3$	X, Y
complex paravector	$\mathbb{C} \oplus \mathbb{C}^3$	Z

Table 4. Components of Z .

scalar:	$S = (Z + Z^{R*})/2$
vector:	$\mathbf{V} \cdot \boldsymbol{\sigma} = (Z - Z^{R*})/2$
real:	$X = (Z + Z^R)/2$
img:	$Y = (Z - Z^R)/(2i)$

Table 5. Real and imaginary scalar/vector parts of Z .

real scalar:	$a = (S + S^R)/2$ $= (X + X^*)/2$
img scalar:	$b = (S - S^R)/(2i)$ $= (Y + Y^*)/2$
real vector:	$\mathbf{A} \cdot \boldsymbol{\sigma} = (\mathbf{V} + \mathbf{V}^R) \cdot \boldsymbol{\sigma}/2$ $= (X - X^*)/2$
img vector:	$\mathbf{B} \cdot \boldsymbol{\sigma} = (\mathbf{V} - \mathbf{V}^R) \cdot \boldsymbol{\sigma}/(2i)$ $= (Y - Y^*)/2$

1.4 Dot products and cross products

For real vectors $\mathbf{A}, \mathbf{B} \in \mathbb{R}^3$, the dot product is defined by

$$\begin{aligned} \mathbf{A} \cdot \mathbf{B} &:= \sum_{n=1}^3 \sum_{m=1}^3 \delta_{nm} A_n B_m \\ &= A_1 B_1 + A_2 B_2 + A_3 B_3. \end{aligned} \quad (28)$$

Define the square of a vector by

$$\mathbf{A}^2 := \mathbf{A} \cdot \mathbf{A}. \quad (29)$$

For the cross product, the Levi-Civita form gives

$$(\mathbf{A} \times \mathbf{B})_k := \sum_{n=1}^3 \sum_{m=1}^3 \epsilon_{nmk} A_n B_m,$$

which gives the vector

$$\mathbf{A} \times \mathbf{B} = \begin{Bmatrix} A_2 B_3 - A_3 B_2, \\ A_3 B_1 - A_1 B_3, \\ A_1 B_2 - A_2 B_1 \end{Bmatrix}. \quad (30)$$

Dot products are commutative, whereas cross products are anti-symmetric:

$$\begin{aligned} \mathbf{A} \cdot \mathbf{B} &= +\mathbf{B} \cdot \mathbf{A}, \\ \mathbf{A} \times \mathbf{B} &= -\mathbf{B} \times \mathbf{A}. \end{aligned}$$

With Eqs. (1) and (2),

$$\begin{aligned} (\mathbf{A} \cdot \boldsymbol{\sigma})(\mathbf{B} \cdot \boldsymbol{\sigma}) &= \sum_{n=1}^3 \sum_{m=1}^3 A_n B_m \sigma_n \sigma_m \\ &= \sum_{n=1}^3 A_n B_n \\ &\quad + \sum_{n < m} (A_n B_m - A_m B_n) \sigma_n \sigma_m \end{aligned}$$

$$\begin{aligned}
&= A_1 B_1 + A_2 B_2 + A_3 B_3 \\
&\quad + (A_2 B_3 - A_3 B_2) i \sigma_1 \\
&\quad - (A_1 B_3 - A_3 B_1) i \sigma_2 \\
&\quad + (A_1 B_2 - A_2 B_1) i \sigma_3.
\end{aligned}$$

This can be efficiently expressed in terms of the dot product and cross product as

$$(\mathbf{A} \cdot \boldsymbol{\sigma})(\mathbf{B} \cdot \boldsymbol{\sigma}) = \mathbf{A} \cdot \mathbf{B} + i(\mathbf{A} \times \mathbf{B}) \cdot \boldsymbol{\sigma}. \quad (31)$$

Table 6 records the same product in component form. These same dot-product and cross-product definitions extend componentwise to complex vectors in $\mathbb{C}^3$, and they apply equally to combinations of vectors, vector-valued functions, and vector operators whenever the component expressions are defined.

The reversed order flips only the antisymmetric term:

$$\begin{aligned}
&(\mathbf{B} \cdot \boldsymbol{\sigma})(\mathbf{A} \cdot \boldsymbol{\sigma}) = \mathbf{A} \cdot \mathbf{B} - i(\mathbf{A} \times \mathbf{B}) \cdot \boldsymbol{\sigma}, \\
&(\mathbf{A} \cdot \boldsymbol{\sigma})(\mathbf{B} \cdot \boldsymbol{\sigma}) + (\mathbf{B} \cdot \boldsymbol{\sigma})(\mathbf{A} \cdot \boldsymbol{\sigma}) = 2 \mathbf{A} \cdot \mathbf{B}, \\
&(\mathbf{A} \cdot \boldsymbol{\sigma})(\mathbf{B} \cdot \boldsymbol{\sigma}) - (\mathbf{B} \cdot \boldsymbol{\sigma})(\mathbf{A} \cdot \boldsymbol{\sigma}) = 2i(\mathbf{A} \times \mathbf{B}) \cdot \boldsymbol{\sigma}.
\end{aligned}$$

The same product immediately gives

$$\begin{aligned}
&(\mathbf{A} \cdot \boldsymbol{\sigma})^2 = \mathbf{A}^2, \\
&(\mathbf{A} \cdot \boldsymbol{\sigma})^{2n} = (\mathbf{A}^2)^n, \\
&(\mathbf{A} \cdot \boldsymbol{\sigma})^{2n+1} = (\mathbf{A}^2)^n (\mathbf{A} \cdot \boldsymbol{\sigma}), \\
&(\mathbf{A} \cdot \boldsymbol{\sigma}) \boldsymbol{\sigma} = \mathbf{A} + i \mathbf{A} \times \boldsymbol{\sigma}.
\end{aligned}$$

These are the basic sigma-product identities from which the later paravector formulas are built.

1.5 Products

With Eq. (31) fixing the scalar/vector split for real vectors, the fully complex product follows as the master formula from which the simpler cases are recovered by restriction.

For Z from Eq. (15) and $Z' = S' + \mathbf{V}' \cdot \boldsymbol{\sigma} \in \mathbb{C} \oplus \mathbb{C}^3$ defined analogously to Eqs. (15) and (17),

$$ZZ' = (SS' + \mathbf{V} \cdot \mathbf{V}') + (S\mathbf{V}' + S'\mathbf{V} + i(\mathbf{V} \times \mathbf{V}')) \cdot \boldsymbol{\sigma}. \quad (32)$$

Equations (15) and (32) fix the generic complex paravector and its master multiplication rule: pure scalar, real vector, real paravector, and mixed products all follow by setting the unused parts to zero.

1.5.1 Products of real scalars, vectors, and paravectors

Let X and Y be the real paravectors from Eqs. (11) and (12). Using Eq. (31),

$$\begin{aligned}
XY &= (a + \mathbf{A} \cdot \boldsymbol{\sigma})(b + \mathbf{B} \cdot \boldsymbol{\sigma}) \\
&= (ab + \mathbf{A} \cdot \mathbf{B}) + (a\mathbf{B} + b\mathbf{A} + i(\mathbf{A} \times \mathbf{B})) \cdot \boldsymbol{\sigma}, \\
YX &= (ab + \mathbf{A} \cdot \mathbf{B}) + (a\mathbf{B} + b\mathbf{A} - i(\mathbf{A} \times \mathbf{B})) \cdot \boldsymbol{\sigma} \\
&= (XY)^R.
\end{aligned}$$

Table 7 gives the same product in component form. The scalar part is symmetric, while the imaginary-vector part changes sign with the order.

Two immediate consequences are

$$\begin{aligned}
[X, Y] &= 2i(\mathbf{A} \times \mathbf{B}) \cdot \boldsymbol{\sigma}, \\
XY \in \mathbb{R} \oplus \mathbb{R}^3 &\iff \mathbf{A} \times \mathbf{B} = 0.
\end{aligned}$$

So two real paravectors commute exactly when their vector parts are parallel.

Setting $Y = X$ removes the antisymmetric term automatically:

$$X^2 = a^2 + \mathbf{A}^2 + 2a \mathbf{A} \cdot \boldsymbol{\sigma}.$$

If $X^2 = b + \mathbf{B} \cdot \boldsymbol{\sigma}$, then matching scalar and vector parts gives

$$\begin{aligned}
b &= a^2 + \mathbf{A}^2, \\
\mathbf{B} &= 2a \mathbf{A},
\end{aligned}$$

hence

$$\begin{aligned}
a^2 &= \frac{1}{2} \left(b \pm \sqrt{b^2 - \mathbf{B}^2} \right), \\
\mathbf{A}^2 &= b - a^2.
\end{aligned}$$

This is the real-paravector square-root relation obtained directly from the product decomposition.

1.5.2 Products of complex paravectors

The master product law (32) is the complex extension of the real formulas above. Writing Z as in Eqs. (15)–(14) and $Z' = X' + iY' = S' + \mathbf{V}' \cdot \boldsymbol{\sigma}$, one obtains every later specialization by setting the unused scalar or vector parts to zero. Table 8 gives the full component decomposition of ZZ' , Table 9 gives Z^2 , and Tables 10–15 record the order-sensitive products built from the conjugate and reverse operators.

The scalar part of a product is cyclic:

$$\text{scalar}(ZZ') = \text{scalar}(Z'Z).$$

In particular, if $Z' = MZ^{*R}$, then

$$\begin{aligned}
\text{scalar}(ZMZ^{*R}) &= \text{scalar}(MZ^{*R}Z) \\
&= \text{scalar}(M)(Z^{*R}Z),
\end{aligned}$$

because $Z^{*R}Z$ is scalar.

The order-sensitive variants fall into three algebraic types:

$$\begin{aligned}
ZZ^* &\text{ is a real scalar plus a complex vector,} \\
ZZ^R &\text{ is a real scalar plus a real vector,} \\
ZZ^{R*} &\text{ is a complex scalar.}
\end{aligned}$$

Those three cases are precisely the ones needed later for norms, metrics, inverses, and transformation formulas.

The following tables collect the basic product decompositions and the order-dependent component expansions used throughout this subsection.

Table 6. Components of $(\mathbf{A} \cdot \boldsymbol{\sigma})(\mathbf{B} \cdot \boldsymbol{\sigma})$.

real scalar:	$\mathbf{A} \cdot \mathbf{B}$
img scalar:	0
real vector:	$\mathbf{0}$
img vector:	$\mathbf{A} \times \mathbf{B}$

Table 7. Components of XY .

real scalar:	$ab + \mathbf{A} \cdot \mathbf{B}$
img scalar:	0
real vector:	$a\mathbf{B} + b\mathbf{A}$
img vector:	$\mathbf{A} \times \mathbf{B}$

Tables 8 and 9 collect the full component expansions for the generic product ZZ' and the special case Z^2 .

Table 8. Components of ZZ' .

real scalar:	$aa' - bb' + \mathbf{A} \cdot \mathbf{A}' - \mathbf{B} \cdot \mathbf{B}'$
img scalar:	$ab' + ba' + \mathbf{A} \cdot \mathbf{B}' + \mathbf{B} \cdot \mathbf{A}'$
real vector:	$a\mathbf{A}' - b\mathbf{B}' + a'\mathbf{A} - b'\mathbf{B} - \mathbf{A} \times \mathbf{B}' - \mathbf{B} \times \mathbf{A}'$
img vector:	$a\mathbf{B}' + b\mathbf{A}' + a'\mathbf{B} + b'\mathbf{A} + \mathbf{A} \times \mathbf{A}' - \mathbf{B} \times \mathbf{B}'$

Table 9. Components of Z^2 .

real scalar:	$a^2 - b^2 + \mathbf{A}^2 - \mathbf{B}^2$
img scalar:	$2ab + 2\mathbf{A} \cdot \mathbf{B}$
real vector:	$2(a\mathbf{A} - b\mathbf{B})$
img vector:	$2(a\mathbf{B} + b\mathbf{A})$

Tables 10 and 11 make the order dependence of ZZ^* and Z^*Z explicit.

Table 10. Components of ZZ^* .

real scalar:	$a^2 + b^2 - \mathbf{A}^2 - \mathbf{B}^2$
img scalar:	0
real vector:	$-2\mathbf{A} \times \mathbf{B}$ ← differs
img vector:	$2(a\mathbf{B} - b\mathbf{A})$

Table 11. Components of Z^*Z .

real scalar:	$a^2 + b^2 - \mathbf{A}^2 - \mathbf{B}^2$
img scalar:	0
real vector:	$2\mathbf{A} \times \mathbf{B}$ ← differs
img vector:	$2(a\mathbf{B} - b\mathbf{A})$

Tables 12 and 13 play the same role for products involving the reverse operator.

Table 12. Components of ZZ^R .

real scalar:	$a^2 + b^2 + \mathbf{A}^2 + \mathbf{B}^2$
img scalar:	0
real vector:	$2(a\mathbf{A} + b\mathbf{B} + \mathbf{A} \times \mathbf{B})$ ← differs
img vector:	$\mathbf{0}$

Table 13. Components of $Z^R Z$.

real scalar:	$a^2 + b^2 + \mathbf{A}^2 + \mathbf{B}^2$
img scalar:	0
real vector:	$2(a\mathbf{A} + b\mathbf{B} - \mathbf{A} \times \mathbf{B})$ ← differs
img vector:	$\mathbf{0}$

Tables 14 and 15 show that the products formed with the reverse-conjugate operator collapse to purely scalar and imaginary-scalar parts.

Table 14. Components of ZZ^{R*} .

real scalar:	$a^2 - b^2 - \mathbf{A}^2 + \mathbf{B}^2$
img scalar:	$2ab - 2\mathbf{A} \cdot \mathbf{B}$
real vector:	$\mathbf{0}$
img vector:	$\mathbf{0}$

Table 15. Components of $Z^{R*}Z$.

real scalar:	$a^2 - b^2 - \mathbf{A}^2 + \mathbf{B}^2$
img scalar:	$2ab - 2\mathbf{A} \cdot \mathbf{B}$
real vector:	$\mathbf{0}$
img vector:	$\mathbf{0}$

1.5.3 Order dependence and commutators

The part of a product that depends on order is its commutator. For generic complex paravectors $Z, Z' \in \mathbb{C} \oplus \mathbb{C}^3$,

$$[Z, Z'] := ZZ' - Z'Z = 2i(\mathbf{V} \times \mathbf{V}') \cdot \boldsymbol{\sigma} \quad (33)$$

Equation (33) is the compact commutator law. In real and imaginary vector components it expands to

$$[Z, Z'] = -2(\mathbf{A} \times \mathbf{B}' + \mathbf{B} \times \mathbf{A}') \cdot \boldsymbol{\sigma} + 2i(\mathbf{A} \times \mathbf{A}' - \mathbf{B} \times \mathbf{B}') \cdot \boldsymbol{\sigma}.$$

The real-paravector commutator

$$[X, Y] = 2i(\mathbf{A} \times \mathbf{B}) \cdot \boldsymbol{\sigma}$$

is the corresponding specialization. In every case the scalar terms cancel, so the commutator isolates the cross-product part of the multiplication law.

The same order dependence appears already in the sigma basis itself. The pseudoscalar commutes with the generators,

$$[i, \sigma_n] = 0, \\ i\boldsymbol{\sigma} = \boldsymbol{\sigma}i = \{\sigma_2\sigma_3, \sigma_3\sigma_1, \sigma_1\sigma_2\},$$

while distinct sigma generators satisfy

$$[\sigma_n, \sigma_m] = 2\sigma_n\sigma_m = 2\sum_{k=1}^3 \epsilon_{nmk} i\sigma_k \quad (n \neq m).$$

Thus the oriented sigma-number-two basis is exactly the commutator closure of the sigma-number-one generators.

Tables 12–15 display the same product types in full component form. Their scalar norm and metric consequences are already fixed by Eqs. (36), (38), and (37), so no new norm or metric definitions are introduced here.

1.6 Norms and Metrics

Using Eq. (29), the product formulas above now collect the norm and scalar paravector metric identities used later in the paper. For a complex scalar $S = a + ib \in \mathbb{C}$, define the scalar norm by

$$\begin{aligned} |S| &:= \sqrt{SS^*}. \\ |S|^2 &= SS^* \\ &= a^2 + b^2. \end{aligned} \quad (34)$$

If $S \in \mathbb{R}$, then Eq. (34) reduces to the ordinary absolute value $|S| = \sqrt{S^2}$. For a complex vector $\mathbf{V} = \mathbf{A} + i\mathbf{B} \in \mathbb{C}^3$, define the vector norm by

$$\begin{aligned} \|\mathbf{V}\| &:= \sqrt{\mathbf{V} \cdot \mathbf{V}^*}. \\ \|\mathbf{V}\|^2 &= \mathbf{V} \cdot \mathbf{V}^* \\ &= \mathbf{A}^2 + \mathbf{B}^2. \end{aligned} \quad (35)$$

The quadratic product $\mathbf{V}^2 = \mathbf{V} \cdot \mathbf{V}$ is not the same object as the positive norm square $\mathbf{V} \cdot \mathbf{V}^*$. For the complex paravector Z from Eqs. (15)–(14), define the paravector norm by

$$\begin{aligned} \|Z\| &:= \sqrt{|S|^2 + \|\mathbf{V}\|^2}, \\ \|Z\|^2 &= |S|^2 + \|\mathbf{V}\|^2 \\ &= a^2 + b^2 + \mathbf{A}^2 + \mathbf{B}^2 \\ &= \text{scalar}(ZZ^R) = \text{scalar}(Z^R Z). \end{aligned} \quad (36)$$

This norm is therefore a nonnegative real scalar, and it is the reverse operator $(\cdot)^R$ that supplies the quadratic form. For the scalar paravector metric, use the complex paravector Z from Eqs. (15)–(14) and the real paravector X from Eq. (11). The scalar paravector metric $\langle\langle \cdot \rangle\rangle$ is defined by the principal square root of a scalar: its argument lies in $\mathbb{C}$ for complex paravectors and in $\mathbb{R}$ for real paravectors. Define the complex scalar paravector metric by

$$\begin{aligned} \langle\langle Z \rangle\rangle &:= \sqrt{S^2 - \mathbf{V}^2}, \\ \langle\langle Z \rangle\rangle^2 &= S^2 - \mathbf{V}^2 \\ &= (a^2 - b^2 - \mathbf{A}^2 + \mathbf{B}^2) + 2i(ab - \mathbf{A} \cdot \mathbf{B}) \\ &= ZZ^{*R} = Z^{*R}Z. \end{aligned} \quad (37)$$

The product order does not matter because $Z^{*R}Z = ZZ^{*R}$ is scalar. Thus the metric is built from the reverse-conjugate operator $(\cdot)^{*R}$, not from the reverse alone. For the real paravector X

from Eq. (11), note that $X^{*R} = X^*$, so the real scalar paravector metric is

$$\begin{aligned} \langle\langle X \rangle\rangle^2 &= a^2 - \mathbf{A}^2 \\ &= XX^* = X^*X. \end{aligned} \quad (38)$$

For $Z, Z' \in \mathbb{C} \oplus \mathbb{C}^3$, the scalar paravector metric is multiplicative:

$$\begin{aligned} \langle\langle ZZ' \rangle\rangle^2 &= ZZ'(ZZ')^{*R} = ZZ'Z'^{*R}Z^{*R} = Z(Z'Z'^{*R})Z^{*R} \\ &= Z\langle\langle Z' \rangle\rangle^2 Z^{*R} = \langle\langle Z' \rangle\rangle^2 ZZ^{*R} = \langle\langle Z \rangle\rangle^2 \langle\langle Z' \rangle\rangle^2 \\ &= \langle\langle Z'Z \rangle\rangle^2. \end{aligned} \quad (39)$$

Here $(ZZ')^{*R} = Z'^{*R}Z^{*R}$, and $\langle\langle Z \rangle\rangle^2$ and $\langle\langle Z' \rangle\rangle^2$ are scalars. Under complex scalar multiplication, the norm and scalar paravector metric scale differently:

$$\|re^{i\theta}Z\|^2 = r^2\|Z\|^2, \quad (40)$$

$$\langle\langle re^{i\theta}Z \rangle\rangle^2 = r^2e^{2i\theta}\langle\langle Z \rangle\rangle^2. \quad (41)$$

The norm is phase-insensitive, while the scalar paravector metric retains the phase square. Equations (29)–(41) therefore collect the full norm and scalar paravector metric structure used later in the paper.

1.7 Inverse formulas

The inverse formulas are the algebraic consequence of the scalar paravector metric identities from Eqs. (37) and (39). For $Z = S + \mathbf{V} \cdot \boldsymbol{\sigma} \in \mathbb{C} \oplus \mathbb{C}^3$, when $\langle\langle Z \rangle\rangle^2 \neq 0$, the inverse may be written in any of the equivalent forms

$$\begin{aligned} Z^{-1} &:= \frac{Z^{*R}}{\langle\langle Z \rangle\rangle^2} = \frac{Z^{*R}}{ZZ^{*R}} = \frac{Z^{*R}}{Z^{*R}Z} \\ &= \frac{S - \mathbf{V} \cdot \boldsymbol{\sigma}}{S^2 - \mathbf{V}^2}. \end{aligned} \quad (42)$$

If the scalar paravector metric vanishes, the inverse does not exist.

Differentiating the identity $Z^{-1}Z = 1$ gives

$$d(Z^{-1}) = -Z^{-1}(dZ)Z^{-1}. \quad (43)$$

For $Z' = S' + \mathbf{V}' \cdot \boldsymbol{\sigma} \in \mathbb{C} \oplus \mathbb{C}^3$, if $\mathbf{V} \times \mathbf{V}' = \mathbf{0}$, then

$$Z^{-1}Z' = Z'Z^{-1}. \quad (44)$$

1.8 Supporting Algebraic Derivations

The supporting derivations record the explicit sigma-basis and operator calculations, together with one generic map decomposition, that support the earlier basis and operator formulas without repeating the product development already collected above.

1.8.1 Basis-Element Classification

Table 1 lists the four sigma numbers and the corresponding basis-element types, while Table 2 lists the signs acquired under the conjugate operator, the reverse operator, and the reverse-conjugate operator.

1.8.2 Complex Paravector-Valued Maps

This notation records how real scalar and vector functions combine into a generic complex paravector-valued map without imposing any physical interpretation:

$$\begin{aligned} Z(t, \mathbf{r}) &= (a(t, \mathbf{r}) + i b(t, \mathbf{r})) \\ &\quad + (\mathbf{A}(t, \mathbf{r}) + i \mathbf{B}(t, \mathbf{r})) \cdot \boldsymbol{\sigma}, \\ a, b &: \mathbb{R} \times \mathbb{R}^3 \Rightarrow \mathbb{R}, \\ \mathbf{A}, \mathbf{B} &: \mathbb{R} \times \mathbb{R}^3 \Rightarrow \mathbb{R}^3, \\ Z &: \mathbb{R} \times \mathbb{R}^3 \Rightarrow \mathbb{C} \oplus \mathbb{C}^3. \end{aligned}$$

1.9 Projector Algebra

Projectors are introduced next because they organize the later function formulas and ladder constructions. Once Eqs. (2) and (31) are fixed, the projector calculus gives the cleanest route from simple unit-direction elements to more general paravector decompositions.

If $\mathbf{A} \in \mathbb{R}^3$ is written as $\mathbf{A} = \theta \mathbf{u}$ with $\theta = \|\mathbf{A}\|$, $\mathbf{u} \in \mathbb{R}^3$, and $\mathbf{u}^2 = 1$, define the idempotents

$$\Pi(\pm) := \frac{1}{2}(1 \pm \mathbf{u} \cdot \boldsymbol{\sigma}) \quad (45)$$

satisfying $\Pi(\pm)^2 = \Pi(\pm)$, $\Pi(+)\Pi(-) = 0$, $\Pi(+)+\Pi(-) = 1$, and $(\mathbf{u} \cdot \boldsymbol{\sigma})\Pi(\pm) = \pm\Pi(\pm)$. Algebraically, $\Pi(+)$ and $\Pi(-)$ isolate the parts aligned and anti-aligned with the direction $\mathbf{u}$. Their role here is purely structural: they split an object into two parts adapted to a chosen axis.

For a scalar-valued function $f : \{\pm 1\} \rightarrow \mathbb{C}$,

$$f(\mathbf{u} \cdot \boldsymbol{\sigma}) = f(+)\Pi(+) + f(-)\Pi(-). \quad (46)$$

This gives a compact projector-based formula for roots, inverses, exponentials, logarithms, and related functions. Once the minimal polynomial of an STA element is known, functions of that element can be defined without leaving the algebraic framework. The detailed formulas begin with projector and eigenvalue identities and continue to ladder operations associated with a chosen axis.

About the z axis, define the ladder elements

$$\begin{aligned} \ell(\pm) &:= \frac{1}{2}(\sigma_1 \pm i\sigma_2), \quad [\sigma_3, \ell(\pm)] = \pm 2\ell(\pm), \\ \ell(\pm)^2 &= 0, \quad \ell(+)\ell(-) = \Pi(+), \quad \ell(-)\ell(+) = \Pi(-). \end{aligned} \quad (47)$$

Equation (47) records the nilpotent combinations of σ_1 and σ_2 that exchange the two projector parts selected by Eq. (45). The projector eigenvalue formulas, mixed projector products, and ladder identities built from Eqs. (45), (46), and (47) are collected next.

1.9.1 Functions of $\mathbf{u} \cdot \boldsymbol{\sigma}$

The projector pair $\Pi(\pm)$ gives the two-root decomposition of the unit-direction element $\mathbf{u} \cdot \boldsymbol{\sigma}$. The corresponding eigenvalue data

are

$$\begin{aligned} \Pi(\pm) &= \frac{1}{2}(1 \pm \mathbf{u} \cdot \boldsymbol{\sigma}), \\ \Pi(\pm)^2 &= \Pi(\pm), \\ \Pi(+)\Pi(-) &= 0, \\ \Pi(+)+\Pi(-) &= 1, \\ (\mathbf{u} \cdot \boldsymbol{\sigma})\Pi(\pm) &= \pm\Pi(\pm), \end{aligned}$$

so $\mathbf{u} \cdot \boldsymbol{\sigma}$ has the two eigenvalues $+1$ and -1 . Any function with a convergent power series therefore collapses to those two scalar values:

$$\begin{aligned} f(\mathbf{u} \cdot \boldsymbol{\sigma}) &= f(+)\Pi(+) + f(-)\Pi(-) \\ &= \frac{1}{2}(f(+)+f(-)) + \frac{1}{2}(f(+)-f(-))\mathbf{u} \cdot \boldsymbol{\sigma}. \end{aligned}$$

These are the basic projector formulas for a unit-direction element. The first examples below collect the square root, logarithm branch data, rational inverse, Cayley transform, and the two exponential families for later reuse:

$$\begin{aligned} \sqrt{\mathbf{u} \cdot \boldsymbol{\sigma}} &= \pm\Pi(+) \pm i\Pi(-), \\ \log(\mathbf{u} \cdot \boldsymbol{\sigma}) &= 2i\pi(n\Pi(+) + (\tfrac{1}{2} + m)\Pi(-)), \\ (z + \mathbf{u} \cdot \boldsymbol{\sigma})^{-1} &= \frac{1}{z+1}\Pi(+) + \frac{1}{z-1}\Pi(-) \\ &= \frac{z - \mathbf{u} \cdot \boldsymbol{\sigma}}{z^2 - 1}, \\ \frac{i - \mathbf{u} \cdot \boldsymbol{\sigma}}{i + \mathbf{u} \cdot \boldsymbol{\sigma}} &= i\mathbf{u} \cdot \boldsymbol{\sigma}, \\ \exp(\theta \mathbf{u} \cdot \boldsymbol{\sigma}) &= \cosh(\theta) + \sinh(\theta)\mathbf{u} \cdot \boldsymbol{\sigma}, \\ \exp(i\theta \mathbf{u} \cdot \boldsymbol{\sigma}) &= \cos(\theta) + i\sin(\theta)\mathbf{u} \cdot \boldsymbol{\sigma}, \end{aligned}$$

where $m, n \in \mathbb{Z}$ index logarithm branches. The square root, inverse, Cayley-type transform, and the two exponential identities are all obtained by direct substitution into Eq. (46). The later exponential and logarithm subsections revisit the same formulas in the broader real- and complex-paravector settings.

1.9.2 Functions of X

Let

$$\begin{aligned} X &= t + \mathbf{r} \cdot \boldsymbol{\sigma}, \\ t &= \text{scalar}(X) \in \mathbb{R}, \\ \mathbf{r} &= \text{vector}(X) \in \mathbb{R}^3, \\ \mathbf{u} &:= \mathbf{r}/\|\mathbf{r}\| \quad \text{when } \mathbf{r} \neq 0. \end{aligned}$$

If $\mathbf{r} \neq 0$, write $\mathbf{r} = \|\mathbf{r}\|\mathbf{u}$ with $\mathbf{u}^2 = 1$. Then the minimal polynomial of X is

$$\begin{aligned} X^2 - 2tX + (t^2 - \mathbf{r}^2) &= 0, \\ \lambda(\pm) &= t \pm \|\mathbf{r}\|, \\ 0 &= (X - \lambda(-))(X - \lambda(+)) \\ &= X^2 - \lambda(-)X \\ &\quad - X\lambda(+) + \lambda(-)\lambda(+). \end{aligned}$$

With

$$a := \lambda(+), \quad b := \lambda(-),$$

the corresponding projector decomposition is

$$\begin{aligned} \Pi(\pm) &= \frac{1}{2}(1 \pm \mathbf{u} \cdot \boldsymbol{\sigma}), \\ X &= a \Pi(+) + b \Pi(-), \\ \Pi(\pm)X &= \lambda(\pm)\Pi(\pm). \end{aligned}$$

If f has a convergent power series, then

$$\begin{aligned} f(X) &= f(a)\Pi(+) + f(b)\Pi(-) \\ &= \frac{1}{2}(f(a) + f(b)) + \frac{1}{2}(f(a) - f(b)) \mathbf{u} \cdot \boldsymbol{\sigma}. \end{aligned}$$

Writing $X' := f(X) = t' + \mathbf{r}' \cdot \boldsymbol{\sigma}$, one finds

$$\begin{aligned} t' &= \frac{1}{2}(f(a) + f(b)), \\ \|\mathbf{r}'\| &= \frac{1}{2}(f(a) - f(b)), \\ \mathbf{r}' &= \|\mathbf{r}'\| \mathbf{u}. \end{aligned}$$

The projector summary may therefore be recorded as

$$\begin{aligned} \lambda(\pm) &= t \pm \|\mathbf{r}\|, \\ \mathbf{u} &= \mathbf{r}/\|\mathbf{r}\|, \\ \Pi(\pm) &= \frac{1}{2}(1 \pm \mathbf{u} \cdot \boldsymbol{\sigma}), \\ \Pi(\pm)X &= \lambda(\pm)\Pi(\pm), \\ X &= \lambda(+)\Pi(+) + \lambda(-)\Pi(-), \\ f(X) &= f(\lambda(+))\Pi(+) + f(\lambda(-))\Pi(-). \end{aligned}$$

The standard examples then follow immediately:

$$\begin{aligned} \exp(X) &= e^t \left(\cosh(\|\mathbf{r}\|) + \sinh(\|\mathbf{r}\|) \mathbf{u} \cdot \boldsymbol{\sigma} \right), \\ \sqrt{X} &= \pm \sqrt{a} \Pi(+) \pm \sqrt{b} \Pi(-), \\ \log(X) &= \log(a) \Pi(+) + \log(b) \Pi(-), \\ X^{-1} &= \frac{\Pi(+)}{a} + \frac{\Pi(-)}{b} \\ &= \frac{X^*}{t^2 - \mathbf{r}^2}, \\ X^n &= a^n \Pi(+) + b^n \Pi(-) \\ &= \frac{1}{2}(a^n + b^n) + \frac{1}{2}(a^n - b^n) \mathbf{u} \cdot \boldsymbol{\sigma}. \end{aligned}$$

For two paravectors diagonalized along possibly different axes,

$$\begin{aligned} f(X)g(X') &= (f(a)\Pi(+) + f(b)\Pi(-)) \\ &\quad \times (g(a')\Pi'(+) + g(b')\Pi'(-)), \end{aligned}$$

so

$$\begin{aligned} f(X)g(X') &= f(a)g(a') \Pi(+) \Pi'(+) + f(a)g(b') \Pi(+) \Pi'(-) \\ &\quad + f(b)g(a') \Pi(-) \Pi'(+) + f(b)g(b') \Pi(-) \Pi'(-) \\ &= \sum_{s, s'=\pm} f(\lambda(s))g(\lambda'(s')) \Pi(s) \Pi'(s'). \end{aligned}$$

The mixed projector products are

$$\begin{aligned} \Pi(+) \Pi'(+)&= \frac{1}{4}(1 + \mathbf{u} \cdot \mathbf{u}') \\ &\quad + \frac{1}{4}(\mathbf{u}' + \mathbf{u} + i(\mathbf{u} \times \mathbf{u}')) \cdot \boldsymbol{\sigma}, \\ \Pi(+) \Pi'(-)&= \frac{1}{4}(1 - \mathbf{u} \cdot \mathbf{u}') \\ &\quad + \frac{1}{4}(-\mathbf{u}' + \mathbf{u} - i(\mathbf{u} \times \mathbf{u}')) \cdot \boldsymbol{\sigma}, \\ \Pi(-) \Pi'(+)&= \frac{1}{4}(1 - \mathbf{u} \cdot \mathbf{u}') \\ &\quad + \frac{1}{4}(\mathbf{u}' - \mathbf{u} - i(\mathbf{u} \times \mathbf{u}')) \cdot \boldsymbol{\sigma}, \\ \Pi(-) \Pi'(-)&= \frac{1}{4}(1 + \mathbf{u} \cdot \mathbf{u}') \\ &\quad + \frac{1}{4}(-\mathbf{u}' - \mathbf{u} + i(\mathbf{u} \times \mathbf{u}')) \cdot \boldsymbol{\sigma}. \end{aligned}$$

As a useful example, the Cayley-type transform is itself a projector exponential:

$$\begin{aligned} \frac{1 - \mathbf{r} \cdot \boldsymbol{\sigma}}{1 + \mathbf{r} \cdot \boldsymbol{\sigma}} &= \exp(-2 \operatorname{arctanh}(\mathbf{r} \cdot \boldsymbol{\sigma})) \\ &= \exp(-2 \operatorname{arctanh}(\|\mathbf{r}\|)) \Pi(+) \\ &\quad + \exp(+2 \operatorname{arctanh}(\|\mathbf{r}\|)) \Pi(-). \end{aligned}$$

1.9.3 Minimal Polynomial and Projector Decomposition

The two-root data of a complex paravector are collected here because the later function formulas, projector identities, and exponential identities all depend on this algebraic factorization; see Eq. (46):

$$\begin{aligned} Z &= S + \mathbf{V} \cdot \boldsymbol{\sigma} \in \mathbb{C} \oplus \mathbb{C}^3, \\ Z^2 - 2SZ + (S^2 - \mathbf{V}^2) &= 0, \\ \lambda(\pm) &= S \pm \sqrt{\mathbf{V}^2}, \\ (Z - \lambda(-))(Z - \lambda(+)) &= 0. \end{aligned}$$

When $\mathbf{V}^2 \neq 0$, these two eigenvalues provide the projector data needed for the same functional calculus used in the real-paravector case.

1.9.4 Spin Projectors

The projector family from Eq. (45) resolves a state into the two spin components along a chosen unit axis $\mathbf{u} \in \mathbb{R}^3$. The basic identities are

$$\begin{aligned} \Pi(\pm) &= \frac{1}{2}(1 \pm \mathbf{u} \cdot \boldsymbol{\sigma}), \\ \Pi(\pm)^2 &= \Pi(\pm), \\ \Pi(+) \Pi(-) &= 0, \\ \Pi(+) + \Pi(-) &= 1, \\ (\mathbf{u} \cdot \boldsymbol{\sigma}) \Pi(\pm) &= \pm \Pi(\pm), \\ \Pi(\pm)^R &= \Pi(\pm), \\ \Pi(\pm)^* &= \Pi(\mp). \end{aligned}$$

Accordingly,

$$\mathbf{u} \cdot \boldsymbol{\sigma} = (+1)\Pi(+) + (-1)\Pi(-),$$

so the two projector parts carry the eigenvalues $+1$ and -1 of the axis operator.

If the axis is rotated by the rotor R from Eq. (52), define

$$\mathbf{u}' \cdot \boldsymbol{\sigma} = R^R(\mathbf{u} \cdot \boldsymbol{\sigma})R, \quad \Pi'(\pm) := \frac{1}{2}(1 \pm \mathbf{u}' \cdot \boldsymbol{\sigma}).$$

Then

$$\mathbf{u}' \cdot \boldsymbol{\sigma} = (+1)\Pi'(+)+(-1)\Pi'(-).$$

For a complex paravector state $\psi \in \mathbb{C} \oplus \mathbb{C}^3$, define the spin-projected parts

$$\psi(\pm) := \Pi(\pm)\psi.$$

These satisfy

$$\begin{aligned} (\mathbf{u} \cdot \boldsymbol{\sigma})\psi(\pm) &= \pm\psi(\pm), \\ \psi &= \psi(+) + \psi(-), \end{aligned}$$

so the state is decomposed into spin-up and spin-down parts along $\mathbf{u}$. Reversion acts as

$$\begin{aligned} \psi(\pm)^R &= \psi^R \Pi(\pm)^R \\ &= \psi^R \Pi(\pm), \end{aligned}$$

and therefore

$$\begin{aligned} \psi(\pm)^R \psi(\pm) &= \psi^R \Pi(\pm)\psi, \\ \psi(\pm)^R \psi(\mp) &= \psi^R \Pi(\pm)\Pi(\mp)\psi = 0, \\ \psi(\mp)^R \psi(\pm) &= \psi^R \Pi(\mp)\Pi(\pm)\psi = 0. \end{aligned}$$

In the bra-ket shorthand

$$\langle \psi | M | \psi \rangle := \text{scalar}(\psi^R M \psi),$$

the expectation of the rotated axis expands as

$$\begin{aligned} \langle \psi | \mathbf{u}' \cdot \boldsymbol{\sigma} | \psi \rangle &= \langle \psi(+) + \psi(-) | \mathbf{u}' \cdot \boldsymbol{\sigma} | \psi(+) + \psi(-) \rangle \\ &= \langle \psi(+) | \mathbf{u}' \cdot \boldsymbol{\sigma} | \psi(+) \rangle \\ &\quad + \langle \psi(+) | \mathbf{u}' \cdot \boldsymbol{\sigma} | \psi(-) \rangle \\ &\quad + \langle \psi(-) | \mathbf{u}' \cdot \boldsymbol{\sigma} | \psi(+) \rangle \\ &\quad + \langle \psi(-) | \mathbf{u}' \cdot \boldsymbol{\sigma} | \psi(-) \rangle. \end{aligned}$$

The algebra governing these same-side and cross-side terms is

$$\begin{aligned} (\mathbf{u}' \cdot \boldsymbol{\sigma})(\mathbf{u} \cdot \boldsymbol{\sigma}) &= \mathbf{u}' \cdot \mathbf{u} + i(\mathbf{u}' \times \mathbf{u}) \cdot \boldsymbol{\sigma}, \\ (\mathbf{u} \cdot \boldsymbol{\sigma})(\mathbf{u}' \cdot \boldsymbol{\sigma}) &= \mathbf{u}' \cdot \mathbf{u} - i(\mathbf{u}' \times \mathbf{u}) \cdot \boldsymbol{\sigma}, \\ (\mathbf{u} \cdot \boldsymbol{\sigma})(\mathbf{u}' \cdot \boldsymbol{\sigma})(\mathbf{u} \cdot \boldsymbol{\sigma}) &= (2(\mathbf{u}' \cdot \mathbf{u})\mathbf{u} - \mathbf{u}') \cdot \boldsymbol{\sigma}. \end{aligned}$$

Write the decomposition of the second axis into parts parallel and perpendicular to $\mathbf{u}$ as

$$\mathbf{u}'_{\parallel} := (\mathbf{u} \cdot \mathbf{u}')\mathbf{u}, \quad \mathbf{u}'_{\perp} := \mathbf{u}' - \mathbf{u}'_{\parallel}.$$

Then the projector products are

$$\begin{aligned} \Pi(\pm)(\mathbf{u}' \cdot \boldsymbol{\sigma})\Pi(\pm) &= \pm(\mathbf{u} \cdot \mathbf{u}')\Pi(\pm), \\ \Pi(\pm)(\mathbf{u}' \cdot \boldsymbol{\sigma})\Pi(\mp) &= \frac{1}{2}(\mathbf{u}'_{\perp} \pm i\mathbf{u} \times \mathbf{u}') \cdot \boldsymbol{\sigma} \Pi(\mp). \end{aligned}$$

The diagonal terms are therefore weighted by the axis overlap $\mathbf{u} \cdot \mathbf{u}'$, while the off-diagonal terms are controlled by the perpendicular and commutator parts of the rotated axis.

If θ is the angle between $\mathbf{u}$ and $\mathbf{u}'$, then

$$\mathbf{u}' \cdot \mathbf{u} = \cos(\theta),$$

so the half-angle transition factors are

$$\frac{1}{2}(1 + \mathbf{u}' \cdot \mathbf{u}) = \cos^2 \frac{\theta}{2}, \quad \frac{1}{2}(1 - \mathbf{u}' \cdot \mathbf{u}) = \sin^2 \frac{\theta}{2}.$$

If $\mathbf{u}'$ is obtained by rotating $\mathbf{u}$ through angle ϕ about a unit axis $\mathbf{n}$, Rodrigues's formula gives

$$\begin{aligned} \mathbf{u}' \cdot \mathbf{u} &= (\mathbf{u} \cdot \mathbf{u})\cos(\phi) + (\mathbf{u} \cdot (\mathbf{n} \times \mathbf{u}))\sin(\phi) \\ &\quad + (\mathbf{n} \cdot \mathbf{u})(1 - \cos(\phi))(\mathbf{n} \cdot \mathbf{u}) \\ &= (\mathbf{n} \cdot \mathbf{u})^2 + (1 - (\mathbf{n} \cdot \mathbf{u})^2)\cos(\phi). \end{aligned}$$

In the perpendicular case $\mathbf{n} \cdot \mathbf{u} = 0$, this reduces to

$$\mathbf{u}' \cdot \mathbf{u} = \cos(\phi),$$

hence

$$\frac{1}{2}(1 + \mathbf{u}' \cdot \mathbf{u}) = \cos^2 \frac{\phi}{2}, \quad \frac{1}{2}(1 - \mathbf{u}' \cdot \mathbf{u}) = \sin^2 \frac{\phi}{2}.$$

These are the projector transition factors for a spin- $\frac{1}{2}$ axis rotation.

1.9.5 Ladder Operations About the z Axis

For raising and lowering about the z axis, choose

$$\mathbf{u} = \mathbf{e}_3 = (0, 0, 1), \quad \Pi(\pm) = \frac{1}{2}(1 \pm \sigma_3).$$

Define the ladder operators by

$$\ell(\pm) := \frac{1}{2}(\sigma_1 \pm i\sigma_2).$$

They satisfy the commutator relation

$$[\sigma_3, \ell(\pm)] = \pm 2\ell(\pm),$$

so $\ell(+)$ raises the σ_3 eigenvalue and $\ell(-)$ lowers it.

Their projector products are

$$\ell(+)\ell(-) = \Pi(+), \quad \ell(-)\ell(+) = \Pi(-),$$

and each ladder operator is nilpotent:

$$\ell(\pm)^2 = 0.$$

The one-sided projector actions are

$$\begin{aligned} \ell(+)\Pi(+) &= 0, & \ell(+)\Pi(-) &= \ell(+), \\ \Pi(+)\ell(+) &= \ell(+), & \Pi(-)\ell(+) &= 0, \\ \ell(-)\Pi(-) &= 0, & \ell(-)\Pi(+) &= \ell(-), \\ \Pi(-)\ell(-) &= \ell(-), & \Pi(+)\ell(-) &= 0, \end{aligned}$$

which make the raising/lowering interpretation explicit. The original sigma generators are recovered from the same ladder/projector data:

$$\begin{aligned} \sigma_1 &= \ell(+) + \ell(-), & \sigma_2 &= \frac{\ell(+) - \ell(-)}{i}, \\ \sigma_3 &= \Pi(+) - \Pi(-). \end{aligned}$$

1.9.6 Spin Uncertainty

For unit directions $\mathbf{u}, \mathbf{u}' \in \mathbb{R}^3$,

$$\mathbf{u} \cdot \mathbf{u} = \mathbf{u}' \cdot \mathbf{u}' = 1.$$

Their product is

$$(\mathbf{u} \cdot \boldsymbol{\sigma})(\mathbf{u}' \cdot \boldsymbol{\sigma}) = \mathbf{u} \cdot \mathbf{u}' + i(\mathbf{u} \times \mathbf{u}') \cdot \boldsymbol{\sigma},$$

so the commutator is

$$[\mathbf{u} \cdot \boldsymbol{\sigma}, \mathbf{u}' \cdot \boldsymbol{\sigma}] = 2i(\mathbf{u} \times \mathbf{u}') \cdot \boldsymbol{\sigma}.$$

Equivalently,

$$(\mathbf{u} \cdot \boldsymbol{\sigma})(\mathbf{u}' \cdot \boldsymbol{\sigma}) = \mathbf{u} \cdot \mathbf{u}' + \frac{1}{2}[\mathbf{u} \cdot \boldsymbol{\sigma}, \mathbf{u}' \cdot \boldsymbol{\sigma}].$$

Using the paravector metric

$$\langle\langle Z \rangle\rangle^2 := ZZ^{*\mathbf{R}},$$

one finds

$$\begin{aligned} \langle\langle (\mathbf{u} \cdot \boldsymbol{\sigma})(\mathbf{u}' \cdot \boldsymbol{\sigma}) \rangle\rangle^2 &= (\mathbf{u} \cdot \mathbf{u}')^2 - (i(\mathbf{u} \times \mathbf{u}') \cdot \boldsymbol{\sigma})^2 \\ &= (\mathbf{u} \cdot \mathbf{u}')^2 + (\mathbf{u} \times \mathbf{u}')^2 \\ &\geq (\mathbf{u} \times \mathbf{u}')^2 \\ &= \left\| \frac{1}{2}[\mathbf{u} \cdot \boldsymbol{\sigma}, \mathbf{u}' \cdot \boldsymbol{\sigma}] \right\|^2. \end{aligned}$$

In this sense the metric of the ordered product controls the size of the noncommuting part of the two spin-axis elements. Written formally for infinitesimal axis variations $d\mathbf{u}$ and $d\mathbf{u}'$, this gives the Robertson-style estimate

$$\langle\langle (d\mathbf{u} \cdot \boldsymbol{\sigma})(d\mathbf{u}' \cdot \boldsymbol{\sigma}) \rangle\rangle \geq \frac{1}{2} \|[d\mathbf{u} \cdot \boldsymbol{\sigma}, d\mathbf{u}' \cdot \boldsymbol{\sigma}]\|.$$

This should not be confused with the full Robertson or Robertson–Schrödinger state-variance inequality [11]; only the operator-level noncommutativity of the spin-axis elements is being recorded here.

1.10 Exponential and Logarithmic Algebra

With the projector calculus fixed, exponential and logarithmic formulas can be developed in one place. The sigma-number-one exponential produces hyperbolic half-angle transform elements, and the sigma-number-two exponential produces the circular half-angle elements used later for rotors and for the quaternion correspondence.

1.10.1 Real Exponential and Half-Angle Elements

With the projector calculus in hand, the sigma-number-one exponential becomes especially transparent. For the real paravector

$$X = a + \mathbf{A} \cdot \boldsymbol{\sigma} \in \mathbb{R} \oplus \mathbb{R}^3,$$

write

$$\begin{aligned} \mathbf{A} &= \theta \mathbf{u}, \quad \theta = \|\mathbf{A}\|, \\ \mathbf{u} &= \frac{\mathbf{A}}{\|\mathbf{A}\|} \quad (\mathbf{A} \neq 0), \quad \mathbf{u} = (u_1, u_2, u_3), \\ \mathbf{u}^2 &= 1. \end{aligned}$$

Then Eq. (46) gives

$$\begin{aligned} \exp(\pm \mathbf{A} \cdot \boldsymbol{\sigma}) &= \exp(\pm \theta \mathbf{u} \cdot \boldsymbol{\sigma}) \\ &= e^{\pm \theta} \Pi(+) + e^{\mp \theta} \Pi(-) \\ &= \cosh(\theta) \pm \sinh(\theta) \mathbf{u} \cdot \boldsymbol{\sigma}. \end{aligned} \quad (48)$$

Equivalently, the same result follows directly from the power series by separating the even and odd powers:

$$\begin{aligned} \exp(\mathbf{A} \cdot \boldsymbol{\sigma}) &= \exp(\theta \mathbf{u} \cdot \boldsymbol{\sigma}) \\ &= \sum_{n=0}^{\infty} \frac{\theta^n (\mathbf{u} \cdot \boldsymbol{\sigma})^n}{n!} \\ &= \sum_{n=0}^{\infty} \frac{\theta^{2n}}{(2n)!} + \sum_{n=0}^{\infty} \frac{\theta^{2n+1}}{(2n+1)!} \mathbf{u} \cdot \boldsymbol{\sigma} \\ &= \cosh(\theta) + \sinh(\theta) \mathbf{u} \cdot \boldsymbol{\sigma}. \end{aligned}$$

With

$$\begin{aligned} \gamma &= \cosh(\theta), \\ \mathbf{V} &= \tanh(\theta) \mathbf{u}, \end{aligned}$$

Eq. (48) becomes the basic hyperbolic parametrization used repeatedly below:

$$\begin{aligned} \exp(\mathbf{A} \cdot \boldsymbol{\sigma}) &= \gamma(1 + \mathbf{V} \cdot \boldsymbol{\sigma}), \\ \exp(-\mathbf{A} \cdot \boldsymbol{\sigma}) &= \gamma(1 - \mathbf{V} \cdot \boldsymbol{\sigma}). \end{aligned}$$

For the real paravector X from Eq. (11), the scalar part commutes with the vector part, so

$$\begin{aligned} \exp(X) &= \exp(a) \exp(\mathbf{A} \cdot \boldsymbol{\sigma}), \\ \exp(X^*) &= \exp(a) \exp(-\mathbf{A} \cdot \boldsymbol{\sigma}) = \exp(X)^*, \\ \exp(-X) &= \exp(-a) \exp(-\mathbf{A} \cdot \boldsymbol{\sigma}) = \exp(X)^{-1}. \end{aligned}$$

The corresponding half-angle element is

$$L(\theta, \mathbf{u}) = \exp\left(\frac{\theta}{2} \mathbf{u} \cdot \boldsymbol{\sigma}\right), \quad X' = L^* X L. \quad (49)$$

Equation (49) is the two-sided hyperbolic transform built from Eq. (48). Its detailed transform consequences are developed in the generalized transform section. Finally, the conjugate operator simply exchanges the two projector weights:

$$\begin{aligned} \exp(\mathbf{A} \cdot \boldsymbol{\sigma})^* &= e^{-\theta} \Pi(+) + e^{\theta} \Pi(-) \\ &= \gamma(1 - \mathbf{V} \cdot \boldsymbol{\sigma}) \\ &= \exp(-\mathbf{A} \cdot \boldsymbol{\sigma}) \\ &= \exp((\mathbf{A} \cdot \boldsymbol{\sigma})^*) \\ &= \exp(\mathbf{A} \cdot \boldsymbol{\sigma})^{-1}. \end{aligned}$$

1.10.2 Real Logarithm of a Paravector

The logarithm is the inverse problem for the same projector formulas. Let Y be the real paravector from Eq. (12), and assume $b > \|\mathbf{B}\|$, so that the two real eigenvalues are positive and the principal logarithm is defined. Writing $\mathbf{B} = \|\mathbf{B}\|\mathbf{u}$ when $\mathbf{B} \neq 0$, one has

$$\begin{aligned}\lambda(\pm) &= b \pm \|\mathbf{B}\|, \\ Y &= \lambda(+)\Pi(+) + \lambda(-)\Pi(-),\end{aligned}$$

with the same projector pair $\Pi(\pm) = \frac{1}{2}(1 \pm \mathbf{u} \cdot \boldsymbol{\sigma})$. The logarithm therefore acts separately on the two eigenvalues:

$$\begin{aligned}\log(Y) &= \log(\lambda(+))\Pi(+) + \log(\lambda(-))\Pi(-) \\ &= \frac{1}{2} \log(\lambda(+)\lambda(-)) + \frac{1}{2} \log(\lambda(+)/\lambda(-)) \mathbf{u} \cdot \boldsymbol{\sigma}. \quad (50)\end{aligned}$$

If $Y = \exp(X)$ with X from Eq. (11) and $\mathbf{A} = \alpha\mathbf{u}$, then $\lambda(\pm) = e^{a \pm \alpha}$, so the recovered parameters are

$$\begin{aligned}Y &= \exp(X) \\ &= \exp(a + \mathbf{A} \cdot \boldsymbol{\sigma}) \\ &= \exp(a) \exp(\mathbf{A} \cdot \boldsymbol{\sigma}) \\ &= \exp(a) (\cosh(\alpha) + \sinh(\alpha) \mathbf{u} \cdot \boldsymbol{\sigma}) \\ &= b + \|\mathbf{B}\| \mathbf{u} \cdot \boldsymbol{\sigma},\end{aligned}$$

so one reads off

$$\begin{aligned}\alpha &= \operatorname{arctanh}(\|\mathbf{B}\|/b), \\ \exp(a) &= \sqrt{b^2 - \mathbf{B}^2}, \\ a &= \frac{1}{2} \log(b^2 - \mathbf{B}^2), \\ \mathbf{A} &= \alpha \mathbf{u} = \operatorname{arctanh}(\|\mathbf{B}\|/b) \frac{\mathbf{B}}{\|\mathbf{B}\|}.\end{aligned}$$

Thus the logarithm of a time-like real paravector is again a real paravector,

$$\log(Y) = a + \mathbf{A} \cdot \boldsymbol{\sigma}.$$

1.10.3 Circular Exponentials and Half-Angle Elements

Replacing θ by $i\theta$ turns the same projector calculation into the trigonometric form of the sigma-number-two exponential:

$$\begin{aligned}\exp(\pm i\mathbf{A} \cdot \boldsymbol{\sigma}) &= \exp(\pm i\theta \mathbf{u} \cdot \boldsymbol{\sigma}) \\ &= e^{\pm i\theta} \Pi(+) + e^{\mp i\theta} \Pi(-) \\ &= \cos(\theta) \pm i \sin(\theta) \mathbf{u} \cdot \boldsymbol{\sigma}. \quad (51)\end{aligned}$$

In particular, the negative-sign form may be written as

$$\exp(-i\mathbf{A} \cdot \boldsymbol{\sigma}) = \cos(\theta) + \sin(\theta) \mathbf{u} \cdot (-i\boldsymbol{\sigma}).$$

Later, after introducing the quaternion basis $\mathbf{q} = -i\boldsymbol{\sigma}$, this same formula becomes

$$\exp(-i\mathbf{A} \cdot \boldsymbol{\sigma}) = \cos(\theta) + \sin(\theta) \mathbf{u} \cdot \mathbf{q}.$$

For a real angle $\theta \in \mathbb{R}$ and a unit real vector $\mathbf{u} \in \mathbb{R}^3$, the corresponding circular half-angle exponential is

$$\begin{aligned}R(\theta, \mathbf{u}) &:= \exp\left(\frac{\theta}{2} \mathbf{u} \cdot i\boldsymbol{\sigma}\right), \\ R^R R &= 1, \quad X' = R^R X R.\end{aligned} \quad (52)$$

Equation (52) is the circular transform built from Eq. (51). Its detailed transform consequences are developed in the generalized transform section. The quaternion restatement of the same sigma-number-two algebra is collected later, after the transform discussion has been completed.

1.10.4 Exponentials of Complex Paravectors

Write the real paravectors

$$\begin{aligned}X &= a + \mathbf{A} \cdot \boldsymbol{\sigma} \in \mathbb{R} \oplus \mathbb{R}^3, \\ Y &= b + \mathbf{B} \cdot \boldsymbol{\sigma} \in \mathbb{R} \oplus \mathbb{R}^3, \\ Z &= S + \mathbf{V} \cdot \boldsymbol{\sigma} \in \mathbb{C} \oplus \mathbb{C}^3, \\ &= X + iY,\end{aligned}$$

where

$$S := a + ib \in \mathbb{C}, \quad \mathbf{V} := \mathbf{A} + i\mathbf{B} \in \mathbb{C}^3.$$

Because the scalar part S commutes with $\mathbf{V} \cdot \boldsymbol{\sigma}$, the exponential separates into scalar and directional factors:

$$\begin{aligned}\exp(Z) &= \exp(S + \mathbf{V} \cdot \boldsymbol{\sigma}), \\ &= \exp(S) \exp(\mathbf{V} \cdot \boldsymbol{\sigma}).\end{aligned}$$

$$\begin{aligned}\exp(Z^*) &= \exp(S^*) \exp(-\mathbf{V}^* \cdot \boldsymbol{\sigma}), \\ \exp(Z)^{-1} &= \exp(-S) \exp(-\mathbf{V} \cdot \boldsymbol{\sigma}), \\ \exp(Z^*)^{-1} &= \exp(-S^*) \exp(\mathbf{V}^* \cdot \boldsymbol{\sigma}), \\ (\mathbf{V} \cdot \boldsymbol{\sigma})^2 &= \mathbf{V} \cdot \mathbf{V} = \mathbf{V}^2, \\ &= \mathbf{A}^2 - \mathbf{B}^2 + 2i\mathbf{A} \cdot \mathbf{B}.\end{aligned}$$

The key simplification is that all higher powers collapse onto the two basis elements 1 and $\mathbf{V} \cdot \boldsymbol{\sigma}$:

$$\begin{aligned}(\mathbf{V} \cdot \boldsymbol{\sigma})^{2n} &= (\mathbf{V}^2)^n, \\ z &= \sqrt{\mathbf{V}^2} \quad \text{complex scalar}, \\ \exp(\mathbf{V} \cdot \boldsymbol{\sigma}) &= \sum_{n=0}^{\infty} \frac{(\mathbf{V} \cdot \boldsymbol{\sigma})^n}{n!}, \\ &= \sum_{n=0}^{\infty} \frac{(\mathbf{V}^2)^n}{(2n)!} + \sum_{n=0}^{\infty} \frac{(\mathbf{V}^2)^n}{(2n+1)!} (\mathbf{V} \cdot \boldsymbol{\sigma}), \\ &= \cosh(z) + \sinh(z) \frac{\mathbf{V} \cdot \boldsymbol{\sigma}}{z}.\end{aligned}$$

1.11 Generalized Transform Algebra

With the projector and exponential formulas fixed, the remaining purely mathematical transform material can be collected in one

place. The order is generic first and special-case second: the similarity law comes first, then the basis-transport identities, then the circular and hyperbolic transforms built from the earlier exponential elements:

$$X' = T^{-1}XT. \quad (53)$$

Here T is a generic invertible algebra element. Equation (53) is the common similarity form behind the transformed-basis and ordered-product constructions that follow.

1.11.1 General Similarity-Transform Identities

Equation (53) gives the differential and commutator forms. In the first display, $t \in \mathbb{R}$, $\mathbf{r} \in \mathbb{R}^3$, $X = t + \mathbf{r} \cdot \boldsymbol{\sigma} \in \mathbb{R} \oplus \mathbb{R}^3$, and T is a generic invertible algebra element:

$$\begin{aligned} X' &= T^{-1}XT, \\ &= T^{-1}(t + \mathbf{r} \cdot \boldsymbol{\sigma})T, \\ &= t + \mathbf{r} \cdot (T^{-1}\boldsymbol{\sigma}T), \\ &= t + \mathbf{r} \cdot \boldsymbol{\sigma}'. \end{aligned}$$

$$T^{-1} = T^{\mathbf{R}*} / (T^{\mathbf{R}*}T).$$

The scalar paravector metric is invariant under this similarity law because Eq. (39) is multiplicative:

$$\begin{aligned} \langle\langle X' \rangle\rangle^2 &= \langle\langle T^{-1}XT \rangle\rangle^2, \\ &= \langle\langle T^{-1} \rangle\rangle^2 \langle\langle X \rangle\rangle^2 \langle\langle T \rangle\rangle^2, \\ &= \langle\langle T^{-1} \rangle\rangle^2 \langle\langle T \rangle\rangle^2 \langle\langle X \rangle\rangle^2, \\ &= \langle\langle T^{-1}T \rangle\rangle^2 \langle\langle X \rangle\rangle^2, \\ &= \langle\langle 1 \rangle\rangle^2 \langle\langle X \rangle\rangle^2, \\ &= \langle\langle X \rangle\rangle^2. \end{aligned}$$

So the transform changes the algebraic representative of X without changing its scalar metric. This identity matters because it expresses the inverse directly in the algebra. For rotors one has $T^{\mathbf{R}*}T = 1$, so the inverse simplifies to the operator form used throughout the rotation and moving-frame calculations:

$$\begin{aligned} T^{\mathbf{R}*}T &= 1, & T^{-1} &= T^{\mathbf{R}*}, \\ T^{-1}T &= 1, & d(T^{-1})T &= -T^{-1}dT, \\ dX' &= d(T^{-1})XT + T^{-1}dXT + T^{-1}XdT, \\ &= -(T^{-1}dT)X' + T^{-1}dXT + X'(T^{-1}dT). \end{aligned}$$

Introducing the differential transform element packages the moving-frame dependence into a single algebra element. Inserting TT^{-1} into the differentiated expression then isolates the commutator contribution:

$$\begin{aligned} d\Theta &:= T^{-1}dT, \\ dX' &= T^{-1}dXT + [X', d\Theta], \\ ddX' &= d(T^{-1})dXT + T^{-1}ddXT + T^{-1}dXdT, \\ &\quad + [dX', d\Theta] + [X', dd\Theta]. \end{aligned}$$

The first derivative already shows the main structure: the carried differential $T^{-1}dXT$ is corrected only by the commutator with the moving-frame element $d\Theta$. Differentiating once more exposes the higher-order transport terms built from the same object:

$$\begin{aligned} d(T^{-1})dXT &= (-T^{-1}dT)(T^{-1}dXT), \\ &= -d\Theta(dX' - [X', d\Theta]), \end{aligned}$$

$$\begin{aligned} T^{-1}dXdT &= (T^{-1}dXT)(T^{-1}dT), \\ &= (dX' - [X', d\Theta])d\Theta. \end{aligned}$$

Substituting these auxiliary identities back into ddX' yields the second-order transport law:

$$\begin{aligned} ddX' &= T^{-1}ddXT, \\ &\quad + 2[dX', d\Theta], \\ &\quad + [X', dd\Theta], \\ &\quad + [d\Theta, [X', d\Theta]]. \end{aligned}$$

This second-order formula is the clean algebraic statement behind second derivatives in a varying frame. Nothing new is added beyond the similarity transform itself; the extra pieces come only from differentiating the transforming element. For a one-parameter family $t \mapsto T(t)$, with $t \in \mathbb{R}$, the same transport equation takes the parameter-derivative form

$$\begin{aligned} d\Theta/dt &= \Omega, \\ dd\Theta/dt^2 &= d\Omega/dt, \\ dX'/dt &= T^{-1}(dX/dt)T + [X', \Omega], \\ ddX'/dt^2 &= T^{-1}(ddX/dt^2)T, \\ &\quad + 2[dX'/dt, \Omega], \\ &\quad + [X', d\Omega/dt], \\ &\quad + [\Omega, [X', \Omega]]. \end{aligned}$$

1.11.2 Transformed Basis

The transformed basis relations below use the conjugate and reverse-operator rules of Eqs. (6) and (7):

$$\begin{aligned} T(x) &= f(x) + \mathbf{g}(x) \cdot \boldsymbol{\sigma}, \\ T^{-1} &= (f - \mathbf{g} \cdot \boldsymbol{\sigma}) / (f^2 - \mathbf{g}^2), \\ dT &= df + d\mathbf{g} \cdot \boldsymbol{\sigma}, \\ \sigma'_n &:= T^{-1}\sigma_n T, \\ \sigma_n &= T\sigma'_n T^{-1}. \end{aligned}$$

Here $f(x)$ is a real scalar-valued function, $\mathbf{g}(x)$ is a real vector-valued function, and $T(x) \in \mathbb{R} \oplus \mathbb{R}^3$ is a real paravector-valued function. These definitions send the original basis into a new basis by similarity transformation. If T is specialized to a boost or rotor, then $T^{\mathbf{R}*}T = 1$ and $T^{-1} = T^{\mathbf{R}*}$. The next identities show that the transformed generators satisfy the same algebra and operator properties.

Applying the conjugate and reverse operators gives

$$\begin{aligned}(\sigma'_n)^* &= (T^{-1})^* \sigma_n^* T^*, \\ &= -(T^*)^{-1} \sigma_n T^*,\end{aligned}$$

$$\begin{aligned}(\sigma'_n)^R &= T^R \sigma_n^R (T^{-1})^R, \\ &= T^R \sigma_n (T^R)^{-1},\end{aligned}$$

$$\begin{aligned}(\sigma'_n)^{*R} &= -T^{*R} \sigma_n (T^{-1})^{*R}, \\ &= -T^{-1} \sigma_n T \text{ if } T^{-1} = T^{*R}, \\ &= -\sigma'_n \text{ if } T^{-1} = T^{*R}.\end{aligned}$$

Thus, whenever $T^{-1} = T^{*R}$, the transformed basis inherits the same parity under the reverse-conjugate operator as the original one:

$$\begin{aligned}(\sigma'_n)^2 &= T^{-1} \sigma_n T T^{-1} \sigma_n T, \\ &= T^{-1} \sigma_n^2 T = 1.\end{aligned}$$

For $n \neq m$, the mixed products transform as follows:

$$\begin{aligned}T^{-1} \sigma_n \sigma_m T &= T^{-1} \sigma_n T T^{-1} \sigma_m T = \sigma'_n \sigma'_m, \\ &= T^{-1} \epsilon_{nmk} i \sigma_k T = \epsilon_{nmk} i \sigma'_k.\end{aligned}$$

$$\begin{aligned}\sigma'_1 \sigma'_2 \sigma'_3 &= T^{-1} \sigma_1 T T^{-1} \sigma_2 T T^{-1} \sigma_3 T, \\ &= T^{-1} \sigma_1 \sigma_2 \sigma_3 T = i T^{-1} T = i.\end{aligned}$$

So the transformed basis obeys the same sigma algebra as the original basis. The only genuinely new content appears when the transforming element depends on the parameter and derivatives of the basis must be taken. Differentiating the transformed basis makes the additional commutator terms explicit. Let $Z_s \in \mathbb{C}$ be a scalar-valued function, let $Z_V \in \mathbb{C}^3$ be a vector-valued function, and write the corresponding complex paravector as $Z = Z_s + Z_V \cdot \sigma'$. Then

$$\begin{aligned}d\sigma'_n &= [\sigma'_n, T^{-1} dT], \\ d\sigma' &= [\sigma', T^{-1} dT] \quad \text{vector}, \\ Z &= Z_s + Z_V \cdot \sigma' \quad \text{paravector}, \\ \langle\langle Z \rangle\rangle^2 &= Z Z^{*R}, \\ &= (Z_s + Z_V \cdot \sigma')(Z_s - Z_V \cdot \sigma') \text{ if } T^{-1} = T^{*R}, \\ &= Z_s^2 - Z_V^2 \text{ if } T^{-1} = T^{*R}.\end{aligned}$$

$$\begin{aligned}dZ &= dZ_s + dZ_V \cdot \sigma' + Z_V \cdot d\sigma', \\ &= dZ_s + dZ_V \cdot \sigma' + Z_V \cdot [\sigma', T^{-1} dT].\end{aligned}$$

$$\begin{aligned}T^{-1} dT &= (f df - \mathbf{g} \cdot d\mathbf{g}) / (f^2 - \mathbf{g}^2), \\ &+ (f d\mathbf{g} - df \mathbf{g} - i \mathbf{g} \times d\mathbf{g}) \cdot \sigma' / (f^2 - \mathbf{g}^2).\end{aligned}$$

Thus $T^{-1} dT$ contains one scalar part and one vector part that drive the transformed basis. The real vector part contributes sigma-number-one terms, while the imaginary vector part contributes sigma-number-two terms:

$$\begin{aligned}\Omega &:= T^{-1} (dT/dt), \\ \Omega' &:= T \Omega T^{-1} = (dT/dt) T^{-1}, \\ \text{scalar}(\sigma'_n) &= \text{scalar}(T^{-1} \sigma_n T), \\ &= \text{scalar}(\sigma_n T T^{-1}), \\ &= \text{scalar}(\sigma_n), \\ &= 0.\end{aligned}$$

$$\begin{aligned}\text{scalar}(\Omega \sigma'_n) &= \text{scalar}(T^{-1} (dT/dt) T^{-1} \sigma_n T) \\ &= \text{scalar}((dT/dt) T^{-1} \sigma_n T T^{-1}) \\ &= \text{scalar}((dT/dt) T^{-1} \sigma_n) \\ &= \text{scalar}(\Omega' \sigma_n).\end{aligned}$$

Decomposing Ω into one scalar part, one real vector part, and one imaginary vector part makes the transformed-basis decomposition explicit. Here $\Omega_s \in \mathbb{C}$, while $k = \{k_n\} \in \mathbb{R}^3$ and $\omega = \{\omega_n\} \in \mathbb{R}^3$ are real vectors:

$$\begin{aligned}k_n + i\omega_n &:= \text{scalar}(2\Omega' \sigma_n), \\ \Omega &= \Omega_s + \frac{1}{2}(k + i\omega) \cdot \sigma', \\ [\sigma', \omega \cdot (i\sigma')] &= -2\omega \times \sigma', \\ [\sigma', \mathbf{k} \cdot \sigma'] &= 2i\mathbf{k} \times \sigma', \\ d\sigma'/dt &= [\sigma', \Omega], \\ &= i(k + i\omega) \times \sigma', \\ &= (-\omega + ik) \times \sigma'.\end{aligned}$$

Substituting this basis motion into dZ/dt produces the transport terms on the vector component:

$$\begin{aligned}Z_V \cdot [\sigma', \Omega] &= (-Z_V \times \omega + i Z_V \times k) \cdot \sigma', \\ dZ/dt &= dZ_s/dt, \\ &+ (dZ_V/dt - Z_V \times \omega + i Z_V \times k) \cdot \sigma'.\end{aligned}$$

This is the moving-basis transport law for a general paravector in the transformed frame.

As a simple non-inertial-frame example, let

$$X = t + \mathbf{r} \cdot \sigma'.$$

Then

$$dX/dt = dt/dt + (d\mathbf{r}/dt - \mathbf{r} \times \omega + i \mathbf{r} \times k) \cdot \sigma'.$$

If one writes the time-parameterized split

$$m dX/dt = E + \mathbf{P}' \cdot \sigma',$$

then

$$\begin{aligned}E/m &= dt/dt, \\ \mathbf{P}'/m &= d\mathbf{r}/dt + \mathbf{r} \times (-\omega + ik).\end{aligned}$$

Whenever $T^{-1} = T^{*\mathbf{R}}$, the transformed interval remains

$$\begin{aligned}\langle\langle dX \rangle\rangle^2 &= dt^2 - (\mathbf{dr} - \mathbf{r} \times \boldsymbol{\omega} dt + i \mathbf{r} \times k dt)^2 \\ &= (1 - (\mathbf{v} - \mathbf{r} \times \boldsymbol{\omega} + i \mathbf{r} \times k)^2) dt^2,\end{aligned}$$

where $\mathbf{v} := d\mathbf{r}/dt$. The induced transformed proper-time element is therefore

$$ds' := \sqrt{1 - (\mathbf{v} - \mathbf{r} \times \boldsymbol{\omega} + i \mathbf{r} \times k)^2} dt.$$

1.11.3 Rotation

The identities in this subsection unpack Eq. (52) as the rotation case of the general transform law, with the transform element specialized to $T = R$. Here the generator has sigma number two, so the exponential is circular rather than hyperbolic:

$$\begin{aligned}R(\theta, \mathbf{u}) &:= \exp(\tfrac{1}{2}\theta(\mathbf{u} \cdot i\boldsymbol{\sigma})), \\ &= \cos(\tfrac{1}{2}\theta) + \sin(\tfrac{1}{2}\theta)(\mathbf{u} \cdot i\boldsymbol{\sigma}) \quad \text{rotor}.\end{aligned}$$

$$R^{-1} = \exp(-\tfrac{1}{2}\theta(\mathbf{u} \cdot i\boldsymbol{\sigma})).$$

Here $\theta \in \mathbb{R}$, $\mathbf{u} \in \mathbb{R}^3$ is a unit real vector, and R is generated by the sigma-number-two element $\mathbf{u} \cdot i\boldsymbol{\sigma}$. Its inverse satisfies $R^{-1} = R^{\mathbf{R}} = R^{*\mathbf{R}}$, and the transformed paravector is

$$\begin{aligned}X' &= R^{\mathbf{R}} X R, \\ &= t' + \mathbf{r}' \cdot \boldsymbol{\sigma}.\end{aligned}$$

$$\begin{aligned}t' &= t, \\ \mathbf{r}' &= \mathbf{r} \cos(\theta) + (\mathbf{u} \times \mathbf{r}) \sin(\theta), \\ &\quad + (\mathbf{u} \cdot \mathbf{r})(1 - \cos(\theta))\mathbf{u},\end{aligned}$$

Rodrigues's formula.

Differentiating the transformed quantities gives the circular-transform analogue of the general transport law:

$$\begin{aligned}d(R^{\mathbf{R}}) &= (dR)^{\mathbf{R}} = dR^{\mathbf{R}}, \\ R^{\mathbf{R}} R &= 1, \\ dR^{\mathbf{R}} R + R^{\mathbf{R}} dR &= 0, \\ dR &= -R dR^{\mathbf{R}} R.\end{aligned}$$

The associated sigma-number-two angular-velocity element is

$$\begin{aligned}\Omega &:= -R(dR^{\mathbf{R}}/ds) = (dR/ds)R^{\mathbf{R}}, \\ \Omega^{\mathbf{R}} &= -\Omega, \\ \Omega' &= R^{\mathbf{R}}\Omega R = R^{\mathbf{R}}(dR/ds), \\ \Omega'^{\mathbf{R}} &= -\Omega', \\ dR/ds &= \Omega R.\end{aligned}$$

Starting from $X' = R^{\mathbf{R}} X R$, one may equivalently write

$$RX' = XR.$$

Differentiating this relation gives the intermediate transport steps

$$\begin{aligned}dR X' + R dX' &= dX R + X dR, \\ R dX' &= dX R + X dR - dR X' .\end{aligned}$$

The resulting derivative formula has the same commutator structure as Eq. (53), with Ω constrained to be reverse-odd:

$$\begin{aligned}dX'/ds &= R^{\mathbf{R}}(dX/ds)R + [X', \Omega'], \\ &= R^{\mathbf{R}}(dX/ds + [X, \Omega])R.\end{aligned}$$

1.11.4 Spin Rotors

The rotor used for spin rotations is built from the pseudoscalar

$$i = \sigma_1 \sigma_2 \sigma_3.$$

For each basis index k , multiplication by i leaves the complementary bivector:

$$i\sigma_k = \sigma_n \sigma_m, \quad n \neq m \neq k.$$

Let $\mathbf{n} = (n_1, n_2, n_3) \in \mathbb{R}^3$ and $\mathbf{u} = (u_1, u_2, u_3) \in \mathbb{R}^3$ be unit vectors, so

$$\mathbf{n}^2 = 1, \quad \mathbf{u}^2 = 1.$$

The corresponding rotor is

$$\begin{aligned}R &:= \exp\left(\frac{\phi}{2}(\mathbf{n} \cdot i\boldsymbol{\sigma})\right) \\ &= \cos\left(\frac{\phi}{2}\right) + \sin\left(\frac{\phi}{2}\right)(\mathbf{n} \cdot i\boldsymbol{\sigma}).\end{aligned}$$

Because of the half-angle form, the rotor is double-valued:

$$R(\phi + 2\pi) = -R(\phi).$$

Its reverse is

$$R^{\mathbf{R}} = \cos\left(\frac{\phi}{2}\right) - \sin\left(\frac{\phi}{2}\right)(\mathbf{n} \cdot i\boldsymbol{\sigma}),$$

and therefore

$$R^{\mathbf{R}} R = \cos^2\left(\frac{\phi}{2}\right) + \sin^2\left(\frac{\phi}{2}\right) = 1,$$

so the inverse is

$$R^{-1} = R^{\mathbf{R}} = R^{*\mathbf{R}}.$$

Acting on a unit-direction element gives

$$R^{\mathbf{R}}(\mathbf{u} \cdot \boldsymbol{\sigma})R = \mathbf{u}' \cdot \boldsymbol{\sigma},$$

where $\mathbf{u}'$ is the rotated unit vector. In vector form,

$$\begin{aligned}\mathbf{u}' &= \mathbf{u} \cos(\phi) + (\mathbf{n} \times \mathbf{u}) \sin(\phi) \\ &\quad + (\mathbf{n} \cdot \mathbf{u})(1 - \cos(\phi))\mathbf{n},\end{aligned}$$

which is Rodrigues's formula.

Now let $\psi \in \mathbb{C} \oplus \mathbb{C}^3$ be a complex paravector:

$$\begin{aligned}\psi &= \psi_s + \psi_v \cdot \sigma, \\ \psi_s &= \text{scalar}(\psi) \in \mathbb{C}, \\ \psi_v &= \text{vector}(\psi) \in \mathbb{C}^3.\end{aligned}$$

Its reverse is

$$\begin{aligned}\psi^R &= \psi_s^R - \psi_v^R \cdot \sigma \\ &= \psi_s^* - \psi_v^* \cdot \sigma.\end{aligned}$$

Here the reverse operator plays the role of the Hermitian adjoint. If the spinor transforms by left multiplication,

$$\psi' = R\psi,$$

then for the bra-ket shorthand

$$\langle \psi' | M | \psi \rangle := \text{scalar}(\psi^R M \psi),$$

one obtains

$$\begin{aligned}\langle \psi' | \mathbf{u} \cdot \sigma | \psi' \rangle &= \langle R\psi | \mathbf{u} \cdot \sigma | R\psi \rangle \\ &= \text{scalar}(\psi^R (R^R (\mathbf{u} \cdot \sigma) R) \psi) \\ &= \text{scalar}(\psi^R (\mathbf{u}' \cdot \sigma) \psi) \\ &= \langle \psi | \mathbf{u}' \cdot \sigma | \psi \rangle.\end{aligned}$$

Finally, decompose the paravector density as

$$\psi\psi^R = \text{scalar}(\psi\psi^R) + \text{vector}(\psi\psi^R) \cdot \sigma.$$

Using the cyclicity of the scalar part,

$$\begin{aligned}\text{scalar}(\psi^R (\mathbf{u}' \cdot \sigma) \psi) &= \text{scalar}((\mathbf{u}' \cdot \sigma) \psi \psi^R) \\ &= \mathbf{u}' \cdot \text{vector}(\psi \psi^R) \\ &= \mathbf{u} \cdot \text{vector}(\psi' \psi'^R).\end{aligned}$$

So the same rotor that rotates the measured axis also rotates the vector part of the spinor density, giving the usual spin- $\frac{1}{2}$ transformation law in paravector form.

1.11.5 Lorentz Boost

The identities in this subsection unpack Eq. (49) as the Lorentz-boost case of the general transform law, with the transform element specialized to $T = L$. They are placed next to the circular identities because both come from the same exponential calculus, with sigma-number-one generators producing hyperbolic functions instead of circular ones. The auxiliary parameters β and γ are introduced first because they shorten the resulting formulas:

$$\begin{aligned}\theta &= \text{arctanh}(\beta), \\ \beta &= \tanh(\theta), \\ \gamma &= \cosh(\theta), \\ \gamma\beta &= \sinh(\theta), \\ \mathbf{u} &= (u_1, u_2, u_3), \mathbf{u}^2 = 1, \\ L(\theta, \mathbf{u}) &:= \exp(\tfrac{1}{2}\theta(\mathbf{u} \cdot \sigma)), \\ &= \cosh(\tfrac{1}{2}\theta) + \sinh(\tfrac{1}{2}\theta)(\mathbf{u} \cdot \sigma).\end{aligned}$$

$$L^{-1} = \exp(-\tfrac{1}{2}\theta(\mathbf{u} \cdot \sigma)).$$

Here $\theta, \beta, \gamma \in \mathbb{R}$, $\mathbf{u} \in \mathbb{R}^3$ is a unit real vector, and $L \in \mathbb{R} \oplus \mathbb{R}^3$ is the hyperbolic transform element. Its inverse satisfies $L^{-1} = L^* = L^{*\mathbb{R}}$. Elements along the same axis commute:

$$[L(\theta, \mathbf{u}), L(\theta', \mathbf{u})] = 0.$$

Consequently, collinear boosts combine by addition of the parameter θ :

$$\begin{aligned}L(\theta', \mathbf{u}) &= L(\theta_1, \mathbf{u})L(\theta_2, \mathbf{u}), \\ \beta' &= (\beta_1 + \beta_2)/(1 + \beta_1\beta_2), \\ \gamma' &= \gamma_1\gamma_2(1 + \beta_1\beta_2).\end{aligned}$$

Re-mapping the real space-time paravector $X = t + \mathbf{r} \cdot \sigma$, with $t \in \mathbb{R}$, $\mathbf{r} \in \mathbb{R}^3$, and $X \in \mathbb{R} \oplus \mathbb{R}^3$, the transform gives

$$\begin{aligned}X &= t + \mathbf{r} \cdot \sigma, \\ X' &= L^* X L, \\ &= t' + \mathbf{r}' \cdot \sigma.\end{aligned}$$

$$\begin{aligned}t' &= t \cosh(\theta) - \sinh(\theta)(\mathbf{u} \cdot \mathbf{r}), \\ &= \gamma(t - \beta(\mathbf{u} \cdot \mathbf{r})).\end{aligned}$$

The vector part is clearest after splitting $\mathbf{r}$ into components parallel and perpendicular to the chosen direction:

$$\begin{aligned}\text{par}(\mathbf{r}) &:= (\mathbf{r} \cdot \mathbf{u})\mathbf{u} \quad \text{parallel to } \mathbf{u}, \\ \text{prp}(\mathbf{r}) &:= \mathbf{r} - \text{par}(\mathbf{r}) \quad \text{perpendicular to } \mathbf{u}, \\ \mathbf{r} &= \text{par}(\mathbf{r}) + \text{prp}(\mathbf{r}), \\ \text{par}(\mathbf{r}') &= \gamma(\text{par}(\mathbf{r}) - \beta t\mathbf{u}), \\ \text{prp}(\mathbf{r}') &= \text{prp}(\mathbf{r}), \\ \mathbf{r}' &= \text{par}(\mathbf{r}') + \text{prp}(\mathbf{r}'), \\ &= \text{prp}(\mathbf{r}) + \gamma(\text{par}(\mathbf{r}) - \beta t\mathbf{u}), \\ &= \mathbf{r} + (\gamma(\mathbf{r} \cdot \mathbf{u} - \beta t) - \mathbf{r} \cdot \mathbf{u})\mathbf{u}.\end{aligned}$$

Thus Eq. (49) gives the hyperbolic split of scalar and vector parts: the component parallel to $\mathbf{u}$ mixes with the scalar part while the perpendicular component is unchanged. In the small-parameter limit $\beta \rightarrow 0$,

$$\gamma \Rightarrow 1, t' \Rightarrow t, \mathbf{r}' \Rightarrow \mathbf{r}.$$

The paravector metric invariance then follows directly from $L^{-1} = L^*$:

$$\begin{aligned}\langle\langle X' \rangle\rangle^2 &= X' X'^*, \\ &= (L^* X L)(L^* X L)^*, \\ &= L^* X L L^* X^* L, \\ &= L^* X X^* L, \\ &= \langle\langle X \rangle\rangle^2 L^* L, \\ &= \langle\langle X \rangle\rangle^2.\end{aligned}$$

1.11.6 Compositions of Hyperbolic and Circular Transforms

The ordered products are listed here with their inverses. The order matters even when each factor separately has a simple inverse rule.

For one hyperbolic element and one circular element,

$$\begin{aligned} T &= LR, \\ T^{-1} &= T^{*R} = R^{*R}L^{*R} = R^RL^*, \\ T &= RL, \\ T^{-1} &= T^{*R} = L^{*R}R^{*R} = L^*R^R. \end{aligned}$$

For two factors of the same type,

$$\begin{aligned} T &= R_1R_2, \\ T^{-1} &= T^{*R} = R_2^RR_1^R, \\ T &= L_1L_2, \\ T^{-1} &= T^{*R} = L_2^*L_1^*. \end{aligned}$$

These formulas are simple order identities, but they matter because they fix the inverse of each ordered product without ambiguity.

1.12 Quaternion Correspondence

Now that the circular exponential and rotor law have both been developed, the same sigma-number-two algebra can be restated in Hamilton's quaternion language without introducing a new multiplication rule. The point of this subsection is to collect that correspondence in one place after the rotation discussion has been completed.

1.12.1 Quaternion Basis and Involutions

The same sigma-number-two basis used in Eq. (51) also reproduces Hamilton's quaternion units, so the rotation algebra can be written in quaternion form without leaving the STA multiplication table [1, 3]:

$$\begin{aligned} q_1 &= -\sigma_2\sigma_3, & q_2 &= -\sigma_3\sigma_1, \\ q_3 &= -\sigma_1\sigma_2, & \mathbf{q} &:= (q_1, q_2, q_3) = -i\sigma. \end{aligned} \quad (54)$$

These basis elements satisfy the standard quaternion identities

$$q_1^2 = q_2^2 = q_3^2 = q_1q_2q_3 = -1.$$

So the quaternion units are not new generators; they are the sigma-number-two basis already present in the STA algebra, only relabeled.

If $\mathbf{B} = (B_1, B_2, B_3) \in \mathbb{R}^3$, then $\mathbf{B} \cdot \mathbf{q} = -i\mathbf{B} \cdot \sigma$. Since $q_k = -i\sigma_k$, the abstract operator rules immediately give

$$\mathbf{q}^* = \mathbf{q}, \quad \mathbf{q}^R = -\mathbf{q}.$$

Thus the conjugate operator fixes the quaternion basis, while the reverse operator changes its sign.

1.12.2 Quaternion Products

For $a \in \mathbb{R}$ and $\mathbf{B} \in \mathbb{R}^3$, define the quaternion

$$Q := a + \mathbf{B} \cdot \mathbf{q} = a - i\mathbf{B} \cdot \sigma. \quad (55)$$

In the manuscript's complex-paravector notation, the same quaternion may be written as

$$Q = X - iY, \quad X = a + \mathbf{A} \cdot \sigma, \quad Y = b + \mathbf{B} \cdot \sigma,$$

with the quaternion specialization

$$\mathbf{A} = 0, \quad b = 0.$$

So here $X = a$ is purely scalar and $Y = \mathbf{B} \cdot \sigma$ is purely vector. Because $\mathbf{q}^* = \mathbf{q}$, one has

$$Q^* = Q.$$

Therefore

$$\begin{aligned} QQ^* &= Q^2 \\ &= (a - i\mathbf{B} \cdot \sigma)^2 \\ &= a^2 - 2ia\mathbf{B} \cdot \sigma + (-i)^2(\mathbf{B} \cdot \sigma)^2 \\ &= a^2 - \mathbf{B}^2 - 2ia\mathbf{B} \cdot \sigma. \end{aligned} \quad (56)$$

The linear term must carry the factor a . This is just the sigma-number-two multiplication law written in quaternion language.

Since $\mathbf{q}^R = -\mathbf{q}$, the reverse of Q is

$$Q^R = a - \mathbf{B} \cdot \mathbf{q} = a + i\mathbf{B} \cdot \sigma.$$

Multiplying Q by its reverse gives

$$QQ^R = a^2 + \mathbf{B}^2. \quad (57)$$

So the reverse product recovers the standard positive quaternion norm square, while Eq. (56) records the ordinary algebra square.

1.12.3 Quaternion Form of the Rotor Law

Now that Eq. (52) and its transform consequences have been developed, the quaternion rotation law can be read off directly. For a unit quaternion Q satisfying $QQ^R = 1$, and for a real vector $\mathbf{r} \in \mathbb{R}^3$, one has

$$\mathbf{r}' \cdot \mathbf{q} = Q(\mathbf{r} \cdot \mathbf{q})Q^R. \quad (58)$$

This is the quaternion version of the rotor action $\mathbf{r}' \cdot \sigma = R^R(\mathbf{r} \cdot \sigma)R$. In particular, choosing

$$Q = \cos\left(\frac{\theta}{2}\right) - \sin\left(\frac{\theta}{2}\right)\mathbf{u} \cdot \mathbf{q}$$

is equivalent to the rotor $R(\theta, \mathbf{u}) = \exp(\frac{\theta}{2}\mathbf{u} \cdot i\sigma)$, because $\mathbf{q} = -i\sigma$. Thus quaternion rotations are not a parallel construction: they are exactly the same circular half-angle algebra already developed in the STA basis.

1.13 Matrix and Auxiliary Realizations

The algebra above is complete on its own, but two familiar realizations are useful to record explicitly. The standard 2×2 matrix model in Eq. (59) and determinant identity (60) give a concrete realization of the generator algebra [9], while the auxiliary 2×2 map in Eq. (61) packages a complex paravector into a diagonalizable 2×2 operator built from the same conjugate and reverse structure [12, 15]. The quaternion correspondence has been collected in the preceding subsection because it uses the same sigma-number-two basis and the same circular exponential structure as Eq. (52).

1.13.1 Standard 2×2 Matrix Realization

Although the algebra has been developed directly in STA form, it is useful to close the algebraic portion with a single self-contained 2×2 matrix model. This realization confirms the earlier product rules in a familiar concrete form without replacing the algebraic viewpoint used throughout the paper [9].

The standard realization is

$$\sigma_1 = \{0, 1; 1, 0\}, \quad \sigma_2 = \{0, -i; i, 0\}, \quad \sigma_3 = \{1, 0; 0, -1\}, \quad (59)$$

Here the i appearing in the entries of σ_2 is strictly the ordinary complex unit $i := \sqrt{-1}$. The matrix entries are not defined recursively from the STA pseudoscalar; rather, the abstract pseudoscalar is represented afterward by the matrix product $\sigma_1 \sigma_2 \sigma_3 = i\mathbf{I}$. In this realization the basis identities become

$$\sigma_n^2 = \mathbf{I}, \quad \sigma_m \sigma_n = -\sigma_n \sigma_m \quad (m \neq n), \quad \sigma_1 \sigma_2 \sigma_3 = i\mathbf{I}.$$

So Eq. (59) gives a concrete 2×2 model for the generators and their pseudoscalar.

For a real paravector $X = t + \mathbf{r} \cdot \boldsymbol{\sigma} \in \mathbb{R} \oplus \mathbb{R}^3$, with $\mathbf{r} = (r_1, r_2, r_3) \in \mathbb{R}^3$, the matrix form is

$$X = t + \mathbf{r} \cdot \boldsymbol{\sigma} = \begin{Bmatrix} t + r_3 & r_1 - ir_2 \\ r_1 + ir_2 & t - r_3 \end{Bmatrix}, \quad (60)$$

$$\det X = t^2 - \mathbf{r}^2.$$

Equation (60) reproduces exactly the real scalar paravector metric already encoded by Eq. (38). The same matrix also carries the familiar trace and characteristic data:

$$\begin{aligned} \text{tr}(X) &= 2t, \\ \lambda^2 - (\text{tr } X)\lambda + \det X &= 0, \\ \lambda(\pm) &= t \pm \|\mathbf{r}\|. \end{aligned}$$

These eigenvalues are the matrix version of the split into scalar part t and vector magnitude $\|\mathbf{r}\|$.

Within the same realization, the matrix involutions can also be written explicitly. Let $(\cdot)^c$ denote entrywise complex conjugation and let $(\cdot)^{cT}$ denote complex conjugate transpose. Then

$$\begin{aligned} i^c &= -i, & \sigma_1^c &= \sigma_1, & \sigma_2^c &= -\sigma_2, & \sigma_3^c &= \sigma_3, \\ (\sigma_n)^{cT} &= \sigma_n, \\ (\sigma_n \sigma_m)^{cT} &= \sigma_m \sigma_n = -\sigma_n \sigma_m \quad (n \neq m), \end{aligned}$$

$$\begin{aligned} (i\mathbf{I})^{cT} &= -i\mathbf{I}, \\ (i\sigma_n)^{cT} &= ((\sigma_1 \sigma_2 \sigma_3) \sigma_n)^{cT} \\ &= \sigma_n^{cT} (\sigma_1 \sigma_2 \sigma_3)^{cT} \\ &= -i\sigma_n, \end{aligned}$$

and therefore

$$\begin{aligned} \mathbf{I}^R &= \mathbf{I}, & \sigma_n^R &= \sigma_n, \\ (i\mathbf{I})^R &= -i\mathbf{I}, & (i\sigma_n)^R &= -i\sigma_n, \\ (\cdot)^R &= (\cdot)^{cT}. \end{aligned}$$

So in the matrix model the reverse operator coincides with the Hermitian adjoint. The sigma conjugation formulas are generated by the fixed matrix

$$\mathbf{M} := \{0, 1; -1, 0\}, \quad \mathbf{M}^{-1} = -\mathbf{M}.$$

With this matrix one finds

$$\begin{aligned} \mathbf{I}^* &= \mathbf{I}, & \sigma_n^* &= -\sigma_n, \\ (i\mathbf{I})^* &= -i\mathbf{I}, & (i\sigma_n)^* &= i\sigma_n, \\ (\cdot)^* &= \mathbf{M}(\cdot)^c \mathbf{M}^{-1}, \\ (\cdot)^{*R} &= (\cdot)^{R*} = \mathbf{M}(\cdot)^T \mathbf{M}^{-1}. \end{aligned}$$

Thus the matrix section now contains, in one place, the basis matrices, the spacetime paravector matrix, the determinant/trace/eigenvalue formulas, complex conjugation, complex-conjugate transpose, reverse, sigma conjugation, and reverse sigma conjugation. These concrete formulas match the abstract operator rules defined earlier in the algebraic development.

1.13.2 Auxiliary 2×2 Realization

It is also useful to package a complex paravector $Z \in \mathbb{C} \oplus \mathbb{C}^3$ into the auxiliary 2×2 map

$$W(Z) := \{0, Z; Z^{*R}, 0\}. \quad (61)$$

$$W(Z)^2 = \text{diag}(ZZ^{*R}, Z^{*R}Z) = \langle\langle Z \rangle\rangle^2 \mathbf{I}. \quad (62)$$

Here $W(Z)$ is the block realization determined by Z and its reverse-conjugate. Equation (62) shows that W converts the scalar paravector metric into a quadratic matrix identity: the square of $W(Z)$ is the scalar $\langle\langle Z \rangle\rangle^2$ times the identity. The result is diagonal because the off-diagonal products reduce to the scalar paravector metric formulas already established in Eqs. (36) and (37) [12, 15].

2 Space-Time Kinematics

The same paravector structure also carries a space-time interpretation: the scalar paravector metric becomes the Minkowski metric of the interval, and normalized time-like paravectors become proper velocities. This interpretation follows the relativistic kinematic development of Lorentz, Einstein, and Minkowski, recast here in paravector form [4–6, 12, 13].

2.1 Differential space-time and proper velocity

Specializing the real paravector X from Eq. (11) to differentials gives

$$dX = dt + d\mathbf{r} \cdot \boldsymbol{\sigma} \quad (63)$$

and the interval is the differential version of Eq. (37):

$$ds^2 = \langle\langle dX \rangle\rangle^2 = \langle\langle dt + \boldsymbol{\sigma} \cdot d\mathbf{r} \rangle\rangle^2 = dX dX^* = dt^2 - d\mathbf{r}^2. \quad (64)$$

Here $t, s, \gamma \in \mathbb{R}$, $\mathbf{r}, \mathbf{v}, \mathbf{P} \in \mathbb{R}^3$, and $X, U \in \mathbb{R} \oplus \mathbb{R}^3$. The scalar part dt is the temporal separation and the vector part $d\mathbf{r} \cdot \boldsymbol{\sigma}$ is the spatial separation. Equation (64) is therefore not an added metric axiom but the direct spacetime interpretation of the scalar paravector metric already introduced in Eq. (37).

If $\mathbf{v} = d\mathbf{r}/dt \in \mathbb{R}^3$ and $\gamma = (1 - \mathbf{v}^2)^{-1/2} \in \mathbb{R}$, then

$$U := \frac{dX}{ds} = \gamma(1 + \mathbf{v} \cdot \boldsymbol{\sigma}), \quad \langle\langle U \rangle\rangle^2 = 1. \quad (65)$$

Multiplying Eq. (65) by the real rest mass $m \in \mathbb{R}$ gives the proper momentum formula

$$mU = E + \mathbf{P} \cdot \boldsymbol{\sigma}, \quad E = m\gamma, \quad \mathbf{P} = m\gamma\mathbf{v}, \quad (66)$$

which immediately yields $E^2 - \mathbf{P}^2 = m^2$. The normalization $\langle\langle U \rangle\rangle^2 = 1$ is the essential point: proper velocity is the unit time-like paravector tangent to the worldline. Multiplying by m then produces the energy-momentum paravector in a way that makes the relativistic mass shell relation appear immediately.

2.2 Proper acceleration and free motion

The proper acceleration is the real paravector

$$A := \frac{dU}{ds} = \frac{d^2 X}{ds^2}, \quad \text{scalar}(U^* A) = 0. \quad (67)$$

The orthogonality condition in Eq. (67) is the paravector analogue of the standard four-acceleration orthogonality relation. It is included here because the Lorentz-force law later in the paper should be read as a specific dynamical realization of this general kinematic structure.

A free-particle Lagrangian is

$$\mathcal{L}_{\text{free}} = -\frac{m}{\gamma}, \quad (68)$$

whose Euler-Lagrange equations reproduce inertial motion and the usual nonrelativistic expansion

$$\mathcal{L}_{\text{free}} = -m + \frac{1}{2}m\mathbf{v}^2 + O(\mathbf{v}^4). \quad (69)$$

Equation (69) is worth displaying explicitly because it shows how the relativistic action reduces to the classical kinetic expression plus an additive rest-energy constant when $\|\mathbf{v}\| \ll 1$.

2.3 Supporting Kinematic Derivations

The supporting derivations expand Eqs. (64), (65), and (66), then record the generalized-coordinate and metric remarks associated with the geometric part of the development.

2.3.1 Space-Time Numbers

A spacetime increment is the real paravector

$$dX = dt + dr_1\sigma_1 + dr_2\sigma_2 + dr_3\sigma_3 = dt + d\mathbf{r} \cdot \boldsymbol{\sigma},$$

with conjugate

$$dX^* = dt - dr_1\sigma_1 - dr_2\sigma_2 - dr_3\sigma_3 = dt - d\mathbf{r} \cdot \boldsymbol{\sigma}.$$

Here $dt \in \mathbb{R}$ is the scalar part and $d\mathbf{r} \in \mathbb{R}^3$ is the spatial vector part. Their recovery formulas are

$$\frac{dX + dX^*}{2} = dt, \quad \frac{dX - dX^*}{2} = d\mathbf{r} \cdot \boldsymbol{\sigma},$$

so

$$\text{scalar}(dX) = dt, \quad \text{vector}(dX) = d\mathbf{r}.$$

The paravector metric is

$$\langle\langle dX \rangle\rangle^2 = dX dX^* = dX^* dX = dt^2 - d\mathbf{r}^2 = ds^2.$$

It obeys the familiar Lorentzian identities

$$\begin{aligned} dt^2 > d\mathbf{r}^2, \quad dX \neq 0 &\Rightarrow \langle\langle dX \rangle\rangle^2 > 0, \\ \langle\langle dX + dX' \rangle\rangle^2 - \langle\langle dX \rangle\rangle^2 &= \langle\langle dX' \rangle\rangle^2 - \langle\langle dX \rangle\rangle^2 \\ &= 2(dt dt' - d\mathbf{r} \cdot d\mathbf{r}'), \\ \langle\langle a dX \rangle\rangle^2 &= a^2 \langle\langle dX \rangle\rangle^2, \quad a \in \mathbb{R}. \end{aligned}$$

The commutator of two spacetime increments is purely spatial:

$$\begin{aligned} [dX, dX'] &= [d\mathbf{r} \cdot \boldsymbol{\sigma}, d\mathbf{r}' \cdot \boldsymbol{\sigma}] \\ &= 2i(d\mathbf{r} \times d\mathbf{r}') \cdot \boldsymbol{\sigma}. \end{aligned}$$

Along a time-like path, the positive arc length satisfies

$$\begin{aligned} ds &= \sqrt{dt^2 - d\mathbf{r}^2} \\ &= \sqrt{1 - d\mathbf{r}^2/dt^2} dt \\ &= \sqrt{1 - \mathbf{v}^2} dt, \end{aligned}$$

where

$$\mathbf{v} := \frac{d\mathbf{r}}{dt} = (v_1, v_2, v_3) \in \mathbb{R}^3, \quad \gamma := \frac{dt}{ds} = \frac{1}{\sqrt{1 - \mathbf{v}^2}}.$$

For a real non-null increment, $dX^R = dX$, so the inverse exists and

$$\begin{aligned} (dX)^{-1} &= \frac{dX^*}{\langle\langle dX \rangle\rangle^2} \\ &= \frac{dX^*}{dX dX^*} \\ &= \frac{dX^*}{ds^2}. \end{aligned}$$

Null displacements are singled out algebraically because this inverse breaks down exactly when $ds^2 = 0$.

2.3.2 Interval Classes and Proper Time

For a spacetime increment $dX = dt + d\mathbf{r} \cdot \boldsymbol{\sigma}$, the interval is

$$\langle\langle dX \rangle\rangle^2 = dX dX^* = dt^2 - d\mathbf{r}^2 = ds^2.$$

This identity gives the three standard interval classes without any additional geometric machinery:

$$\begin{aligned} ds^2 > 0 &\Rightarrow \text{time-like,} \\ ds^2 = 0 &\Rightarrow \text{null,} \\ ds^2 < 0 &\Rightarrow \text{space-like.} \end{aligned}$$

If $\langle\langle dX \rangle\rangle^2 > 0$, the interval is time-like. Then the arc length s equals the proper time τ measured along the path:

$$\begin{aligned} \tau = s &= \int ds = \int d\tau = \int \sqrt{1 - \mathbf{v}^2} dt \\ &= \int \frac{1}{\gamma} dt. \end{aligned}$$

This is the time-dilation relation in natural units. In SI units the same statement is

$$ds = c d\tau.$$

If $\langle\langle dX \rangle\rangle^2 = 0$, the interval is null or light-like. Then

$$dt^2 = d\mathbf{r}^2 \quad \Rightarrow \quad dt = \pm \|d\mathbf{r}\|,$$

and $(dX)^{-1}$ does not exist.

If $\langle\langle dX \rangle\rangle^2 < 0$, the interval is space-like.

The advantage of the paravector notation is that the interval classification, the proper-time formula, and the null singularity of $(dX)^{-1}$ all come from the single algebraic identity $dX dX^* = dt^2 - d\mathbf{r}^2$.

2.3.3 4-Velocity (Proper Velocity)

The proper-velocity paravector is the normalized tangent to a time-like worldline:

$$\begin{aligned} \frac{dX}{ds} &= \frac{dt}{ds} + \frac{d\mathbf{r}}{ds} \cdot \boldsymbol{\sigma} \\ &= \frac{dt + \boldsymbol{\sigma} \cdot d\mathbf{r}}{\sqrt{dt^2 - d\mathbf{r}^2}} \\ &= \frac{dt/dt + \boldsymbol{\sigma} \cdot (d\mathbf{r}/dt)}{\sqrt{(dt/dt)^2 - (d\mathbf{r}/dt)^2}} \\ &= \frac{1 + \mathbf{v} \cdot \boldsymbol{\sigma}}{\sqrt{1 - \mathbf{v}^2}} \\ &= \gamma(1 + \mathbf{v} \cdot \boldsymbol{\sigma}). \end{aligned}$$

Its scalar and vector parts are therefore

$$\text{scalar}\left(\frac{dX}{ds}\right) = \gamma, \quad \text{vector}\left(\frac{dX}{ds}\right) = \gamma\mathbf{v}.$$

Its norm is unity:

$$\begin{aligned} \left\langle\left\langle \frac{dX}{ds} \right\rangle\right\rangle^2 &= \gamma^2(1 + \mathbf{v} \cdot \boldsymbol{\sigma})(1 - \mathbf{v} \cdot \boldsymbol{\sigma}) \\ &= \gamma^2(1 - \mathbf{v}^2) \\ &= 1. \end{aligned}$$

Also,

$$\frac{dX}{dt} = \frac{dX}{ds} \frac{ds}{dt} = \frac{1}{\gamma} \frac{dX}{ds}.$$

Define

$$\begin{aligned} U &:= \frac{dX}{ds} \\ &= \gamma + \gamma\mathbf{v} \cdot \boldsymbol{\sigma} \\ &= U_s + U_v \cdot \boldsymbol{\sigma}, \\ U_s &= \gamma, \quad U_v = \gamma\mathbf{v}. \end{aligned}$$

Since $ds^2 = dX dX^*$ with $ds > 0$,

$$\begin{aligned} \frac{dX^*}{ds} &= (dX)^{-1} ds \\ &= \left(\frac{dX}{ds}\right)^{-1}, \end{aligned}$$

so $U^{-1} = U^*$ and

$$\langle\langle U \rangle\rangle^2 = UU^* = 1.$$

Differentiating this normalization gives

$$\begin{aligned} \frac{d}{ds} \langle\langle U \rangle\rangle^2 &= \frac{dU}{ds} U^* + U \left(\frac{dU}{ds}\right)^* \\ &= \left(\frac{dU}{ds} U^*\right) + \left(\frac{dU}{ds} U^*\right)^{*R}, \end{aligned}$$

hence

$$\text{scalar}\left(U^* \frac{dU}{ds}\right) = 0.$$

Finally,

$$\begin{aligned} \langle\langle dX \rangle\rangle^2 &= dX dX^* \\ &= \left(\frac{dX}{ds} ds\right) \left(\left(\frac{dX}{ds}\right)^* ds\right) \\ &= UU^* ds^2 \\ &= \langle\langle U \rangle\rangle^2 ds^2 \\ &= ds^2. \end{aligned}$$

This normalization and orthogonality structure is the starting point for the proper-acceleration formulas below. The following table resolves the proper-velocity paravector into its scalar and spatial components.

Table 16. Components of dX/ds .

scalar:	γ
vector:	$\gamma\mathbf{v}$

2.3.4 4-Acceleration (Proper Acceleration)

Let

$$\mathbf{a} := \frac{d\mathbf{v}}{dt} = \{a_1, a_2, a_3\} \in \mathbb{R}^3.$$

With $\gamma = (1 - \mathbf{v}^2)^{-1/2}$, one has

$$\begin{aligned}\frac{d\gamma}{dt} &= \gamma^3(\mathbf{v} \cdot \mathbf{a}), \\ \frac{d\gamma}{ds} &= \gamma^4(\mathbf{v} \cdot \mathbf{a}).\end{aligned}$$

The velocity derivatives needed below are

$$\begin{aligned}\partial_{\mathbf{v}}\left(-\frac{1}{\gamma}\right) &= \gamma\mathbf{v}, \\ \partial_{\mathbf{v}}(\gamma\mathbf{v}) &= \gamma\mathbf{I}_3 + \gamma^3\mathbf{v}\mathbf{v}^T,\end{aligned}$$

so acting on $d\mathbf{v}/dt = \mathbf{a}$ gives

$$\begin{aligned}\frac{d(\gamma\mathbf{v})}{dt} &= (\partial_{\mathbf{v}}(\gamma\mathbf{v}))\frac{d\mathbf{v}}{dt} \\ &= \gamma\mathbf{a} + \gamma^3(\mathbf{v} \cdot \mathbf{a})\mathbf{v}, \\ \frac{d(\gamma\mathbf{v})}{ds} &= \frac{d(\gamma\mathbf{v})}{dt} \frac{dt}{ds} \\ &= \gamma^2\mathbf{a} + \gamma^4(\mathbf{v} \cdot \mathbf{a})\mathbf{v}.\end{aligned}$$

Recall that the proper-velocity paravector decomposes as

$$\begin{aligned}U &= U_s + U_{\mathbf{v}} \cdot \boldsymbol{\sigma}, \\ U_s &= \gamma, \quad U_{\mathbf{v}} = \gamma\mathbf{v}.\end{aligned}$$

Therefore the 4-acceleration, also called proper acceleration, is

$$\begin{aligned}A &:= \frac{dU}{ds} \\ &= \frac{d\gamma}{ds} + \frac{d(\gamma\mathbf{v})}{ds} \cdot \boldsymbol{\sigma} \\ &= \gamma^4(\mathbf{v} \cdot \mathbf{a}) + \left(\gamma^4(\mathbf{v} \cdot \mathbf{a})\mathbf{v} + \gamma^2\mathbf{a}\right) \cdot \boldsymbol{\sigma}.\end{aligned}$$

Equivalently,

$$A = A_s + A_{\mathbf{v}} \cdot \boldsymbol{\sigma}, \quad A_s = \gamma^4(\mathbf{v} \cdot \mathbf{a}), \quad A_{\mathbf{v}} = \gamma^4(\mathbf{v} \cdot \mathbf{a})\mathbf{v} + \gamma^2\mathbf{a}.$$

For $\mathbf{v} \neq 0$, let $\mathbf{u} = \mathbf{v}/\|\mathbf{v}\|$ with $\mathbf{u}^2 = 1$, and decompose

$$\begin{aligned}\text{par}(\mathbf{a}) &:= (\mathbf{a} \cdot \mathbf{u})\mathbf{u}, \\ \text{prp}(\mathbf{a}) &:= \mathbf{a} - \text{par}(\mathbf{a}).\end{aligned}$$

Then

$$\begin{aligned}\mathbf{a}^2 &= \text{par}(\mathbf{a})^2 + \text{prp}(\mathbf{a})^2, \\ (\mathbf{v} \cdot \mathbf{a})^2 &= \mathbf{v}^2 \text{par}(\mathbf{a})^2,\end{aligned}$$

so the proper-acceleration metric is

$$\begin{aligned}\langle\langle A \rangle\rangle^2 &= -\gamma^4(\mathbf{a}^2 + \gamma^2(\mathbf{v} \cdot \mathbf{a})^2) \\ &= -\gamma^4(\text{prp}(\mathbf{a})^2 + \gamma^2\text{par}(\mathbf{a})^2).\end{aligned}$$

This is strictly space-like unless the acceleration vanishes.

The products with U are

$$\begin{aligned}UA &= (U_s A_s + U_{\mathbf{v}} \cdot A_{\mathbf{v}}) \\ &\quad + (U_s A_{\mathbf{v}} + A_s U_{\mathbf{v}} + i(U_{\mathbf{v}} \times A_{\mathbf{v}})) \cdot \boldsymbol{\sigma}, \\ U^* A &= (U_s A_s - U_{\mathbf{v}} \cdot A_{\mathbf{v}}) \\ &\quad + (U_s A_{\mathbf{v}} - A_s U_{\mathbf{v}} - i(U_{\mathbf{v}} \times A_{\mathbf{v}})) \cdot \boldsymbol{\sigma}.\end{aligned}$$

Their component forms are

$$\begin{aligned}UA &= 2\gamma^5(\mathbf{v} \cdot \mathbf{a}) \\ &\quad + (\gamma^3\mathbf{a} + 2\gamma^5(\mathbf{v} \cdot \mathbf{a})\mathbf{v}) \cdot \boldsymbol{\sigma} \\ &\quad + i\gamma^3(\mathbf{v} \times \mathbf{a}) \cdot \boldsymbol{\sigma}, \\ U^* A &= \gamma^3\mathbf{a} \cdot \boldsymbol{\sigma} - i\gamma^3(\mathbf{v} \times \mathbf{a}) \cdot \boldsymbol{\sigma}.\end{aligned}$$

In particular,

$$\text{scalar}(U^* A) = 0,$$

which is the orthogonality relation between proper velocity and proper acceleration. Table 16 records the scalar and vector parts of U , while Tables 17 and 18 compare the corresponding scalar and vector parts of UA and U^*A . The following two tables compare the products UA and U^*A , making the orthogonality structure of proper acceleration explicit.

Table 17. Components of UA .

real scalar:	$2\gamma^5(\mathbf{v} \cdot \mathbf{a})$
img scalar:	0
real vector:	$\gamma^3\mathbf{a} + 2\gamma^5(\mathbf{v} \cdot \mathbf{a})\mathbf{v}$
img vector:	$\gamma^3(\mathbf{v} \times \mathbf{a})$

Table 18. Components of U^*A .

real scalar:	0
img scalar:	0
real vector:	$\gamma^3\mathbf{a}$
img vector:	$-\gamma^3(\mathbf{v} \times \mathbf{a})$

2.3.5 4-Momentum (Proper Momentum)

Multiplying proper velocity by the rest mass produces the energy-momentum paravector,

$$mU = E + \mathbf{P} \cdot \boldsymbol{\sigma}.$$

Its scalar part is the relativistic energy,

$$\begin{aligned}E &:= \text{scalar}(mU) \\ &= m\gamma.\end{aligned}$$

If $\mathbf{v} = 0$, then

$$E = m \quad \text{rest energy} \quad (= mc^2 \text{ in SI units}).$$

The kinetic energy is therefore $E - m$.

Its vector part is the relativistic momentum,

$$\begin{aligned}\mathbf{P} &:= \text{vector}(mU) \\ &= m\gamma\mathbf{v}.\end{aligned}$$

Because $\langle\langle U \rangle\rangle^2 = 1$, the mass shell follows immediately:

$$\begin{aligned}\langle\langle mU \rangle\rangle^2 &= m^2 \\ &= (E + \mathbf{P} \cdot \boldsymbol{\sigma})(E - \mathbf{P} \cdot \boldsymbol{\sigma}) \\ &= E^2 - \mathbf{P}^2.\end{aligned}$$

Hence

$$E^2 = \mathbf{P}^2 + m^2.$$

For $m = 0$, this reduces to $E^2 = \mathbf{P}^2$, which is the null case appropriate to lightlike or massless motion.

2.3.6 Generalized Spacetime

Let

$$\begin{aligned}f &: \mathbb{R} \times \mathbb{R}^3 \rightarrow \mathbb{R}, \\ \mathbf{g} &: \mathbb{R} \times \mathbb{R}^3 \rightarrow \mathbb{R}^3,\end{aligned}$$

and define the generalized spacetime paravector

$$X = f(t, \mathbf{r}) + \mathbf{g}(t, \mathbf{r}) \cdot \boldsymbol{\sigma}.$$

Its differential is

$$\begin{aligned}dX &= df + d\mathbf{g} \cdot \boldsymbol{\sigma} \\ &= (\partial_t f + \partial_t \mathbf{g} \cdot \boldsymbol{\sigma})dt + \sum_{a=1}^3 (\partial_{r_a} f + \partial_{r_a} \mathbf{g} \cdot \boldsymbol{\sigma})dr_a \\ &=: A dt + \mathbf{B} \cdot d\mathbf{r},\end{aligned}$$

where $A := \partial_t f + \partial_t \mathbf{g} \cdot \boldsymbol{\sigma}$ and $B_a := \partial_{r_a} f + \partial_{r_a} \mathbf{g} \cdot \boldsymbol{\sigma}$. Hence

$$\frac{dX}{dt} = A + \mathbf{B} \cdot \frac{d\mathbf{r}}{dt}.$$

The conjugate differential is

$$\begin{aligned}dX^* &= df - d\mathbf{g} \cdot \boldsymbol{\sigma} \\ &= (\partial_t f - \partial_t \mathbf{g} \cdot \boldsymbol{\sigma})dt + \sum_{a=1}^3 (\partial_{r_a} f - \partial_{r_a} \mathbf{g} \cdot \boldsymbol{\sigma})dr_a \\ &=: A^* dt + \mathbf{B}^* \cdot d\mathbf{r},\end{aligned}$$

so

$$\frac{dX^*}{dt} = A^* + \mathbf{B}^* \cdot \frac{d\mathbf{r}}{dt}.$$

The paravector metric is still determined by the scalar-vector split:

$$\begin{aligned}\langle\langle dX \rangle\rangle^2 &= dX dX^* \\ &= \left(\frac{dX}{dt}\right) \left(\frac{dX^*}{dt}\right) dt^2 \\ &= (df + d\mathbf{g} \cdot \boldsymbol{\sigma})(df - d\mathbf{g} \cdot \boldsymbol{\sigma}) \\ &= df^2 - d\mathbf{g}^2.\end{aligned}$$

Therefore the proper-time element is

$$\begin{aligned}ds &= \sqrt{df^2 - d\mathbf{g}^2} \\ &= \sqrt{1 - \left(\frac{d\mathbf{g}}{df}\right)^2} df.\end{aligned}$$

The corresponding proper velocity is

$$\begin{aligned}U &:= \frac{dX}{ds} \\ &= \frac{df + d\mathbf{g} \cdot \boldsymbol{\sigma}}{\sqrt{df^2 - d\mathbf{g}^2}} \\ &= \frac{1 + \left(\frac{d\mathbf{g}}{df}\right) \cdot \boldsymbol{\sigma}}{\sqrt{1 - \left(\frac{d\mathbf{g}}{df}\right)^2}} \\ &= \frac{1 + \mathbf{v} \cdot \boldsymbol{\sigma}}{\sqrt{1 - \mathbf{v}^2}} \\ &= \gamma(1 + \mathbf{v} \cdot \boldsymbol{\sigma}),\end{aligned}$$

where the geometric velocity and geometric Lorentz factor are

$$\begin{aligned}\mathbf{v} &:= \frac{d\mathbf{g}}{df} \\ &= \frac{d\mathbf{g}/dt}{df/dt} \\ &= \frac{d\mathbf{g}/ds}{df/ds}, \\ \gamma &:= \frac{df}{ds} \\ &= \frac{df}{\sqrt{df^2 - d\mathbf{g}^2}} \\ &= \frac{df/dt}{ds/dt} \\ &= \frac{1}{\sqrt{1 - \mathbf{v}^2}}.\end{aligned}$$

The timelike condition is

$$\mathbf{v}^2 < 1.$$

With this reparametrization the energy-momentum relation is unchanged:

$$mU = E + \mathbf{P} \cdot \boldsymbol{\sigma}.$$

The geometric acceleration is

$$\begin{aligned}\mathbf{a} &:= \frac{d\mathbf{v}}{ds} \\ &= \frac{d}{ds} \left(\frac{d\mathbf{g}}{df}\right) \\ &= \frac{df}{ds} \frac{d}{df} \left(\frac{d\mathbf{g}}{df}\right) \\ &= \gamma \frac{d^2 \mathbf{g}}{df^2}.\end{aligned}$$

Because the proper-velocity algebra is unchanged, the proper acceleration still has the form

$$\frac{dU}{ds} = \gamma^4(\mathbf{v} \cdot \mathbf{a}) + (\gamma^4(\mathbf{v} \cdot \mathbf{a})\mathbf{v} + \gamma^2 \mathbf{a}) \cdot \boldsymbol{\sigma}.$$

If

$$f' := \frac{df}{dt}, \quad \mathbf{g}' := \frac{d\mathbf{g}}{dt}, \quad f'' := \frac{df'}{dt}, \quad \mathbf{g}'' := \frac{d\mathbf{g}'}{dt},$$

then

$$\frac{d\mathbf{v}}{dt} = \frac{d}{dt} \left(\frac{\mathbf{g}'}{f'} \right) = \frac{\mathbf{g}'' f' - \mathbf{g}' f''}{(f')^2}.$$

For comparison, a plane curve $(x(t), y(t))$ has curvature

$$k = \frac{|x' y'' - y' x''|}{(x'^2 + y'^2)^{3/2}},$$

so the same derivative numerator controls bending in the generalized coordinates. The commutator of proper velocity and proper acceleration is

$$\begin{aligned} \left[U, \frac{dU}{ds} \right] &= \left[U, \frac{dU}{dt} \right] \frac{dt}{ds} \\ &= \gamma \left[U, \frac{dU}{dt} \right] \\ &= 2\gamma^3 (\mathbf{v} \times \mathbf{a}) \cdot (i\boldsymbol{\sigma}). \end{aligned}$$

For inertial motion,

$$m \frac{dU}{ds} = 0,$$

which is the geodesic equation, equivalently

$$\frac{d^2 X}{ds^2} = 0.$$

Its scalar and vector parts are

$$\begin{aligned} \gamma^3 (\mathbf{v} \cdot \mathbf{a}) &= 0, \\ \gamma^3 (\mathbf{v} \cdot \mathbf{a}) \mathbf{v} + \gamma \mathbf{a} &= 0. \end{aligned}$$

The second relation may be written as

$$\gamma M \mathbf{a} = 0, \quad M := \mathbf{I}_3 + \gamma^2 \mathbf{v} \mathbf{v}^T = \mathbf{I}_3 + \frac{\mathbf{v} \mathbf{v}^T}{1 - \mathbf{v}^2}.$$

For $\mathbf{v}^2 < 1$, the matrix M is positive definite, so $\gamma M \mathbf{a} = 0$ implies $\mathbf{a} = 0$. Hence

$$\frac{d^2 \mathbf{g}}{df^2} = 0,$$

and along an inertial worldline

$$\mathbf{g} = \mathbf{c}_1 f + \mathbf{c}_2, \quad \mathbf{v} = \frac{d\mathbf{g}}{df} = \mathbf{c}_1,$$

so inertial motion is linear in the $(f, \mathbf{g})$ coordinates with constant geometric velocity.

2.3.7 Lagrangian Mechanics and Conserved Quantities

For a particle trajectory written in terms of time, position, and velocity, let the action be

$$S = \int \mathcal{L}(t, \mathbf{r}, \mathbf{v}) dt.$$

The Euler–Lagrange equation is then

$$\begin{aligned} \left(\boldsymbol{\nabla} - \frac{d}{dt} \partial_{\mathbf{v}} \right) \mathcal{L} &= 0, \\ \frac{d}{dt} \partial_{\mathbf{v}} \mathcal{L} &= \boldsymbol{\nabla} \mathcal{L}. \end{aligned}$$

This makes the standard conserved quantities immediate. The canonical momentum is

$$\partial_{\mathbf{v}} \mathcal{L},$$

so if $\boldsymbol{\nabla} \mathcal{L} = 0$, then

$$\frac{d}{dt} \partial_{\mathbf{v}} \mathcal{L} = 0,$$

and the canonical momentum is constant.

The Hamiltonian is defined by

$$\begin{aligned} \mathcal{H} &= \mathbf{v} \cdot (\partial_{\mathbf{v}} \mathcal{L}) - \mathcal{L} \\ &= (\mathbf{v} \cdot \partial_{\mathbf{v}} - 1) \mathcal{L}. \end{aligned}$$

Its time derivative is

$$\begin{aligned} \frac{d\mathcal{H}}{dt} &= \frac{d}{dt} (\partial_{\mathbf{v}} \mathcal{L}) \cdot \mathbf{v} + (\partial_{\mathbf{v}} \mathcal{L}) \cdot \frac{d\mathbf{v}}{dt} - \frac{d\mathcal{L}}{dt} \\ &= (\boldsymbol{\nabla} \mathcal{L}) \cdot \mathbf{v} + (\partial_{\mathbf{v}} \mathcal{L}) \cdot \frac{d\mathbf{v}}{dt} - \frac{d\mathcal{L}}{dt} \\ &= (\boldsymbol{\nabla} \mathcal{L}) \cdot \mathbf{v} + (\partial_{\mathbf{v}} \mathcal{L}) \cdot \frac{d\mathbf{v}}{dt} \\ &\quad - \left(\partial_t \mathcal{L} + (\boldsymbol{\nabla} \mathcal{L}) \cdot \mathbf{v} + (\partial_{\mathbf{v}} \mathcal{L}) \cdot \frac{d\mathbf{v}}{dt} \right) \\ &= -\partial_t \mathcal{L}. \end{aligned}$$

Therefore, if $\partial_t \mathcal{L} = 0$, then

$$\frac{d\mathcal{H}}{dt} = 0,$$

so the Hamiltonian is constant. The free-particle and minimally coupled electromagnetic examples below are direct applications of these identities.

2.3.8 Action of a Free Particle

The free-particle action is the mass-weighted proper time along the world line. With $\tau = \int ds$, $U = dX/ds$, and $\gamma = dt/ds$,

$$\begin{aligned} S_{\text{free}} &= \int \mathcal{L}_{\text{free}} dt = -m\tau \\ &= -m \int \langle\langle dX \rangle\rangle \\ &= -m \int ds \\ &= -m \int \left\langle\left\langle \frac{dX}{dt} \right\rangle\right\rangle dt \\ &= -m \int \frac{ds}{dt} dt \\ &= -m \int \left\langle\left\langle \frac{U}{\gamma} \right\rangle\right\rangle dt \end{aligned}$$

$$\begin{aligned}
&= -m \int \frac{1}{\gamma} dt \\
&= -m \int \sqrt{1 - \mathbf{v}^2} dt.
\end{aligned}$$

Therefore

$$\mathcal{L}_{\text{free}} = -\frac{m}{\gamma}.$$

Differentiating with respect to velocity gives the canonical momentum,

$$\begin{aligned}
\partial_{\mathbf{v}} \mathcal{L}_{\text{free}} &= m\gamma \mathbf{v} \\
&= \mathbf{P}, \\
(\partial_{\mathbf{v}} \cdot \boldsymbol{\sigma}) \mathcal{L}_{\text{free}} &= \mathbf{P} \cdot \boldsymbol{\sigma}.
\end{aligned}$$

The Hamiltonian is

$$\begin{aligned}
\mathcal{H} &= (\partial_{\mathbf{v}} \mathcal{L}_{\text{free}}) \cdot \mathbf{v} - \mathcal{L}_{\text{free}} \\
&= m\gamma \mathbf{v}^2 + \frac{m}{\gamma} \\
&= m\gamma \left(\mathbf{v}^2 + \frac{1}{\gamma^2} \right) \\
&= m\gamma \\
&= E.
\end{aligned}$$

Because the free Lagrangian has no explicit time or position dependence,

$$\partial_t \mathcal{L}_{\text{free}} = 0, \quad \nabla \mathcal{L}_{\text{free}} = 0,$$

so the conserved quantities are the relativistic energy $E = m\gamma$ and momentum $\mathbf{P} = m\gamma \mathbf{v}$.

2.3.9 Non-Relativistic Approximation

For the non-relativistic approximation, expand the free Lagrangian directly:

$$\begin{aligned}
\mathcal{L}_{\text{free}} &= -\frac{m}{\gamma} \\
&= -m + \frac{1}{2}m\mathbf{v}^2 + O(\mathbf{v}^4).
\end{aligned}$$

Then the canonical momentum becomes

$$\begin{aligned}
\partial_{\mathbf{v}} \mathcal{L}_{\text{free}} &= \frac{\partial}{\partial \mathbf{v}} \left(\frac{1}{2}m\mathbf{v}^2 + O(\mathbf{v}^4) \right) \\
&= m\mathbf{v} + O(\mathbf{v}^3) \\
&\approx m\mathbf{v}.
\end{aligned}$$

The total energy is the kinetic term plus the rest-mass term:

$$\begin{aligned}
\mathcal{H} &= (m\mathbf{v} + O(\mathbf{v}^3)) \cdot \mathbf{v} + m - \frac{1}{2}m\mathbf{v}^2 - O(\mathbf{v}^4) \\
&= m + \frac{1}{2}m\mathbf{v}^2 + O(\mathbf{v}^4) \\
&\approx m + \frac{1}{2}m\mathbf{v}^2.
\end{aligned}$$

Equivalently, the low-velocity series can be summarized as

$$\begin{aligned}
\mathcal{L}_{\text{free}} &= -m + \frac{1}{2}m\mathbf{v}^2 + O(\mathbf{v}^4), \\
\mathbf{P} &= m\mathbf{v} + O(\mathbf{v}^3), \\
\mathcal{H} &= m + \frac{1}{2}m\mathbf{v}^2 + O(\mathbf{v}^4)
\end{aligned}$$

shows exactly how the relativistic action contains the Newtonian kinetic term together with the additive rest-energy constant. Classical mechanics therefore appears as the approximation, not as the starting point.

2.3.10 Generalized Coordinate Fields

The preceding derivations depend only on the paravector algebra and the Minkowski metric, not on the particular choice of scalar coordinate t and vector coordinate $\mathbf{r}$. For a more general real paravector coordinate field,

$$X = f(t, \mathbf{r}) + \mathbf{g}(t, \mathbf{r}) \cdot \boldsymbol{\sigma}, \quad dX = df + d\mathbf{g} \cdot \boldsymbol{\sigma},$$

with $f \in \mathbb{R}$, $\mathbf{g} \in \mathbb{R}^3$, and $X \in \mathbb{R} \oplus \mathbb{R}^3$, one still has the same interval formula

$$\langle\langle dX \rangle\rangle^2 = dX dX^* = df^2 - d\mathbf{g}^2.$$

If df is time-like, the same algebraic steps as before give

$$\begin{aligned}
ds &= \sqrt{df^2 - d\mathbf{g}^2}, \quad \mathbf{v} = \frac{d\mathbf{g}}{df}, \\
U &= \frac{dX}{ds} = \gamma(1 + \mathbf{v} \cdot \boldsymbol{\sigma}), \quad \gamma = \frac{df}{ds}.
\end{aligned}$$

Thus the proper-velocity formula is a structural consequence of the paravector interval itself; changing coordinates changes only the functions f and $\mathbf{g}$, not the algebraic form of the result.

2.3.11 Metric and Diagonalization

For comparison with matrix notation, write the coordinate tuple as

$$x = (t, r_1, r_2, r_3)^T, \quad dx = (dt, dr_1, dr_2, dr_3)^T.$$

Then the interval may be written as

$$ds^2 = dx^T Q(x) dx$$

for a symmetric matrix $Q(x) = Q(x)^T$. The eigenvalue problem is

$$\det(Q - \lambda_n \mathbf{I}) = 0, \quad (Q - \lambda_n \mathbf{I})u_n = 0,$$

so the spectral data may be assembled as

$$D = \text{diag}(\lambda_0, \lambda_1, \lambda_2, \lambda_3), \quad N = (u_0 \ u_1 \ u_2 \ u_3).$$

Because Q is symmetric, the eigenvalues are real and the eigenvectors may be chosen orthonormally, giving

$$N^T N = N N^T = \mathbf{I}, \quad N^{-1} = N^T.$$

When Q is invertible, this yields

$$Q^{-1} = ND^{-1}N^T,$$

and in general

$$\begin{aligned} Q &= NDN^T \\ &= \sum_n \lambda_n u_n u_n^T. \end{aligned}$$

Therefore

$$\begin{aligned} ds^2 &= dx^T Q dx \\ &= \sum_n \lambda_n (u_n \cdot dx)^2. \end{aligned}$$

Formally, one may absorb the eigenvalues into rescaled coordinates,

$$dx'_n := \sqrt{\lambda_n} (u_n \cdot dx),$$

so that

$$ds^2 = dx' \cdot dx'.$$

This diagonalized rewrite can be useful, but it does not erase the Lorentzian sign structure already built into $dXdX^*$. For an indefinite metric some λ_n carry negative sign, so the interval is not thereby converted into a positive-definite Euclidean norm. Diagonalization reorganizes the quadratic form; the Minkowski-signature content remains the primary statement.

3 Covariant Transform Applications

Section 1.11 established the generic similarity law (53), the circular rotor law (52), the hyperbolic boost law (49), and the ordered-product identities that accompany them. Here those formulas are applied to the interval geometry of Section 2 to obtain their spacetime interpretation.

When the real paravector $X = t + \mathbf{r} \cdot \boldsymbol{\sigma}$ is read as the spacetime object defined by Eqs. (63) and (64), Eq. (49) is the Lorentz transformation and the invariance of $\langle\langle X \rangle\rangle^2$ becomes the invariance of the spacetime interval. The same earlier transform formulas are then reused in Section 5 for field transformations and in Section 6 for ordered operator constructions, so the physical sections can refer back to one common algebraic source rather than restating it.

4 Differential Operators

The electromagnetic and wave-equation sections use one fixed set of differential conventions. They are collected here once so the later sections can refer back to the same numbered equations instead of redefining the operators in multiple places.

For spatial derivatives with respect to $\mathbf{r} = (r_1, r_2, r_3)$, define

$$\partial_{\mathbf{r}} := \left\{ \frac{\partial}{\partial r_1}, \frac{\partial}{\partial r_2}, \frac{\partial}{\partial r_3} \right\}. \quad (70)$$

The gradient notation is then

$$\boldsymbol{\nabla} := \partial_{\mathbf{r}} = \left\{ \frac{\partial}{\partial r_1}, \frac{\partial}{\partial r_2}, \frac{\partial}{\partial r_3} \right\} = \{\partial_1, \partial_2, \partial_3\}. \quad (71)$$

For shorthand, the coordinate symbol $\mathbf{r}$ is dropped in the last form, so $\partial_i := \partial/\partial r_i$ for $i \in \{1, 2, 3\}$. The corresponding spatial Laplacian is

$$\boldsymbol{\nabla}^2 := \partial_1^2 + \partial_2^2 + \partial_3^2. \quad (72)$$

The STA differential paravector and its conjugate are

$$\partial := \partial_t + \boldsymbol{\sigma} \cdot \boldsymbol{\nabla} = \partial_t + \sigma_1 \partial_1 + \sigma_2 \partial_2 + \sigma_3 \partial_3. \quad (73)$$

$$\partial^* := \partial_t - \boldsymbol{\sigma} \cdot \boldsymbol{\nabla} = \partial_t - \sigma_1 \partial_1 - \sigma_2 \partial_2 - \sigma_3 \partial_3. \quad (74)$$

Their product gives the d'Alembertian

$$\begin{aligned} \square &:= \langle\langle \partial \rangle\rangle^2 \\ &= \partial \partial^* \end{aligned} \quad (75)$$

$$\begin{aligned} &= \partial^* \partial \\ &= \partial_t^2 - \boldsymbol{\nabla}^2. \end{aligned} \quad (76)$$

4.1 Complex Fields

For a generic complex paravector field, write

$$\begin{aligned} \partial &= \partial_t + \boldsymbol{\nabla} \cdot \boldsymbol{\sigma}, \\ F &= (a + ib) + (\mathbf{A} + i\mathbf{B}) \cdot \boldsymbol{\sigma}. \end{aligned}$$

Assume

$$\begin{aligned} a &: \mathbb{R} \times \mathbb{R}^3 \rightarrow \mathbb{R}, & \mathbf{A} &: \mathbb{R} \times \mathbb{R}^3 \rightarrow \mathbb{R}^3, \\ b &: \mathbb{R} \times \mathbb{R}^3 \rightarrow \mathbb{R}, & \mathbf{B} &: \mathbb{R} \times \mathbb{R}^3 \rightarrow \mathbb{R}^3, \end{aligned}$$

so that

$$F : \mathbb{R} \times \mathbb{R}^3 \rightarrow \mathbb{C} \oplus \mathbb{C}^3.$$

The real/imaginary scalar/vector parts of ∂F and $i\partial F$ are then: Tables 19 and 20 record the scalar and vector parts of ∂F and $i\partial F$ for the generic complex paravector field $F = (a + ib) + (\mathbf{A} + i\mathbf{B}) \cdot \boldsymbol{\sigma}$.

Table 19. Components of ∂F .

real scalar:	$\partial_t a + \boldsymbol{\nabla} \cdot \mathbf{A}$
img scalar:	$\partial_t b + \boldsymbol{\nabla} \cdot \mathbf{B}$
real vector:	$\boldsymbol{\nabla} a + \partial_t \mathbf{A} - \boldsymbol{\nabla} \times \mathbf{B}$
img vector:	$\boldsymbol{\nabla} b + \boldsymbol{\nabla} \times \mathbf{A} + \partial_t \mathbf{B}$

Table 20. Components of $i\partial F$.

real scalar:	$-\partial_t b - \boldsymbol{\nabla} \cdot \mathbf{B}$
img scalar:	$\partial_t a + \boldsymbol{\nabla} \cdot \mathbf{A}$
real vector:	$-\boldsymbol{\nabla} b - \boldsymbol{\nabla} \times \mathbf{A} - \partial_t \mathbf{B}$
img vector:	$\boldsymbol{\nabla} a + \partial_t \mathbf{A} - \boldsymbol{\nabla} \times \mathbf{B}$

5 Electro-Magnetic Fields

Electro-magnetic fields, gauge structure, and the Lorentz force admit compact STA forms. In physical content this is Maxwell theory [2], while the paravector packaging follows the later STA presentations of Hestenes, Baylis, and Doran–Lasenby [12, 13, 15].

5.1 Fields, potentials, and Maxwell equations

The differential conventions are fixed once in Section 4; in particular, Eqs. (70)–(76) define the spatial derivative vector, the short-hand ∂_i , the paravector derivative, its conjugate, and the d'Alembertian. With those operators fixed, and with $V, \rho, \lambda, S, q \in \mathbb{R}$, $\mathbf{E}, \mathbf{B}, \mathbf{A}, \mathbf{J} \in \mathbb{R}^3$, the electromagnetic field vector and potential are written as

$$\mathbf{F} := \mathbf{E} + i\mathbf{B}, \quad F := \mathbf{F} \cdot \boldsymbol{\sigma}, \quad \Phi = V + \mathbf{A} \cdot \boldsymbol{\sigma}. \quad (77)$$

Here $\Phi \in \mathbb{R} \oplus \mathbb{R}^3$, $\mathbf{F} \in \mathbb{C}^3$ is the complex field vector, and $F \in \mathbb{O} \oplus \mathbb{C}^3$ is the corresponding STA field paravector. Equation (77) packages electric and magnetic fields into one object $\mathbf{F}$, scalar and vector potentials into one object Φ , and then lets the single symbol F carry the STA field notation used below.

5.2 Boost transformation of electro-magnetic fields

The same boost element L from Eq. (49) acts on the field paravector by

$$F' = L^* F L. \quad (78)$$

Equation (78) is the direct field analogue of the event transform $L^* X L$. It shows that F is not an extra object appended to the mechanics, but part of the same algebraic structure: boosts act on the field paravector by the same two-sided rule that acts on space-time paravectors, and the usual parallel/perpendicular split of $\mathbf{E}$ and $\mathbf{B}$ follows from the same boost algebra.

The identity

$$\partial\Phi = S + F^*, \quad S := \partial_t V + \boldsymbol{\nabla} \cdot \mathbf{A} \quad (79)$$

encodes

$$\mathbf{E} = -\partial_t \mathbf{A} - \boldsymbol{\nabla} V, \quad \mathbf{B} = \boldsymbol{\nabla} \times \mathbf{A}. \quad (80)$$

The compact Maxwell equation is

$$\partial F = \rho - \mathbf{J} \cdot \boldsymbol{\sigma}. \quad (81)$$

Splitting real/imaginary scalar/vector parts of (81) reproduces

$$\boldsymbol{\nabla} \cdot \mathbf{E} = \rho, \quad \boldsymbol{\nabla} \cdot \mathbf{B} = 0, \quad (82)$$

$$\boldsymbol{\nabla} \times \mathbf{B} = \mathbf{J} + \partial_t \mathbf{E}, \quad \boldsymbol{\nabla} \times \mathbf{E} = -\partial_t \mathbf{B}. \quad (83)$$

Equations (82) and (83) are standard STA/GA electrodynamics up to sign conventions fixed by Eqs. (73), (74), (76), and (77); component by component they recover Maxwell's field equations in the present signature and field convention [2, 12, 13, 16]. Equation (81) avoids taking divergence and curl as separate primitive operators at this stage. They emerge automatically when the scalar and vector parts of the single STA equation are separated.

5.3 Wave equations, gauge, and Lorentz force

Applying ∂^* to (81) gives

$$\square F = \partial^*(\rho - \mathbf{J} \cdot \boldsymbol{\sigma}), \quad (84)$$

whose scalar part yields charge conservation,

$$\partial_t \rho + \boldsymbol{\nabla} \cdot \mathbf{J} = 0. \quad (85)$$

Equation (85) is the continuity equation written in the present notation. For a real scalar gauge function $\lambda \in \mathbb{R}$,

$$\Phi' = \Phi + \partial^* \lambda, \quad F' = F. \quad (86)$$

This gauge transformation shows why the potential Φ is retained even when the field is physically primary: Φ carries the redundancy, while F carries the gauge-invariant derivative content.

For a particle of real mass $m \in \mathbb{R}$, real charge $q \in \mathbb{R}$, and proper velocity $U \in \mathbb{R} \oplus \mathbb{R}^3$, the proper-force formula is

$$m \frac{dU}{ds} = q \operatorname{real}(FU) = q\gamma(\mathbf{v} \cdot \mathbf{E} + (\mathbf{E} + \mathbf{v} \times \mathbf{B}) \cdot \boldsymbol{\sigma}), \quad (87)$$

which packages both relativistic power and the Lorentz three-force. The real part in Eq. (87) extracts the physically relevant part of the product FU . The scalar term gives power transfer and the vector term gives the ordinary Lorentz force, so the standard split between energy change and momentum change is preserved inside one STA formula.

5.4 Lagrangian and energy-momentum remarks

The electromagnetic particle Lagrangian is

$$\mathcal{L} = -\frac{m}{\gamma} - q(V - \mathbf{v} \cdot \mathbf{A}), \quad (88)$$

and its Euler–Lagrange equations return $d\mathbf{P}/dt = q(\mathbf{E} + \mathbf{v} \times \mathbf{B})$. Here the free action from Eq. (68) is supplemented by the potential term, and the field-strength form of the force law reappears after variation.

A natural paravector candidate for field energy density and flux is

$$T := \frac{1}{2} F F^R = \frac{1}{2} (\mathbf{E}^2 + \mathbf{B}^2) + (\mathbf{E} \times \mathbf{B}) \cdot \boldsymbol{\sigma}. \quad (89)$$

Caution. Equation (89) is a useful frame-dependent energy-density/Poynting paravector, but by itself it is not the full electromagnetic stress-energy tensor. In STA language the full stress-energy object is more naturally a linear map such as $T(a) = \frac{1}{2} F a F^R$ [12, 15].

5.5 Supporting Electromagnetic Derivations

The supporting derivations write out the field decomposition, potential reconstruction, source formulas, gauge transformations, force-law expansions, and variational identities underlying Eqs. (70)–(89).

5.5.1 Electricity and Magnetism

This subsection collects the differential notation, the electromagnetic field paravector, its simplest quadratic combinations, and the component form of Maxwell's equation in one place. Here $\rho, S \in \mathbb{R}$, $\mathbf{E}, \mathbf{B}, \mathbf{J} \in \mathbb{R}^3$, $\mathbf{F} = \mathbf{E} + i\mathbf{B} \in \mathbb{C}^3$, and $F = \mathbf{F} \cdot \boldsymbol{\sigma} \in \mathbb{O} \oplus \mathbb{C}^3$. The spatial derivative conventions are

$$\boldsymbol{\nabla} = \{\partial_1, \partial_2, \partial_3\},$$

$$\partial_i := \partial/\partial r_i,$$

$$\nabla^2 = \partial_1^2 + \partial_2^2 + \partial_3^2,$$

and the differential paravector is

$$\begin{aligned}\partial &= \partial_t + \boldsymbol{\sigma} \cdot \boldsymbol{\nabla} \\ &= \partial_t + \sigma_1 \partial_1 + \sigma_2 \partial_2 + \sigma_3 \partial_3, \\ \partial^* &= \partial_t - \boldsymbol{\sigma} \cdot \boldsymbol{\nabla}.\end{aligned}$$

The generic real/imaginary scalar/vector decomposition of ∂F and $i\partial F$ has already been recorded in Subsection 4.1. With that operator bookkeeping fixed, the electric and magnetic fields can be combined without introducing any new multiplication rule.

Collect the electric and magnetic fields into the complex three-vector $\mathbf{F}$ and the field paravector F :

$$\begin{aligned}\mathbf{E} &= \{E_1, E_2, E_3\} && \text{real vector electric field,} \\ \mathbf{B} &= \{B_1, B_2, B_3\} && \text{real vector magnetic field,} \\ \mathbf{F} &= \mathbf{E} + i\mathbf{B} && \text{electromagnetic field,} \\ F &= \mathbf{F} \cdot \boldsymbol{\sigma} = (\mathbf{E} + i\mathbf{B}) \cdot \boldsymbol{\sigma}, && i = \sigma_1 \sigma_2 \sigma_3.\end{aligned}$$

$$\begin{aligned}F^* &= (-\mathbf{E} + i\mathbf{B}) \cdot \boldsymbol{\sigma}, \\ &= -\mathbf{F}^* \cdot \boldsymbol{\sigma}.\end{aligned}$$

The two simplest quadratic field combinations are

$$\begin{aligned}\frac{1}{2}F^2 &= \frac{1}{2}(\mathbf{E}^2 - \mathbf{B}^2) + i\mathbf{E} \cdot \mathbf{B}, \\ \frac{1}{2}FF^R &= \frac{1}{2}(\mathbf{E}^2 + \mathbf{B}^2) + (\mathbf{E} \times \mathbf{B}) \cdot \boldsymbol{\sigma}.\end{aligned}$$

Thus $\frac{1}{2}F^2$ contains the Lorentz invariants, while $\frac{1}{2}FF^R$ separates into the energy density and the Poynting vector.

Define the real source paravector by

$$\begin{aligned}D &= \rho + \mathbf{J} \cdot \boldsymbol{\sigma} \in \mathbb{R} \oplus \mathbb{R}^3, \\ D^* &= \rho - \mathbf{J} \cdot \boldsymbol{\sigma}.\end{aligned}$$

Here $\rho \in \mathbb{R}$ is the scalar charge density and $\mathbf{J} \in \mathbb{R}^3$ is the current-density vector. Maxwell's equations then read

$$\begin{aligned}\partial F &= D^* && \text{Maxwell's equations,} \\ &= \rho - \mathbf{J} \cdot \boldsymbol{\sigma}.\end{aligned}$$

$$\partial^* F^* = D.$$

In this form, the scalar, sigma-number-one, and sigma-number-two parts of ∂F reproduce the four classical Maxwell equations in one step. Expanding the product shows where each component sits.

Expanding ∂F into scalar, vector, and bivector parts gives

$$\begin{aligned}\partial F &= (\partial_t + \boldsymbol{\sigma} \cdot \boldsymbol{\nabla})((\mathbf{E} + i\mathbf{B}) \cdot \boldsymbol{\sigma}), \\ &= (\boldsymbol{\nabla} \cdot \mathbf{E}), \\ &\quad + (\boldsymbol{\nabla} \cdot \mathbf{B})i, \\ &\quad + (\partial_t \mathbf{E} - \boldsymbol{\nabla} \times \mathbf{B}) \cdot \boldsymbol{\sigma}, \\ &\quad + (\partial_t \mathbf{B} + \boldsymbol{\nabla} \times \mathbf{E}) \cdot (i\boldsymbol{\sigma}).\end{aligned}$$

The first two lines are the scalar and pseudoscalar parts. The last two lines are the sigma-number-one and sigma-number-two

vector parts. Matching these components with $D^* = \rho - \mathbf{J} \cdot \boldsymbol{\sigma}$ immediately forces the magnetic-divergence term and Faraday term to vanish while fixing the electric source terms.

Accordingly, the component equations of $\partial F = \rho - \mathbf{J} \cdot \boldsymbol{\sigma}$ are

$$\begin{aligned}\boldsymbol{\nabla} \cdot \mathbf{E} &= \rho, \\ \boldsymbol{\nabla} \cdot \mathbf{B} &= 0, \\ \partial_t \mathbf{E} - \boldsymbol{\nabla} \times \mathbf{B} &= -\mathbf{J}, \\ \partial_t \mathbf{B} + \boldsymbol{\nabla} \times \mathbf{E} &= 0.\end{aligned}$$

Separating the vector components then gives

$$\begin{aligned}\boldsymbol{\nabla} \times \mathbf{B} &= \mathbf{J} + \partial_t \mathbf{E}, \\ \boldsymbol{\nabla} \times \mathbf{E} &= -\partial_t \mathbf{B}.\end{aligned}$$

5.5.2 Lorentz Transform of Fields

This subsection specializes the boost law to the electromagnetic field paravector

$$F = (\mathbf{E} + i\mathbf{B}) \cdot \boldsymbol{\sigma}.$$

For a boost $L = \exp(\frac{1}{2}\theta \mathbf{u} \cdot \boldsymbol{\sigma})$ with $\mathbf{u}^2 = 1$, $\beta = \tanh(\theta)$, and $\gamma = \cosh(\theta)$, the transformed field and its parallel/perpendicular parts relative to $\mathbf{u}$ obey

$$\begin{aligned}F' &= L^* F L, \\ \text{par}(\mathbf{E}') &= \text{par}(\mathbf{E}), \\ \text{par}(\mathbf{B}') &= \text{par}(\mathbf{B}), \\ \text{prp}(\mathbf{E}') &= \gamma(\text{prp}(\mathbf{E}) + \beta \mathbf{u} \times \mathbf{B}), \\ \text{prp}(\mathbf{B}') &= \gamma(\text{prp}(\mathbf{B}) - \beta \mathbf{u} \times \mathbf{E}).\end{aligned}$$

Equivalently, the transformed fields can be written in the compact form

$$\begin{aligned}\mathbf{E}' &= \text{par}(\mathbf{E}) + \gamma(\text{prp}(\mathbf{E}) + \beta \mathbf{u} \times \mathbf{B}), \\ \mathbf{B}' &= \text{par}(\mathbf{B}) + \gamma(\text{prp}(\mathbf{B}) - \beta \mathbf{u} \times \mathbf{E}), \\ \partial' &= L^* \partial L, \\ \partial' F' &= L^*(\rho - \mathbf{J} \cdot \boldsymbol{\sigma})L.\end{aligned}$$

The parallel components are unchanged, while the perpendicular components mix by the standard electric-magnetic cross terms. This is the component form of the single algebraic boost law $F' = L^* F L$, together with the induced transformed operator relation $\partial' F' = L^*(\rho - \mathbf{J} \cdot \boldsymbol{\sigma})L$.

5.5.3 Wave Operator and d'Alembertian

This subsection collects the differential-operator conventions and their sourced wave consequences in one place:

$$\begin{aligned}\partial &= \partial_t + \boldsymbol{\sigma} \cdot \boldsymbol{\nabla}, \\ \partial^* &= \partial_t - \boldsymbol{\sigma} \cdot \boldsymbol{\nabla}, \\ \square &:= \partial_t^2 - \boldsymbol{\nabla}^2 \\ &= \partial \partial^* = \partial^* \partial \\ &= \langle\langle \partial \rangle\rangle^2.\end{aligned}$$

With the sigma algebra one also has

$$\partial^2 = \partial_t^2 + \nabla^2 + 2\partial_t \nabla \cdot \sigma,$$

so $\partial^2 \neq \square$. Different sources may instead adopt the metric signature $(-, +, +, +)$ or use a different box notation, including $\square^2$, for related wave operators. The sourced wave equations are

$$\square F = \partial^* \partial F = \partial^* (\rho - \mathbf{J} \cdot \sigma).$$

$$\begin{aligned} \square F^* &= \partial(\rho + \mathbf{J} \cdot \sigma), \\ &= (\square \mathbf{E} + i \square \mathbf{B}) \cdot \sigma, \\ &= (\partial_t \rho + \nabla \cdot \mathbf{J}) + (-\partial_t \mathbf{J} - \nabla \rho + i(\nabla \times \mathbf{J})) \cdot \sigma. \\ \partial_t \rho &= -\nabla \cdot \mathbf{J} \quad \text{conservation of charge.} \end{aligned}$$

The scalar part of $\square F^*$ is therefore the continuity equation, while Eq. (84) and the vector part of $\square F^*$ give the sourced wave equations for $\mathbf{E}$ and $\mathbf{B}$. In vacuum both source terms disappear and each field component satisfies the ordinary wave equation.

In vacuum, where $\rho = 0$ and $\mathbf{J} = 0$, this reduces to

$$\begin{aligned} \square \mathbf{E} &= 0 \Rightarrow \partial_t^2 \mathbf{E} = \nabla^2 \mathbf{E}, \\ \square \mathbf{B} &= 0 \Rightarrow \partial_t^2 \mathbf{B} = \nabla^2 \mathbf{B}. \end{aligned}$$

Tables 21 and 22 record the scalar and vector parts of Maxwell's equation and its sourced wave consequence.

Table 21. Components of $\partial F = \rho - \mathbf{J} \cdot \sigma$.

real scalar:	$\nabla \cdot \mathbf{E} = \rho$
img scalar:	$\nabla \cdot \mathbf{B} = 0$
real vector:	$\partial_t \mathbf{E} - \nabla \times \mathbf{B} = -\mathbf{J}$
img vector:	$\partial_t \mathbf{B} + \nabla \times \mathbf{E} = 0$

Table 22. Components of $\square F = \partial^* (\rho - \mathbf{J} \cdot \sigma)$.

real scalar:	$\partial_t \rho + \nabla \cdot \mathbf{J} = 0$
img scalar:	$0 = 0$
real vector:	$\square \mathbf{E} = -\partial_t \mathbf{J} - \nabla \rho$
img vector:	$\square \mathbf{B} = \nabla \times \mathbf{J}$

5.5.4 Potential Fields

Eqs. (79) and (80) put gauge freedom, source reconstruction, and the electric/magnetic field split in one place. Here $V, S, \rho, \lambda \in \mathbb{R}$, $\mathbf{A}, \mathbf{E}, \mathbf{B}, \mathbf{J} \in \mathbb{R}^3$, $\Phi \in \mathbb{R} \oplus \mathbb{R}^3$, $\mathbf{F} \in \mathbb{C}^3$, and $F \in \emptyset \oplus \mathbb{C}^3$:

$$\begin{aligned} F^* &= (-\mathbf{E} + i\mathbf{B}) \cdot \sigma, \\ \Phi &= V + \mathbf{A} \cdot \sigma, \\ \partial \Phi &= \partial_t V + \nabla \cdot \mathbf{A}, \\ &\quad + (\partial_t \mathbf{A} + \nabla V) \cdot \sigma, \\ &\quad + i(\nabla \times \mathbf{A}) \cdot \sigma. \end{aligned}$$

This expansion separates the potential gradient into one real scalar part and one pure-vector field part. The scalar piece becomes the Lorenz-gauge scalar S , and the remaining vector piece becomes the electromagnetic field:

$$\begin{aligned} S &:= \text{scalar}(\partial \Phi), \\ &= \partial_t V + \nabla \cdot \mathbf{A}. \end{aligned}$$

The component identities extracted from $\partial \Phi$ are

$$\begin{aligned} \mathbf{E} &= -\partial_t \mathbf{A} - \nabla V, \\ \mathbf{B} &= \nabla \times \mathbf{A}. \end{aligned}$$

The identity is explicit:

$$\begin{aligned} \partial \Phi &= S + F^*, \\ F^* &= \partial \Phi - S, \\ \text{scalar}(F^*) &= \text{scalar}(\partial \Phi - S) = 0, \\ \text{vector}(F^*) &= \text{vector}(\partial \Phi). \end{aligned}$$

The field is therefore the non-scalar part of $\partial \Phi$. Written this way, gauge freedom changes only the scalar part S and leaves the field paravector itself unchanged. Table 23 records the component content of the potential identity $\partial \Phi = S + F^*$.

Table 23. Components of $\partial \Phi = S + F^*$.

real scalar:	$\partial_t V + \nabla \cdot \mathbf{A} = S$
img scalar:	$0 = 0$
real vector:	$\partial_t \mathbf{A} + \nabla V = -\mathbf{E}$
img vector:	$\nabla \times \mathbf{A} = \mathbf{B}$

The differential operator acts before the conjugate and reverse operators are applied, so

$$(\partial \Phi)^{*R} \neq \Phi^{*R} \partial^{*R}.$$

Equivalently, without introducing S , one may write

$$F^* = (\partial \Phi - (\partial \Phi)^{*R})/2.$$

5.5.5 Density from Potential

This subsection collects the derivation of the charge density and current density directly from the potential field in one place. Using the previously defined potential $\Phi = V + \mathbf{A} \cdot \sigma$, scalar part $S = \partial_t V + \nabla \cdot \mathbf{A}$, and identity $F^* = \partial \Phi - S$, one obtains

$$\begin{aligned} F &= \partial^* \Phi^* - S, \\ \partial F &= \partial \partial^* \Phi^* - \partial S, \\ &= \square \Phi^* - \partial S, \\ &= (\square V - \partial_t S) + (-\square \mathbf{A} - \nabla S) \cdot \sigma, \\ &= \rho - \mathbf{J} \cdot \sigma. \end{aligned}$$

$$\begin{aligned} \rho &= \square V - \partial_t S, \\ \mathbf{J} &= \square \mathbf{A} + \nabla S. \end{aligned}$$

These are the component source-reconstruction formulas hidden inside the compact Maxwell equation. The scalar equation determines charge density from the potential data, and the vector equation determines current density from the same data.

In Lorenz gauge, where $S = 0$, these relations reduce to

$$\begin{aligned}\partial_t V &= -\nabla \cdot \mathbf{A}, \\ \rho &= \square V, \\ \mathbf{J} &= \square \mathbf{A}.\end{aligned}$$

5.5.6 Gauge Transform of the Potential

$$\begin{aligned}\text{scalar}(\partial\Phi) &= S, \\ S &= \partial_t V + \nabla \cdot \mathbf{A}.\end{aligned}$$

Let λ be a scalar gauge function. Then

$$\begin{aligned}\Phi' &= \Phi + \partial^* \lambda, \\ \partial\Phi' &= \partial(\Phi + \partial^* \lambda), \\ &= \partial\Phi + \partial(\partial^* \lambda), \\ &= \partial\Phi + \square \lambda.\end{aligned}$$

Since $\square \lambda$ is again a scalar field, only the scalar part changes:

$$\begin{aligned}\text{scalar}(\partial\Phi') &= S + \square \lambda, \\ \text{vector}(\partial\Phi') &= \text{vector}(\partial\Phi), \\ F' &= F.\end{aligned}$$

This makes the gauge split explicit: Φ changes by a derivative of a real scalar, the scalar part of $\partial\Phi$ shifts by $\square \lambda$, and the field paravector remains unchanged.

5.5.7 4-Force (Proper Force)

This subsection collects the relativistic work law, the Lorentz-force law, the component split of FU/γ , and the final proper-force equation in one place.

$$\begin{aligned}mU &= mdX/ds, \\ &= m\gamma + m\gamma \mathbf{v} \cdot \boldsymbol{\sigma}, \\ d/ds &= \gamma d/dt.\end{aligned}$$

The scalar part of $d(mU)/dt$ is the power equation and the vector part is the three-force equation. Writing both pieces in one paravector equation makes the usual split visible without abandoning the relativistic form.

The time component gives the relativistic work law,

$$d/dt(m\gamma) = q\mathbf{v} \cdot \mathbf{E}.$$

Here $\mathbf{E}$ denotes the electric field, not the energy operator E , and the magnetic field contributes no work term in the scalar component. The spatial component gives the relativistic Lorentz force law,

$$\begin{aligned}\mathbf{P} &:= m\gamma \mathbf{v}, \\ d\mathbf{P}/dt &= q\mathbf{E} + q\mathbf{v} \times \mathbf{B}.\end{aligned}$$

Together these assemble into

$$d/dt(mU) = q\mathbf{v} \cdot \mathbf{E} + (q\mathbf{E} + q\mathbf{v} \times \mathbf{B}) \cdot \boldsymbol{\sigma}.$$

To expose the paravector content explicitly, expand

$$\begin{aligned}FU/\gamma &= ((\mathbf{E} + i\mathbf{B}) \cdot \boldsymbol{\sigma})(1 + \mathbf{v} \cdot \boldsymbol{\sigma}), \\ &= \mathbf{E} \cdot \boldsymbol{\sigma} + i\mathbf{B} \cdot \boldsymbol{\sigma} + (\mathbf{E} \cdot \boldsymbol{\sigma})(\mathbf{v} \cdot \boldsymbol{\sigma}), \\ &\quad + i(\mathbf{B} \cdot \boldsymbol{\sigma})(\mathbf{v} \cdot \boldsymbol{\sigma}), \\ &= \mathbf{E} \cdot \boldsymbol{\sigma} + i\mathbf{B} \cdot \boldsymbol{\sigma} + \mathbf{E} \cdot \mathbf{v}, \\ &\quad + i(\mathbf{E} \times \mathbf{v}) \cdot \boldsymbol{\sigma} + i\mathbf{B} \cdot \mathbf{v} - (\mathbf{B} \times \mathbf{v}) \cdot \boldsymbol{\sigma}.\end{aligned}$$

The product FU naturally separates into real and imaginary parts. The real part reproduces the physical Lorentz-force content, while the imaginary part exchanges electric and magnetic roles and is therefore not used in the force law itself. Table 24 gives the component form of the product FU/γ used in the Lorentz-force law.

Table 24. Components of FU/γ .

real scalar:	$\mathbf{E} \cdot \mathbf{v}$
img scalar:	$\mathbf{B} \cdot \mathbf{v}$
real vector:	$\mathbf{E} + \mathbf{v} \times \mathbf{B}$
img vector:	$\mathbf{B} + \mathbf{E} \times \mathbf{v}$

$$\begin{aligned}\text{real}(FU) &= \frac{1}{2}(FU + UF^R), \\ &= \gamma(\mathbf{v} \cdot \mathbf{E} + (\mathbf{E} + \mathbf{v} \times \mathbf{B}) \cdot \boldsymbol{\sigma}).\end{aligned}$$

The imaginary part of FU exchanges the electric and magnetic roles and is therefore not the part used in the physical force law:

$$\begin{aligned}mdU/ds &= q\text{real}(FU), \\ &= \frac{1}{2}q(FU + UF^R).\end{aligned}$$

$$mddX/ds^2 = q\text{real}(FdX/ds) \quad \text{4-force.}$$

5.5.8 Lagrangian Derivation of the Lorentz Force

This subsection collects the minimally coupled action, the Euler-Lagrange derivation of the Lorentz force, the Hamiltonian, and the corresponding power identity in one place. Starting from the Lagrangian of Eq. (88), together with the potential $\Phi = V + \mathbf{A} \cdot \boldsymbol{\sigma}$, proper velocity $U = dX/ds$, and $\gamma = dt/ds$, the action takes the form

$$\begin{aligned}\text{action} &= -m \int ds - q \int \text{scalar}(\Phi^* U) ds, \\ &= \int (-m/\gamma - q\text{scalar}(\Phi^* U/\gamma)) dt, \\ &= \int (-m/\gamma, \\ &\quad - q\text{scalar}((V - \mathbf{A} \cdot \boldsymbol{\sigma})(1 + \mathbf{v} \cdot \boldsymbol{\sigma}))) dt, \\ &= \int (-m/\gamma - qV + q\mathbf{v} \cdot \mathbf{A}) dt.\end{aligned}$$

$$\mathcal{L} = -m/\gamma - q(V - \mathbf{v} \cdot \mathbf{A}) \quad \text{Lagrangian.}$$

This is the standard minimal-coupling Lagrangian written in the present paravector notation. The canonical momentum is shifted

by the vector potential, exactly as in the ordinary component derivation.

The canonical momentum is

$$\begin{aligned}\partial_{\mathbf{v}}\mathcal{L} &= m\gamma\mathbf{v} + q\mathbf{A}, \\ &= \mathbf{P} + q\mathbf{A}.\end{aligned}$$

$$\begin{aligned}d/dt\partial_{\mathbf{v}}\mathcal{L} &= d\mathbf{P}/dt + q\partial_t\mathbf{A} + q(\mathbf{v} \cdot \nabla)\mathbf{A}, \\ \nabla\mathcal{L} &= -q\nabla V + q\nabla(\mathbf{v} \cdot \mathbf{A}).\end{aligned}$$

Using the vector-calculus identity

$$(\mathbf{v}(t) \cdot \nabla)\mathbf{A}(t, \mathbf{r}) = \nabla(\mathbf{v} \cdot \mathbf{A}) - \mathbf{v} \times (\nabla \times \mathbf{A}).$$

the Euler–Lagrange equation $d/dt\partial_{\mathbf{v}}\mathcal{L} = \nabla\mathcal{L}$ yields

$$\begin{aligned}d\mathbf{P}/dt &= -q\nabla V - q\partial_t\mathbf{A}, \\ &\quad + q(\nabla(\mathbf{v} \cdot \mathbf{A}) - (\mathbf{v} \cdot \nabla)\mathbf{A}), \\ &= q(-\nabla V - \partial_t\mathbf{A} + \mathbf{v} \times (\nabla \times \mathbf{A})), \\ &= q(\mathbf{E} + \mathbf{v} \times \mathbf{B}), \\ &= m\text{vector}(dU/ds)/\gamma.\end{aligned}$$

$$\begin{aligned}\mathcal{H} &= (\partial_{\mathbf{v}}\mathcal{L}) \cdot \mathbf{v} - \mathcal{L}, \\ &= m\gamma\mathbf{v} \cdot \mathbf{v} + q\mathbf{v} \cdot \mathbf{A} + m/\gamma.\end{aligned}$$

Adding the explicit potential contribution $qV - q\mathbf{v} \cdot \mathbf{A}$ gives

$$\begin{aligned}&= m\gamma(\mathbf{v}^2 + 1/\gamma^2) + qV, \\ &= m\gamma(\mathbf{v}^2 + 1 - \mathbf{v}^2) + qV, \\ &= m\gamma + qV.\end{aligned}$$

The resulting Hamiltonian is the relativistic energy plus the scalar potential term. Using the Hamiltonian identity $dH/dt = -\partial_t\mathcal{L}$,

$$\begin{aligned}dH/dt &= -\partial_t\mathcal{L}, \\ d\mathcal{H}/dt &= d(m\gamma)/dt + qdV/dt, \\ &= d(m\gamma)/dt + q\partial_t V + q\mathbf{v} \cdot \nabla V. \\ \partial_t H &= q\partial_t V - q\mathbf{v} \cdot \partial_t\mathbf{A}, \\ d(m\gamma)/dt &= q\mathbf{v} \cdot (-\nabla V - \partial_t\mathbf{A}), \\ &= q\mathbf{v} \cdot \mathbf{E}.\end{aligned}$$

5.5.9 Stress-Energy of the Electromagnetic Field

This subsection collects the frame-dependent energy-density/Poynting paravector built from F , together with the differentiated and source-coupled identities that accompany it. The caution in the main text remains in force: $T = \frac{1}{2}FF^R$ is a useful paravector, but it is not by itself the full electromagnetic stress-energy tensor:

$$\begin{aligned}F &= (\mathbf{E} + i\mathbf{B}) \cdot \boldsymbol{\sigma}, \\ F^R &= (\mathbf{E} - i\mathbf{B}) \cdot \boldsymbol{\sigma}, \\ T &= \frac{1}{2}FF^R, \\ &= \frac{1}{2}(\mathbf{E}^2 + \mathbf{B}^2) + (\mathbf{E} \times \mathbf{B}) \cdot \boldsymbol{\sigma}, \\ &= T_s + T_{\mathbf{v}} \cdot \boldsymbol{\sigma}.\end{aligned}$$

$$\partial = \partial_t + \boldsymbol{\sigma} \cdot \nabla.$$

This is a frame-dependent energy-density/Poynting paravector, not by itself the full electromagnetic stress-energy tensor.

Using the generic paravector product rule,

$$\begin{aligned}\partial(ZZ') &= (\partial Z)Z' + Z(\partial Z') + [\boldsymbol{\sigma}, Z] \cdot \nabla Z', \\ \partial T &= \frac{1}{2}\partial(FF^R), \\ &= \frac{1}{2}(\partial F)F^R + \frac{1}{2}F(\partial F^R) \\ &\quad + \frac{1}{2}[\boldsymbol{\sigma}, F] \cdot \nabla F^R.\end{aligned}$$

$$D = \rho + \mathbf{J} \cdot \boldsymbol{\sigma} \quad \text{density paravector,}$$

$$D^* = \rho - \mathbf{J} \cdot \boldsymbol{\sigma},$$

$$\partial = \partial_t + \boldsymbol{\sigma} \cdot \nabla,$$

$$\partial F = D^*,$$

$$F^2 = (\mathbf{E}^2 - \mathbf{B}^2) + 2i(\mathbf{E} \cdot \mathbf{B}) \quad \text{scalar.}$$

Applying the same product rule to F^2 gives

$$\begin{aligned}\partial(ZZ') &= (\partial Z)Z' + Z(\partial Z') + [\boldsymbol{\sigma}, Z] \cdot \nabla Z', \\ \partial(F^2) &= (\partial F)F + F(\partial F) \\ &\quad + [\boldsymbol{\sigma}, F] \cdot \nabla F, \\ &= D^*F + FD^* + [\boldsymbol{\sigma}, F] \cdot \nabla F.\end{aligned}$$

$$[\boldsymbol{\sigma}, F] = 2i\mathbf{F} \times \boldsymbol{\sigma},$$

$$\begin{aligned}[\boldsymbol{\sigma}, F] \cdot \nabla F &= -2i\mathbf{F} \cdot (\nabla \times \mathbf{F}), \\ &\quad - 2((\nabla \cdot \mathbf{F})\mathbf{F} - \frac{1}{2}\nabla F^2) \cdot \boldsymbol{\sigma}.\end{aligned}$$

$$\square F^2 = \partial^* \partial(F^2) \quad \text{scalar.}$$

The component decompositions needed in this discussion are collected directly below: Tables 27, 28, 29, 30, and 31 give the components of ∂T , ∂F^2 , FD^* , D^*F , and $D^*F + FD^*$, respectively. Together with F^2 and $T = \frac{1}{2}FF^R$, these formulas isolate the standard electromagnetic invariants $\mathbf{E}^2 - \mathbf{B}^2$ and $\mathbf{E} \cdot \mathbf{B}$, the energy-density/Poynting split, and the source-coupled terms generated by Maxwell's equation $\partial F = D^*$. Tables 25 and 26 summarize the two simplest invariants built from F and the frame-dependent energy-density/Poynting paravector $T = \frac{1}{2}FF^R$.

Table 25. Components of $\frac{1}{2}F^2$.

real scalar:	$\frac{1}{2}(\mathbf{E}^2 - \mathbf{B}^2)$
img scalar:	$i\mathbf{E} \cdot \mathbf{B}$
real vector:	$\mathbf{0}$
img vector:	$\mathbf{0}$

Table 26. Components of $\frac{1}{2}FF^R$.

scalar:	$\frac{1}{2}(\mathbf{E}^2 + \mathbf{B}^2)$	energy density
vector:	$\mathbf{E} \times \mathbf{B}$	Poynting vector

The remaining tables record the differentiated and source-coupled field combinations used in the energy-momentum discussion.

Table 27. Components of ∂T .

real scalar:	$\partial_t T_s + \nabla \cdot T_v$
img scalar:	0
real vector:	$\nabla T_s + \partial_t T_v$
img vector:	$\nabla \times T_v$

Table 28. Components of ∂F^2 .

real scalar:	$\partial_t (\mathbf{E}^2 - \mathbf{B}^2)$
img scalar:	$\partial_t (2 \mathbf{E} \cdot \mathbf{B})$
real vector:	$\nabla (\mathbf{E}^2 - \mathbf{B}^2)$
img vector:	$\nabla (2 \mathbf{E} \cdot \mathbf{B})$

Table 29. Components of $F D^*$.

real scalar:	$-\mathbf{E} \cdot \mathbf{J}$
img scalar:	$-\mathbf{B} \cdot \mathbf{J}$
real vector:	$\rho \mathbf{E} + \mathbf{B} \times \mathbf{J}$
img vector:	$\rho \mathbf{B} - \mathbf{E} \times \mathbf{J}$

Table 30. Components of $D^* F$.

real scalar:	$-\mathbf{E} \cdot \mathbf{J}$
img scalar:	$-\mathbf{B} \cdot \mathbf{J}$
real vector:	$\rho \mathbf{E} - \mathbf{B} \times \mathbf{J}$
img vector:	$\rho \mathbf{B} + \mathbf{E} \times \mathbf{J}$

Table 31. Components of $D^* F + F D^*$.

real scalar:	$-2 \mathbf{E} \cdot \mathbf{J}$
img scalar:	$-2 \mathbf{B} \cdot \mathbf{J}$
real vector:	$2 \rho \mathbf{E}$
img vector:	$2 \rho \mathbf{B}$

6 Relativistic Wave Equations and Spin Dynamics

The relativistic wave-equation development is a second application of the same algebra: first the scalar wave equation, then minimal coupling and factorization, and only then the two-component paravector factorization, the projector split, and the Pauli limit. The sequence follows the historical progression from Klein and Gordon to Dirac and Pauli, but it is rewritten throughout in the common paravector language established earlier [7–10].

6.1 Klein–Gordon and frequency splitting

For a complex scalar field $\Psi \in \mathbb{C}$,

$$(\partial \partial^* + m^2) \Psi = 0 \quad (90)$$

is the free Klein–Gordon equation [7, 8]. Define also

$$\begin{aligned} \Omega &:= \langle\langle m + \sigma \cdot \nabla \rangle\rangle = \sqrt{m^2 - \nabla^2}, \\ (\partial_t \pm i\Omega) \Psi(\pm) &= 0, \quad \Psi = \Psi(+) + \Psi(-), \end{aligned} \quad (91)$$

which is the positive/negative-frequency factorization of (90). This factorization is useful both conceptually and technically. It separates opposite-frequency branches of the scalar theory, and it also prepares the reader for the later factorizations in which second-order relativistic equations are rewritten as first-order systems.

6.2 Minimal coupling and operator factorization

Minimal coupling enters directly at the operator level. Rather than modifying each component equation separately, one defines energy and momentum operators that already incorporate the potential Φ established in Section 5.

Here $\Psi(\pm) \in \mathbb{C}$, while Ω is a scalar differential operator built from $m \in \mathbb{R}$ and ∇ . Define also the scalar differential operator E and vector differential operator $\mathbf{P}$ by

$$E := i\partial_t - qV, \quad \mathbf{P} := -i\nabla - q\mathbf{A}. \quad (92)$$

Then

$$(-E^2 + \mathbf{P}^2 + m^2) \Psi = 0 \quad (93)$$

expands to the minimal-coupling Klein–Gordon operator

$$\left(\square + iqS - q^2 \langle\langle \Phi \rangle\rangle^2 + m^2 + 2iq(V\partial_t + \mathbf{A} \cdot \nabla) \right) \Psi = 0. \quad (94)$$

The basic covariant-momentum commutators are

$$[E, P_m] = iqE_m, \quad [P_m, P_n] = iq(\partial_m A_n - \partial_n A_m), \quad (95)$$

and using Eq. (95) one gets the useful factorization

$$(E + \sigma \cdot \mathbf{P})(E - \sigma \cdot \mathbf{P}) = E^2 - \mathbf{P}^2 - iqF. \quad (96)$$

Equation (96) is the bridge between the Klein–Gordon and Dirac-type formulas below. The square of a first-order operator built from E and $\mathbf{P}$ produces the second-order relativistic operator together with the field term, which is the structural fact needed before introducing a Dirac-type equation.

6.3 Dirac Two-Component Form, Gauge Covariance, and Pauli Limit

Using the auxiliary 2×2 map from Eq. (61), the free and coupled Dirac operators act on a two-component field $\Psi = \{\Psi_1; \Psi_2\}$ whose components $\Psi_1, \Psi_2 \in \mathbb{C} \oplus \mathbb{C}^3$ are complex paravectors:

$$D_0 := W(i\partial) - m\mathbf{I}, \quad D_0 \Psi = 0, \quad (97)$$

$$D := W(i\partial - q\Phi^*) - m\mathbf{I}, \quad D \Psi = 0. \quad (98)$$

Multiplying (98) by $D' := W(i\partial - q\Phi^*) + m\mathbf{I}$ yields

$$D' D \Psi = 0 \Rightarrow \begin{cases} (-E^2 + \mathbf{P}^2 + m^2 + iqF^*) \Psi_1 = 0, \\ (-E^2 + \mathbf{P}^2 + m^2 + iqF) \Psi_2 = 0. \end{cases} \quad (99)$$

The calculation in Eq. (99) explains why the two-component construction is more than formal packaging. Squaring the first-order system returns Klein–Gordon-type equations with the expected field couplings, so the Dirac form is seen as a controlled

factorization of the relativistic wave operator. For the ordering $Z = i\partial - q\Phi^* = E - \boldsymbol{\sigma} \cdot \mathbf{P}$ inside $W(Z)$, the upper component carries $+iqF^*$ and the lower component carries $+iqF$.

Gauge covariance combines the electromagnetic gauge law (86) with the phase transformation of that same two-component field,

$$\Psi = e^{iq\lambda}\Psi', \quad (100)$$

Equation (100) leaves the form of (98) unchanged.

For the low-energy reduction, using the operators from Eq. (92), one obtains the leading-order Pauli equation for the same two-component field Ψ ,

$$2mE\Psi = (\mathbf{P}^2 - q\mathbf{B} \cdot \boldsymbol{\sigma})\Psi. \quad (101)$$

Solving Eq. (101) for $i\partial_t\Psi$ gives the more familiar Schrödinger-type form

$$i\partial_t\Psi \approx \left(\frac{(-i\nabla - q\mathbf{A})^2}{2m} + qV \right) \Psi - \frac{q}{2m}(\mathbf{B} \cdot \boldsymbol{\sigma})\Psi. \quad (102)$$

This matches the usual Zeeman term [9, 18], provided one remembers that (102) is a leading-order $1/m$ approximation and that later specialization to uniform static $\mathbf{B}$ is used when decoupling the spin-up and spin-down parts. The nonrelativistic theory therefore appears as an approximation internal to the relativistic one, not as an unrelated starting point. The Pauli equation is the leading low-energy shadow of the Dirac two-component field once the rest-energy scale has been separated and higher-order corrections are neglected.

If $\mathbf{B} = \|\mathbf{B}\|\mathbf{u}$ with $\mathbf{B}, \mathbf{u} \in \mathbb{R}^3$, $\|\mathbf{u}\| = 1$, and $\mathbf{B}$ uniform and constant, decompose

$$\begin{aligned} \Psi &= \Psi(+) + \Psi(-), & \Psi(\pm) &= \Pi(\pm)\Psi, \\ \Pi(\pm) &= \frac{1}{2}(1 \pm \mathbf{u} \cdot \boldsymbol{\sigma}). \end{aligned} \quad (103)$$

With the projector decomposition (103) in place, Eq. (102) separates into

$$i\partial_t\Psi(\pm) = ((\mathbf{P}^2 \mp q\|\mathbf{B}\|)/(2m) + qV)\Psi(\pm). \quad (104)$$

In this special case the projector language acquires an immediate physical meaning: the field direction selects a preferred axis, and the two projector parts evolve with opposite Zeeman shifts. If the field direction or magnitude varies, the same projector family from Section 1.9 still organizes the state but need not decouple the dynamics. Writing

$$H := \frac{\mathbf{P}^2}{2m} + qV - \frac{q\|\mathbf{B}\|}{2m}\mathbf{u} \cdot \boldsymbol{\sigma}, \quad i\partial_t\Psi = H\Psi,$$

the projector-selected parts obey

$$\begin{aligned} i\partial_t\Psi(\pm) &= i(\partial_t\Pi(\pm))\Psi + \Pi(\pm)(i\partial_t\Psi) \\ &= H\Psi(\pm) + S(\pm), \end{aligned}$$

with coupling term

$$S(\pm) := (i\partial_t\Pi(\pm) + [\Pi(\pm), H])\Psi.$$

The uniform static-field split in Eq. (104) is precisely the case $S(\pm) = 0$.

6.4 4-State Solution to the Dirac Equation

For a free on-shell Dirac state, start from the unit momentum paravector

$$U = \frac{dX}{ds} = \frac{E + \mathbf{P} \cdot \boldsymbol{\sigma}}{m}.$$

On the mass shell,

$$\langle\langle U \rangle\rangle^2 = \frac{E^2 - \mathbf{P}^2}{m^2} = 1.$$

Using the auxiliary map from Eq. (61),

$$W(U)^2 = \text{diag}(\langle\langle U \rangle\rangle^2, \langle\langle U \rangle\rangle^2) = \mathbf{I}.$$

The corresponding energy projectors are

$$\Lambda(\pm) := \frac{1}{2}(\mathbf{I} \pm W(U)),$$

where $\Lambda(+)$ selects the positive-energy (matter) branch and $\Lambda(-)$ the negative-energy (antimatter) branch. Since $W(U)^2 = \mathbf{I}$, they satisfy

$$\Lambda(\pm)^2 = \Lambda(\pm), \quad \Lambda(+)\Lambda(-) = 0, \quad \Lambda(+) + \Lambda(-) = \mathbf{I}.$$

For an arbitrary spin axis $\mathbf{u} \in \mathbb{R}^3$ with $\mathbf{u}^2 = 1$, use also the spin projectors

$$\Pi(\pm) := \frac{1}{2}(1 \pm \mathbf{u} \cdot \boldsymbol{\sigma}).$$

In the free operator picture, the unit momentum is represented by

$$U \longrightarrow \frac{i\partial}{m}.$$

The mass-normalized positive-energy projector condition may therefore be written as

$$\begin{aligned} \left(W\left(\frac{i\partial}{m}\right) - \mathbf{I} \right) \Psi &= 0, \\ (W(U) - \mathbf{I})\Psi &= 0. \end{aligned} \quad (105)$$

Equation (105) is not a replacement for the Dirac equation; it is the free on-shell projector relation obtained after normalizing momentum by m so that $W(U)^2 = \mathbf{I}$.

Combining the energy split with the spin split gives the four sectors

$$\Psi(\epsilon, s) := \Lambda(\epsilon)\Pi(s)\Psi, \quad \epsilon, s \in \{+, -\}, \quad (106)$$

where ϵ labels the energy sector and s the spin along $\mathbf{u}$. The full Dirac state therefore decomposes as

$$\Psi = \Psi(+, +) + \Psi(+, -) + \Psi(-, +) + \Psi(-, -).$$

This is the four-state decomposition of the free Dirac solution: two energy sectors and, within each, two spin sectors.

For a normalized complex paravector state $\psi \in \mathbb{C} \oplus \mathbb{C}^3$, write the scalar projections

$$\begin{aligned} \langle \psi | \psi \rangle &:= \text{scalar}(\psi^R \psi) = 1, \\ \langle \psi | M | \psi \rangle &:= \text{scalar}(\psi^R M \psi). \end{aligned}$$

Here M may be any algebra element with a well-defined product on ψ . The bracket notation is only a compact mnemonic for scalar extraction on normalized paravector states; the underlying objects remain paravectors throughout.

6.5 Supporting Wave-Equation Derivations

The supporting derivations expand the scalar and two-component paravector wave equations into their intermediate operator identities, explicit component forms, Fourier interpretations, low-energy limits, and spin decompositions.

6.5.1 Relativistic Quantum-Mechanical Setting

The wave-equation section uses one fixed operator language throughout. With the differential conventions from Section 4,

$$\begin{aligned}\partial &= \partial_t + \boldsymbol{\sigma} \cdot \boldsymbol{\nabla}, \\ \partial^* &= \partial_t - \boldsymbol{\sigma} \cdot \boldsymbol{\nabla}, \\ (i\partial)^* &= -i\partial^*.\end{aligned}$$

Recall also that the d'Alembertian is

$$\begin{aligned}\square &:= \partial_t^2 - \boldsymbol{\nabla}^2 \\ &= \partial\partial^* = \partial^*\partial \\ &= \langle\langle\partial\rangle\rangle^2.\end{aligned}$$

All differential operators act left to right.

6.5.2 Complex-Paravector Bilinears and Bra-Ket Extraction

To make the bra-ket notation precise in the paravector setting, write

$$\begin{aligned}\psi &= \psi_s + \psi_v \cdot \boldsymbol{\sigma}, & O &= O_s + O_v \cdot \boldsymbol{\sigma}, \\ Z &= Z_s + Z_v \cdot \boldsymbol{\sigma}, & X &= X_s + X_v \cdot \boldsymbol{\sigma},\end{aligned}$$

with $\psi_s, Z_s \in \mathbb{C}$, $\psi_v, Z_v \in \mathbb{C}^3$, $X_s \in \mathbb{R}$, $X_v \in \mathbb{R}^3$, and with operator-valued components acting left to right. The three involutions on the state are

$$\begin{aligned}\psi^* &= \psi_s^* - \psi_v^* \cdot \boldsymbol{\sigma}, \\ \psi^R &= \psi_s^* + \psi_v^* \cdot \boldsymbol{\sigma}, \\ \psi^{*R} &= \psi_s - \psi_v \cdot \boldsymbol{\sigma}.\end{aligned}$$

The basic component products are collected below.

Table 32 records the direct action of a complex paravector operator on a complex paravector state.

Table 32. Components of $O\psi$.

scalar:	$O_s\psi_s + O_v \cdot \psi_v$
vector:	$O_s\psi_v + O_v\psi_s + i(O_v \times \psi_v)$

Tables 33–35 give the three basic bilinear sandwiches with a generic complex paravector $Z = Z_s + Z_v \cdot \boldsymbol{\sigma}$.

Table 33. Components of $\psi^*Z\psi$.

scalar:	$Z_s(\psi_s ^2 - \psi_v^* \cdot \psi_v)$ $+ \psi_s^*(Z_v \cdot \psi_v) - \psi_s(Z_v \cdot \psi_v^*)$ $- i\psi_v^* \cdot (Z_v \times \psi_v)$
vector:	$Z_s(\psi_s^*\psi_v - \psi_s\psi_v^* - i\psi_v^* \times \psi_v)$ $+ (\psi_s ^2 + \psi_v^* \cdot \psi_v)Z_v$ $- (Z_v \cdot \psi_v^*)\psi_v - (Z_v \cdot \psi_v)\psi_v^*$ $+ i(\psi_s^*Z_v \times \psi_v + \psi_sZ_v \times \psi_v^*)$

Table 34. Components of $\psi^R Z \psi$.

scalar:	$Z_s(\psi_s ^2 + \psi_v^* \cdot \psi_v)$ $+ \psi_s^*(Z_v \cdot \psi_v) + \psi_s(Z_v \cdot \psi_v^*)$ $+ i\psi_v^* \cdot (Z_v \times \psi_v)$
vector:	$Z_s(\psi_s^*\psi_v + \psi_s\psi_v^* + i\psi_v^* \times \psi_v)$ $+ (\psi_s ^2 - \psi_v^* \cdot \psi_v)Z_v$ $+ (Z_v \cdot \psi_v^*)\psi_v + (Z_v \cdot \psi_v)\psi_v^*$ $+ i(\psi_s^*Z_v \times \psi_v - \psi_sZ_v \times \psi_v^*)$

Table 35. Components of $\psi^{*R} Z \psi$.

scalar:	$Z_s(\psi_s^2 - \psi_v \cdot \psi_v)$
vector:	$(\psi_s^2 + \psi_v \cdot \psi_v)Z_v$ $- 2(Z_v \cdot \psi_v)\psi_v + 2i\psi_sZ_v \times \psi_v$

Specializing to a real paravector $X = X_s + X_v \cdot \boldsymbol{\sigma}$, with $X_s \in \mathbb{R}$ and $X_v \in \mathbb{R}^3$, gives Tables 36–38.

Table 36. Components of $\psi^* X \psi$.

scalar:	$X_s(\psi_s ^2 - \psi_v^* \cdot \psi_v)$ $+ \psi_s^*(X_v \cdot \psi_v) - \psi_s(X_v \cdot \psi_v^*)$ $- i\psi_v^* \cdot (X_v \times \psi_v)$
vector:	$X_s(\psi_s^*\psi_v - \psi_s\psi_v^* - i\psi_v^* \times \psi_v)$ $+ (\psi_s ^2 + \psi_v^* \cdot \psi_v)X_v$ $- (X_v \cdot \psi_v^*)\psi_v - (X_v \cdot \psi_v)\psi_v^*$ $+ i(\psi_s^*X_v \times \psi_v + \psi_sX_v \times \psi_v^*)$

Table 37. Components of $\psi^R X \psi$.

scalar:	$X_s(\psi_s ^2 + \psi_v^* \cdot \psi_v)$ $+ \psi_s^*(X_v \cdot \psi_v) + \psi_s(X_v \cdot \psi_v^*)$ $+ i\psi_v^* \cdot (X_v \times \psi_v)$
vector:	$X_s(\psi_s^*\psi_v + \psi_s\psi_v^* + i\psi_v^* \times \psi_v)$ $+ (\psi_s ^2 - \psi_v^* \cdot \psi_v)X_v$ $+ (X_v \cdot \psi_v^*)\psi_v + (X_v \cdot \psi_v)\psi_v^*$ $+ i(\psi_s^*X_v \times \psi_v - \psi_sX_v \times \psi_v^*)$

Table 38. Components of $\psi^{*R} X \psi$.

scalar:	$X_s(\psi_s^2 - \psi_v \cdot \psi_v)$
vector:	$(\psi_s^2 + \psi_v \cdot \psi_v)X_v$ $- 2(X_v \cdot \psi_v)\psi_v + 2i\psi_sX_v \times \psi_v$

For a scalar function S , the same three sandwiches reduce to Tables 39–41.

Table 39. Components of $\psi^* S \psi$.

scalar:	$S(\psi_s ^2 - \psi_v^* \cdot \psi_v)$
vector:	$S(\psi_s^*\psi_v - \psi_s\psi_v^* - i\psi_v^* \times \psi_v)$

Table 40. Components of $\psi^R S \psi$.

scalar:	$S(\psi_s ^2 + \psi_v^* \cdot \psi_v)$
vector:	$S(\psi_s^*\psi_v + \psi_s\psi_v^* + i\psi_v^* \times \psi_v)$

Table 41. Components of $\psi^{*\mathbf{R}}S\psi$.

scalar:	$S(\psi_s^2 - \psi_{\mathbf{v}} \cdot \psi_{\mathbf{v}})$
vector:	0

The reverse sandwich with a real scalar is the key probabilistic object. For $S \in \mathbb{R}$,

$$\psi^{\mathbf{R}}S\psi = S(\rho_\psi + \mathbf{j}_\psi \cdot \boldsymbol{\sigma}),$$

where

$$\rho_\psi := |\psi_s|^2 + \psi_{\mathbf{v}}^* \cdot \psi_{\mathbf{v}}, \quad \mathbf{j}_\psi := \psi_s^* \psi_{\mathbf{v}} + \psi_s \psi_{\mathbf{v}}^* + i \psi_{\mathbf{v}}^* \times \psi_{\mathbf{v}}.$$

Thus $\psi^{\mathbf{R}}\psi$ is naturally a density-current paravector: the scalar part is the positive density, while the vector part carries the associated directional information. More generally, the raw bilinear $\psi^{\mathbf{R}}O\psi$ is a complex paravector, so before any extraction it contains one complex scalar and one complex vector, that is, four real component channels. The scalar channel gives the ordinary expectation value; the remaining channels can encode currents, fluxes, spin-orientation data, or other geometric bilinears depending on the operator.

At fixed time, the natural normalization is therefore

$$\langle \psi | \psi \rangle := \int_{\mathbb{R}^3} \text{scalar}(\psi^{\mathbf{R}}\psi) d^3r = \int_{\mathbb{R}^3} \rho_\psi d^3r = 1.$$

For a complex-paravector operator O , define

$$\langle \psi | O | \psi \rangle := \int_{\mathbb{R}^3} \text{scalar}(\psi^{\mathbf{R}}O\psi) d^3r.$$

If the state is not already normalized, the corresponding expectation map is

$$\mathbb{E}_\psi[O] := \frac{\int_{\mathbb{R}^3} \text{scalar}(\psi^{\mathbf{R}}O\psi) d^3r}{\int_{\mathbb{R}^3} \text{scalar}(\psi^{\mathbf{R}}\psi) d^3r}.$$

If $O\psi = \phi_s + \phi_{\mathbf{v}} \cdot \boldsymbol{\sigma}$, then the extracted scalar channel is simply

$$\text{scalar}(\psi^{\mathbf{R}}O\psi) = \psi_s^* \phi_s + \psi_{\mathbf{v}}^* \cdot \phi_{\mathbf{v}}.$$

When O is reverse-self-adjoint under the spatial integral and the boundary terms are benign, this scalar expectation is real.

By contrast,

$$\psi^{*\mathbf{R}}\psi = \psi_s^2 - \psi_{\mathbf{v}} \cdot \psi_{\mathbf{v}} = \langle\langle \psi \rangle\rangle^2$$

is the STA metric bilinear. It is algebraically useful, but it is not positive-definite and so it should not be used as the probabilistic normalization. In this framework the single bra-ket bracket $\langle \cdot | \cdot \rangle$ is therefore best reserved for the scalar expectation map built from reversion, while $\langle\langle \cdot \rangle\rangle^2$ continues to denote the paravector metric.

6.5.3 Spin-0 Wave Mechanics

The scalar relativistic wave equation can be read directly from the mass-shell identity developed earlier in the kinematic section.

Here $\Psi \in \mathbb{C}$ is a complex scalar field, $m \in \mathbb{R}$, and the free-particle operators are defined using the differential conventions from Section 4. Start from

$$\langle\langle mU \rangle\rangle^2 = m^2 \quad \Rightarrow \quad -E^2 + \mathbf{P}^2 + m^2 = 0.$$

For a free particle,

$$E = i\partial_t, \quad \mathbf{P} = -i\nabla,$$

and equivalently

$$E + \mathbf{P} \cdot \boldsymbol{\sigma} = i\partial^*.$$

Acting on a scalar field Ψ , the operator chain is

$$\begin{aligned} (-E^2 + \mathbf{P}^2 + m^2)\Psi &= (-(i\partial_t)^2 - \nabla^2 + m^2)\Psi \\ &= (\partial_t^2 - \nabla^2 + m^2)\Psi \\ &= (\partial\partial^* + m^2)\Psi \\ &= (\square + m^2)\Psi. \end{aligned}$$

Thus the free spin-0 relativistic wave equation is

$$(\square + m^2)\Psi = 0 \quad \text{Klein-Gordon equation.}$$

This is the scalar wave equation recorded in Eq. (90). The dispersion relation $\omega^2 = \mathbf{k}^2 + m^2$ and the later positive/negative-frequency split are consequences of this same operator identity.

6.5.4 Positive and Negative Frequency Sectors

The frequency decomposition is one of the central interpretive steps in the scalar theory. Here $\Psi, \Psi(\pm), c(\pm) \in \mathbb{C}$, while $m \in \mathbb{R}$ and ∇ is the spatial derivative operator fixed earlier in Section 4. The frequency operator is

$$\begin{aligned} \Omega &= \sqrt{m^2 - \nabla^2} \\ &= \langle\langle m + \boldsymbol{\sigma} \cdot \nabla \rangle\rangle. \end{aligned}$$

Accordingly,

$$\begin{aligned} \square + m^2 &= \partial_t^2 + \Omega^2 \\ &= \partial_t^2 + (m^2 - \nabla^2) \\ &= (\partial_t - i\Omega)(\partial_t + i\Omega) \\ &= (\partial_t + i\Omega)(\partial_t - i\Omega). \end{aligned}$$

The positive- and negative-frequency sectors are then defined by

$$\begin{aligned} (\partial_t \pm i\Omega)\Psi(\pm) &= 0, \\ \Psi(\pm) &= c(\pm) \exp(\mp i\Omega t), \\ \Psi &= \Psi(+) + \Psi(-). \end{aligned}$$

This decomposition is not merely notation: it separates the two energy branches of the scalar theory in the same way that the Fourier transform separates positive and negative frequencies [17].

6.5.5 Interpreting Root Operators with Fourier

Square-root operators are easiest to interpret one Fourier mode at a time. In this subsection $\Psi \in \mathbb{C}$, the coefficients $c_n \in \mathbb{C}$, the frequencies $\omega_n \in \mathbb{R}$, and the wave vectors $\mathbf{k}_n \in \mathbb{R}^3$. Write

$$\Psi = \sum_n c_n f_n, \\ f_n = \exp(i(\omega_n t + \mathbf{k}_n \cdot \mathbf{r})).$$

Then the d'Alembertian and its square root act modewise:

$$\square \Psi = \sum_n (\mathbf{k}_n^2 - \omega_n^2) c_n f_n, \\ \pm \sqrt{\square} \Psi = \sum_n \pm \sqrt{\mathbf{k}_n^2 - \omega_n^2} c_n f_n, \\ (\square + m^2) \Psi = \sum_n (\mathbf{k}_n^2 + m^2 - \omega_n^2) c_n f_n.$$

For on-shell Klein–Gordon modes,

$$\mathbf{k}_n^2 + m^2 = \omega_n^2, \quad \omega_n = \pm \sqrt{\mathbf{k}_n^2 + m^2},$$

so the operator factorization can be read as the modewise identity

$$\square + m^2 = (m \mp i\sqrt{\square})(m \pm i\sqrt{\square}).$$

Likewise, if $\Omega = \sqrt{m^2 - \nabla^2}$ as in Eq. (91), then

$$\Omega^2 \Psi = \sum_n (m^2 + \mathbf{k}_n^2) c_n f_n, \\ \Omega \Psi = \sum_n \sqrt{m^2 + \mathbf{k}_n^2} c_n f_n.$$

This is the concrete meaning of the root operators used above: they do not represent a formal symbolic manipulation, but multiplication of each Fourier mode by the corresponding dispersion function.

6.5.6 Klein–Gordon with Potential Fields

To keep the minimally coupled scalar theory in one self-contained place, start from the charged-particle Lagrangian

$$\mathcal{L} = -\frac{m}{\gamma} - q(V - \mathbf{v} \cdot \mathbf{A}).$$

Here E denotes the energy operator, not the electric field $\mathbf{E}$. The Hamiltonian and canonical momentum are then

$$i\partial_t = \mathcal{H} = E + qV, \quad -i\nabla = \partial_{\mathbf{v}} \mathcal{L} = \mathbf{P} + q\mathbf{A},$$

so the covariant energy and momentum operators are

$$E = i\partial_t - qV, \quad \mathbf{P} = -i\nabla - q\mathbf{A}.$$

Acting on a smooth test field Ψ , the Leibniz rules needed below are

$$\partial_t(V\Psi) = (\partial_t V)\Psi + V\partial_t \Psi, \\ \partial_m(A_n \Psi) = (\partial_m A_n)\Psi + A_n \partial_m \Psi.$$

From these one obtains

$$E^2 \Psi = E(i\partial_t \Psi - qV\Psi) \\ = (-\partial_t^2 - iq\partial_t V - 2iqV\partial_t + q^2 V^2) \Psi, \\ \mathbf{P}^2 \Psi = \mathbf{P} \cdot (-i\nabla \Psi - q\mathbf{A}\Psi) \\ = (-\nabla^2 + iq\nabla \cdot \mathbf{A} + 2iq\mathbf{A} \cdot \nabla + q^2 \mathbf{A}^2) \Psi.$$

Recall also

$$\square = \partial_t^2 - \nabla^2, \\ \Phi = V + \mathbf{A} \cdot \boldsymbol{\sigma}, \\ S = \partial_t V + \nabla \cdot \mathbf{A}, \\ \langle\langle \partial \rangle\rangle^2 = \square, \\ \langle\langle \Phi \rangle\rangle^2 = V^2 - \mathbf{A}^2.$$

Substituting the operator expansions gives the minimally coupled Klein–Gordon equation in explicit form:

$$0 = (-E^2 + \mathbf{P}^2 + m^2) \Psi \\ = \left(\square + iq(\partial_t V + \nabla \cdot \mathbf{A}) + 2iq(V\partial_t + \mathbf{A} \cdot \nabla) \right. \\ \left. - q^2(V^2 - \mathbf{A}^2) + m^2 \right) \Psi \\ = \left(\square + iqS - q^2 \langle\langle \Phi \rangle\rangle^2 + m^2 + 2iq(V\partial_t + \mathbf{A} \cdot \nabla) \right) \Psi.$$

The electromagnetic fields and commutators enter through

$$\mathbf{E} = -\partial_t \mathbf{A} - \nabla V, \\ \mathbf{B} = \nabla \times \mathbf{A}, \\ F = (\mathbf{E} + i\mathbf{B}) \cdot \boldsymbol{\sigma},$$

and

$$[E, P_m] \Psi = (i\partial_t - qV)(-i\partial_m \Psi - qA_m \Psi) \\ - (-i\partial_m - qA_m)(i\partial_t \Psi - qV\Psi) \\ = -iq((\partial_t A_m)\Psi + A_m \partial_t \Psi) \\ - iq((\partial_m V)\Psi + V\partial_m \Psi) + iqV\partial_m \Psi + iqA_m \partial_t \Psi \\ = -iq(\partial_t A_m + \partial_m V)\Psi \\ = iqE_m \Psi, \\ [P_m, P_n] \Psi = (-i\partial_m - qA_m)(-i\partial_n \Psi - qA_n \Psi) \\ - (-i\partial_n - qA_n)(-i\partial_m \Psi - qA_m \Psi) \\ = iq(\partial_m A_n - \partial_n A_m)\Psi.$$

Hence

$$[E, \mathbf{P}] = iq\mathbf{E}, \quad \mathbf{P} \times \mathbf{P} = iq\mathbf{B}.$$

Substituting those commutators into the ordered products yields

$$(E + \boldsymbol{\sigma} \cdot \mathbf{P})(E - \boldsymbol{\sigma} \cdot \mathbf{P}) = E^2 - \mathbf{P}^2 \\ + \boldsymbol{\sigma} \cdot (-[E, \mathbf{P}] - i\mathbf{P} \times \mathbf{P}) \\ = E^2 - \mathbf{P}^2 - iqF, \\ (E - \boldsymbol{\sigma} \cdot \mathbf{P})(E + \boldsymbol{\sigma} \cdot \mathbf{P}) = E^2 - \mathbf{P}^2 \\ + \boldsymbol{\sigma} \cdot ([E, \mathbf{P}] - i\mathbf{P} \times \mathbf{P}) \\ = E^2 - \mathbf{P}^2 - iqF^*.$$

Since

$$E + \boldsymbol{\sigma} \cdot \mathbf{P} = i\partial^* - q\Phi, \quad E - \boldsymbol{\sigma} \cdot \mathbf{P} = i\partial - q\Phi^* = (E + \boldsymbol{\sigma} \cdot \mathbf{P})^{*\mathbf{R}},$$

their scalar paravector metric is

$$\langle\langle E \pm \boldsymbol{\sigma} \cdot \mathbf{P} \rangle\rangle^2 = E^2 - \mathbf{P}^2 + \boldsymbol{\sigma} \cdot (\pm[E, \mathbf{P}] - i\mathbf{P} \times \mathbf{P}).$$

This is not an ordinary scalar operator: operator ordering still matters, so the two first-order products remain distinct. Consequently,

$$\begin{aligned} -(E + \boldsymbol{\sigma} \cdot \mathbf{P})(E + \boldsymbol{\sigma} \cdot \mathbf{P})^{*\mathbf{R}} &= (\partial^* + iq\Phi)(\partial + iq\Phi^*), \\ -(E + \boldsymbol{\sigma} \cdot \mathbf{P})^{*\mathbf{R}}(E + \boldsymbol{\sigma} \cdot \mathbf{P}) &= (\partial + iq\Phi^*)(\partial^* + iq\Phi). \end{aligned}$$

The same factorizations can be rewritten directly in derivative form:

$$\begin{aligned} (E + \boldsymbol{\sigma} \cdot \mathbf{P})(E - \boldsymbol{\sigma} \cdot \mathbf{P})\Psi &= -(\partial^* + iq\Phi)(\partial + iq\Phi^*)\Psi \\ &= -\square\Psi - iq\partial^*(\Phi^*\Psi) \\ &\quad - iq\Phi\partial\Psi \\ &\quad + q^2\langle\langle\Phi\rangle\rangle^2\Psi \\ &= -\square\Psi - iqS\Psi + q^2\langle\langle\Phi\rangle\rangle^2\Psi \\ &\quad - 2iq(V\partial_t + \mathbf{A} \cdot \nabla)\Psi - iqF\Psi, \\ (E - \boldsymbol{\sigma} \cdot \mathbf{P})(E + \boldsymbol{\sigma} \cdot \mathbf{P})\Psi &= -(\partial + iq\Phi^*)(\partial^* + iq\Phi)\Psi \\ &= -\square\Psi - iq\partial(\Phi\Psi) \\ &\quad - iq\Phi^*\partial^*\Psi \\ &\quad + q^2\langle\langle\Phi\rangle\rangle^2\Psi \\ &= -\square\Psi - iqS\Psi + q^2\langle\langle\Phi\rangle\rangle^2\Psi \\ &\quad - 2iq(V\partial_t + \mathbf{A} \cdot \nabla)\Psi - iqF^*\Psi. \end{aligned}$$

Equivalently,

$$\begin{aligned} \partial^*(\Phi^*\Psi) + \Phi\partial\Psi &= S\Psi + 2(V\partial_t + \mathbf{A} \cdot \nabla)\Psi + F\Psi, \\ \partial(\Phi\Psi) + \Phi^*\partial^*\Psi &= S\Psi + 2(V\partial_t + \mathbf{A} \cdot \nabla)\Psi + F^*\Psi, \end{aligned}$$

and their difference isolates the electric contribution:

$$\begin{aligned} (\partial^*(\Phi^*\Psi) + \Phi\partial\Psi) - (\partial(\Phi\Psi) + \Phi^*\partial^*\Psi) &= ([\Phi, \partial] + [\partial, \Phi^*])\Psi \\ &= (F - F^*)\Psi \\ &= 2(\mathbf{E} \cdot \boldsymbol{\sigma})\Psi. \end{aligned}$$

In summary,

$$\begin{aligned} E &:= i\partial_t - qV, & \mathbf{P} &:= -i\nabla - q\mathbf{A}, \\ \partial &:= \partial_t + \boldsymbol{\sigma} \cdot \nabla, & \partial^* &:= \partial_t - \boldsymbol{\sigma} \cdot \nabla, \\ \Phi &:= V + \mathbf{A} \cdot \boldsymbol{\sigma}, & F &:= (\mathbf{E} + i\mathbf{B}) \cdot \boldsymbol{\sigma}, \\ S &:= \partial_t V + \nabla \cdot \mathbf{A}, & \mathbf{E} &:= -\partial_t \mathbf{A} - \nabla V, \\ \mathbf{B} &:= \nabla \times \mathbf{A}, \end{aligned}$$

and the scalar Klein–Gordon operator can be recovered from either ordered first-order product:

$$\begin{aligned} 0 &= (-E^2 + \mathbf{P}^2 + m^2)\Psi \\ &= \left(\square + iqS - q^2\langle\langle\Phi\rangle\rangle^2 + m^2 + 2iq(V\partial_t + \mathbf{A} \cdot \nabla)\right)\Psi \\ &= (\partial^* + iq\Phi)(\partial + iq\Phi^*)\Psi - iqF\Psi + m^2\Psi \\ &= (\partial + iq\Phi^*)(\partial^* + iq\Phi)\Psi - iqF^*\Psi + m^2\Psi. \end{aligned}$$

The ordered first-order products themselves obey

$$\begin{aligned} (E \pm \boldsymbol{\sigma} \cdot \mathbf{P})(E \mp \boldsymbol{\sigma} \cdot \mathbf{P}) &= E^2 - (\boldsymbol{\sigma} \cdot \mathbf{P})^2 \mp [E, \boldsymbol{\sigma} \cdot \mathbf{P}] \\ &= E^2 - \mathbf{P}^2 - iq(\pm\mathbf{E} + i\mathbf{B}) \cdot \boldsymbol{\sigma}. \end{aligned}$$

Taking the scalar part therefore recovers

$$0 = \text{scalar}((\partial + iq\Phi^*)(\partial^* + iq\Phi))\Psi + m^2\Psi.$$

Only the scalar part reproduces the Klein–Gordon operator itself; the full ordered products retain the non-scalar field terms $-iqF$ and $-iqF^*$ that distinguish the two factorizations.

6.5.7 Spin- $\frac{1}{2}$ Wave Mechanics

Before the electromagnetic field is turned on, the same auxiliary 2×2 map from Eq. (61) gives a self-contained first-order version of the relativistic mass shell. Consider the two-component field

$$\Psi = \begin{Bmatrix} \Psi_1 \\ \Psi_2 \end{Bmatrix}.$$

For any complex paravector or paravector-valued operator Z , define

$$W(Z) := \begin{bmatrix} 0 & Z \\ Z^{*\mathbf{R}} & 0 \end{bmatrix},$$

$$W(Z)^2 = \text{diag}(ZZ^{*\mathbf{R}}, Z^{*\mathbf{R}}Z).$$

If $a, b \in \mathbb{R}$, then $W((a + ib)Z) = (a + ib)W(Z)$. In particular,

$$W(i\partial) = i \begin{bmatrix} 0 & \partial \\ \partial^* & 0 \end{bmatrix} = iW(\partial),$$

$$\begin{aligned} W(i\partial)^2 &= -\text{diag}(\partial\partial^*, \partial^*\partial) \\ &= -\square\mathbf{I}. \end{aligned}$$

Define the free Dirac operator matrix and its mass conjugate by

$$D := W(i\partial) - m\mathbf{I}, \quad D' := W(i\partial) + m\mathbf{I}.$$

Then

$$\begin{aligned} D'D &= (W(i\partial) + m\mathbf{I})(W(i\partial) - m\mathbf{I}) \\ &= W(i\partial)^2 - m^2\mathbf{I} \\ &= -(\square + m^2)\mathbf{I}. \end{aligned}$$

Thus the free Dirac equation

$$D\Psi = 0$$

forces each component to satisfy the Klein–Gordon operator. Formally, its characteristic polynomial is

$$\begin{aligned} 0 &= \det(D - sI) \\ &= \det \begin{bmatrix} -(s+m) & i\partial \\ i\partial^* & -(s+m) \end{bmatrix} \\ &= (s+m)^2 + \square, \end{aligned}$$

so the modewise eigenvalue branches are $s = -m \pm i\sqrt{\square}$, with the square-root operator interpreted in the Fourier sense from the previous subsection.

Writing out $D\Psi = 0$ gives the two coupled equations

$$\begin{aligned} i\partial\Psi_2 - m\Psi_1 &= 0, \\ i\partial^*\Psi_1 - m\Psi_2 &= 0. \end{aligned}$$

Solving the first equation for the upper component and substituting into the second yields

$$\begin{aligned} \Psi_1 &= \frac{i\partial\Psi_2}{m}, \\ 0 &= i\partial^*\Psi_1 - m\Psi_2 \\ &= -\frac{\partial^*\partial\Psi_2}{m} - m\Psi_2 \\ &= -\frac{(\square + m^2)\Psi_2}{m}. \end{aligned}$$

Hence the lower component is a Klein–Gordon solution, and the upper component is obtained from it by one first-order derivative.

Conversely, let Ψ' be any scalar Klein–Gordon solution and let $a, b \in \mathbb{C}$ be constants. Define

$$\Psi_1 = \frac{(ma + ib\partial)\Psi'}{m}, \quad \Psi_2 = \frac{(ia\partial^* + mb)\Psi'}{m}.$$

Because a and b are constant and $(\square + m^2)\Psi' = 0$,

$$\begin{aligned} i\partial\Psi_2 - m\Psi_1 &= \frac{(-\partial a\partial^* + im\partial b - m^2a - imb\partial)\Psi'}{m} \\ &= -\frac{a}{m}(\square + m^2)\Psi' = 0, \\ i\partial^*\Psi_1 - m\Psi_2 &= \frac{(mi\partial^*a - \partial^*b\partial - mi a\partial^* - m^2b)\Psi'}{m} \\ &= -\frac{b}{m}(\square + m^2)\Psi' = 0. \end{aligned}$$

The family generated from a scalar Klein–Gordon solution therefore solves the free two-component system for any constant a and b . The simplest choice is $a = 1$ and $b = 0$, for which

$$\Psi = \begin{Bmatrix} \Psi' \\ \frac{i}{m}\partial^*\Psi' \end{Bmatrix}.$$

This is the free spin- $\frac{1}{2}$ wave-mechanics factorization in its most explicit form: the Dirac operator linearizes the Klein–Gordon mass shell, and scalar solutions generate two-component spinor solutions by a first-order lift.

6.5.8 Dirac with Potential Field

Consider again the two-component field

$$\Psi = \begin{Bmatrix} \Psi_1 \\ \Psi_2 \end{Bmatrix}.$$

For any complex paravector or paravector-valued operator Z , define

$$\begin{aligned} W(Z) &:= \begin{bmatrix} 0 & Z \\ Z^{*\mathbf{R}} & 0 \end{bmatrix}, \\ W(Z)^2 &= \text{diag}(ZZ^{*\mathbf{R}}, Z^{*\mathbf{R}}Z), \\ W(Z)^* &= W(Z^*), \\ W(Z)^{\mathbf{R}} &= W(Z^{\mathbf{R}}). \end{aligned}$$

The Dirac operator with potential is

$$\begin{aligned} D &:= W(i\partial - q\Phi^*) - mI \\ &= iW(\partial) - qW(\Phi^*) - mI. \end{aligned}$$

Because

$$\begin{aligned} W(\partial)^2 &= \square I, \\ W(\partial)W(\Phi^*) &= \begin{bmatrix} 0 & \partial \\ \partial^* & 0 \end{bmatrix} \begin{bmatrix} 0 & \Phi^* \\ \Phi & 0 \end{bmatrix} = \text{diag}(\partial\Phi, \partial^*\Phi^*), \\ W(\Phi^*)W(\partial) &= \begin{bmatrix} 0 & \Phi^* \\ \Phi & 0 \end{bmatrix} \begin{bmatrix} 0 & \partial \\ \partial^* & 0 \end{bmatrix} = \text{diag}(\Phi^*\partial^*, \Phi\partial), \\ W(\Phi^*)^2 &= \begin{bmatrix} 0 & \Phi^* \\ \Phi & 0 \end{bmatrix}^2 = \text{diag}(\Phi^*\Phi, \Phi\Phi^*) \\ &= \langle\langle\Phi\rangle\rangle^2 I, \end{aligned}$$

one obtains

$$\begin{aligned} W(i\partial - q\Phi^*)^2 &= -W(\partial)^2 \\ &\quad - iq(W(\partial)W(\Phi^*) + W(\Phi^*)W(\partial)) \\ &\quad + q^2W(\Phi^*)^2 \\ &= -\square I - iq \text{diag}(\partial\Phi + \Phi^*\partial^*, \partial^*\Phi^* + \Phi\partial) \\ &\quad + q^2\langle\langle\Phi\rangle\rangle^2 I \\ &= -\text{diag}\left((\partial + iq\Phi^*)(\partial^* + iq\Phi), \right. \\ &\quad \left. (\partial^* + iq\Phi)(\partial + iq\Phi^*)\right). \end{aligned}$$

The Dirac equation

$$D\Psi = 0$$

therefore becomes the two coupled equations

$$\begin{aligned} -m\Psi_1 + (i\partial - q\Phi^*)\Psi_2 &= 0, \\ (i\partial^* - q\Phi)\Psi_1 - m\Psi_2 &= 0. \end{aligned}$$

Solving the first equation for the upper component and substituting into the second gives

$$\begin{aligned} \Psi_1 &= \frac{(i\partial - q\Phi^*)\Psi_2}{m}, \\ 0 &= (i\partial^* - q\Phi)\Psi_1 - m\Psi_2 \\ &= -\frac{((\partial^* + iq\Phi)(\partial + iq\Phi^*) + m^2)\Psi_2}{m}. \end{aligned}$$

Thus each component satisfies a Klein–Gordon-type operator with the appropriate ordered field coupling.

The mass-conjugate operator is

$$D' := W(i\partial - q\Phi^*) + m\mathbf{I}.$$

Multiplying by D' yields

$$\begin{aligned} 0 &= D'D\Psi = (W(i\partial - q\Phi^*)^2 - m^2\mathbf{I})\Psi \\ &= -\text{diag}\left((\partial + iq\Phi^*)(\partial^* + iq\Phi) + m^2, \right. \\ &\quad \left. (\partial^* + iq\Phi)(\partial + iq\Phi^*) + m^2\right)\Psi. \end{aligned}$$

Equivalently,

$$\begin{aligned} 0 &= ((\partial + iq\Phi^*)(\partial^* + iq\Phi) + m^2)\Psi_1 \\ &= (-E^2 + \mathbf{P}^2 + m^2 + iqF^*)\Psi_1, \\ 0 &= ((\partial^* + iq\Phi)(\partial + iq\Phi^*) + m^2)\Psi_2 \\ &= (-E^2 + \mathbf{P}^2 + m^2 + iqF)\Psi_2. \end{aligned}$$

Once the square is taken, the coupled Dirac system separates into two Klein–Gordon-type equations whose field terms differ only by the conjugation pattern fixed by the operator ordering.

6.5.9 Gauge Transform of the Dirac System

Let $\lambda \in \mathbb{R}$ be a real scalar gauge field and define

$$\Phi' = \Phi + \partial^*\lambda.$$

Because λ is real,

$$\partial\lambda = (\Phi' - \Phi)^*.$$

Transform the two-component field by

$$\Psi = e^{iq\lambda}\Psi'.$$

Then the derivative of the phase factor is

$$\partial(e^{iq\lambda}\Psi') = e^{iq\lambda}(iq\partial\lambda + \partial)\Psi'.$$

Therefore,

$$\begin{aligned} 0 &= (W(i\partial - q\Phi^*) - m\mathbf{I})\Psi \\ &= (W(i\partial - q\Phi^*) - m\mathbf{I})e^{iq\lambda}\Psi' \\ &= e^{iq\lambda}(W(i\partial - q\partial\lambda - q\Phi^*) - m\mathbf{I})\Psi' \\ &= e^{iq\lambda}(W(i\partial - q(\Phi' - \Phi)^* - q\Phi^*) - m\mathbf{I})\Psi' \\ &= e^{iq\lambda}(W(i\partial - q\Phi'^*) - m\mathbf{I})\Psi'. \end{aligned}$$

Hence

$$(W(i\partial - q\Phi'^*) - m\mathbf{I})\Psi' = 0.$$

This is the gauge-covariance statement: the phase change of the Dirac field is exactly compensated by the shift of the potential, so the coupled Dirac equation keeps the same form in the primed gauge.

6.5.10 Dirac Equation Components

For the component form of the coupled Dirac equation, write

$$\begin{aligned} \partial &:= \partial_t + \boldsymbol{\sigma} \cdot \boldsymbol{\nabla}, \\ \partial^* &:= \partial_t - \boldsymbol{\sigma} \cdot \boldsymbol{\nabla}, \\ \Phi &:= V + \mathbf{A} \cdot \boldsymbol{\sigma} \in \mathbb{R} \oplus \mathbb{R}^3, \\ \Phi^* &:= V - \mathbf{A} \cdot \boldsymbol{\sigma}, \\ \psi &:= \psi_s + \psi_{\mathbf{v}} \cdot \boldsymbol{\sigma} \in \mathbb{C} \oplus \mathbb{C}^3. \end{aligned}$$

For the field term, use

$$\begin{aligned} F &:= \mathbf{F} \cdot \boldsymbol{\sigma}, \\ \text{scalar}(F) &= 0, \\ \mathbf{F} &:= \mathbf{E} + i\mathbf{B} \in \mathbb{C}^3, \\ F^* &= -\mathbf{F}^* \cdot \boldsymbol{\sigma}. \end{aligned}$$

The supporting tables then give the scalar and vector parts of $\partial\psi$, $\partial^*\psi$, $\Phi\psi$, $\Phi^*\psi$, $iqF\psi$, and $iqF^*\psi$ explicitly. Tables 42 and 43 record the direct component action of ∂ and ∂^* on a complex paravector test field.

Table 42. Components of $\partial\psi$.

scalar:	$\partial_t\psi_s + \boldsymbol{\nabla} \cdot \psi_{\mathbf{v}}$
vector:	$\partial_t\psi_{\mathbf{v}} + \boldsymbol{\nabla}\psi_s + i(\boldsymbol{\nabla} \times \psi_{\mathbf{v}})$

Table 43. Components of $\partial^*\psi$.

scalar:	$\partial_t\psi_s - \boldsymbol{\nabla} \cdot \psi_{\mathbf{v}}$
vector:	$\partial_t\psi_{\mathbf{v}} - \boldsymbol{\nabla}\psi_s - i(\boldsymbol{\nabla} \times \psi_{\mathbf{v}})$

Tables 44 and 45 then show how the potential Φ and its conjugate act on the same field in components.

Table 44. Components of $\Phi\psi$.

scalar:	$V\psi_s + \mathbf{A} \cdot \psi_{\mathbf{v}}$
vector:	$V\psi_{\mathbf{v}} + \psi_s\mathbf{A} + i(\mathbf{A} \times \psi_{\mathbf{v}})$

Table 45. Components of $\Phi^*\psi$.

scalar:	$V\psi_s - \mathbf{A} \cdot \psi_{\mathbf{v}}$
vector:	$V\psi_{\mathbf{v}} - \psi_s\mathbf{A} - i(\mathbf{A} \times \psi_{\mathbf{v}})$

Tables 46 and 47 complete the two-component operator tables by recording the field-coupling terms $iqF\psi$ and $iqF^*\psi$.

Table 46. Components of $iqF\psi$.

scalar:	$iq(\mathbf{F} \cdot \psi_{\mathbf{v}})$
vector:	$iq\psi_s\mathbf{F} - q(\mathbf{F} \times \psi_{\mathbf{v}})$

Table 47. Components of $iqF^*\psi$.

scalar:	$-iq(\mathbf{F}^* \cdot \psi_{\mathbf{v}})$
vector:	$-iq\psi_s\mathbf{F}^* + q(\mathbf{F}^* \times \psi_{\mathbf{v}})$

6.5.11 Low-Energy Spin-0 Limit

For the minimally coupled scalar theory, use the covariant operators

$$E = i\partial_t - qV, \quad \mathbf{P} = -i\boldsymbol{\nabla} - q\mathbf{A}.$$

Remove the fast rest-mass oscillation by writing

$$\Psi' = e^{-imt}\Psi.$$

Then

$$\begin{aligned} E^2\Psi' &= e^{-imt}(E + m)^2\Psi, \\ \mathbf{P}^2\Psi' &= e^{-imt}\mathbf{P}^2\Psi. \end{aligned}$$

Substituting into the minimally coupled Klein–Gordon equation gives

$$\begin{aligned} 0 &= (-E^2 + \mathbf{P}^2 + m^2)\Psi' \\ &= e^{-imt}(-(E + m)^2 + \mathbf{P}^2 + m^2)\Psi, \end{aligned}$$

hence

$$0 = (-E^2 - 2mE + \mathbf{P}^2)\Psi.$$

Formally iterating this relation gives

$$\begin{aligned} E &= \frac{\mathbf{P}^2 - E^2}{2m}, \\ &= \frac{\mathbf{P}^2}{2m} - \frac{\mathbf{P}^4}{8m^3} + \cdots. \end{aligned}$$

At low energy,

$$E\Psi \approx \frac{\mathbf{P}^2}{2m}\Psi.$$

Solving for $i\partial_t\Psi$ yields the Schrödinger equation

$$i\partial_t\Psi = \left(\frac{(i\boldsymbol{\nabla} + q\mathbf{A})^2}{2m} + qV \right) \Psi.$$

If $\mathbf{A} = 0$, this reduces to

$$i\partial_t\Psi = \left(-\frac{\nabla^2}{2m} + qV \right) \Psi.$$

The scalar low-energy limit is therefore already encoded in the relativistic Klein–Gordon theory.

6.5.12 Low-Energy Spin- $\frac{1}{2}$ and Pauli Limit

Start from the two-component field

$$\Psi' = \begin{Bmatrix} \Psi'_1 \\ \Psi'_2 \end{Bmatrix}$$

and the potential-coupled Dirac equation

$$(W(i\partial - q\Phi^*) - m\mathbf{I})\Psi' = 0.$$

Its two component equations are

$$\begin{aligned} (i\partial - q\Phi^*)\Psi'_2 - m\Psi'_1 &= 0, \\ (i\partial^* - q\Phi)\Psi'_1 - m\Psi'_2 &= 0. \end{aligned}$$

Write

$$\begin{aligned} \partial &:= \partial_t + \boldsymbol{\sigma} \cdot \boldsymbol{\nabla}, & \partial^* &:= \partial_t - \boldsymbol{\sigma} \cdot \boldsymbol{\nabla}, \\ \Phi &:= V + \mathbf{A} \cdot \boldsymbol{\sigma}, & \Phi^* &:= V - \mathbf{A} \cdot \boldsymbol{\sigma}. \end{aligned}$$

Then the two equations become

$$\begin{aligned} (i\partial_t - qV)\Psi'_2 + ((i\boldsymbol{\nabla} + q\mathbf{A}) \cdot \boldsymbol{\sigma})\Psi'_2 - m\Psi'_1 &= 0, \\ (i\partial_t - qV)\Psi'_1 - ((i\boldsymbol{\nabla} + q\mathbf{A}) \cdot \boldsymbol{\sigma})\Psi'_1 - m\Psi'_2 &= 0. \end{aligned}$$

Introduce the covariant operators

$$E := i\partial_t - qV, \quad \mathbf{P} := -i\boldsymbol{\nabla} - q\mathbf{A}.$$

The system is then

$$\begin{aligned} (E - \mathbf{P} \cdot \boldsymbol{\sigma})\Psi'_2 - m\Psi'_1 &= 0, \\ (E + \mathbf{P} \cdot \boldsymbol{\sigma})\Psi'_1 - m\Psi'_2 &= 0. \end{aligned}$$

Peel off the fast rest-mass oscillation with

$$\Psi'_n = e^{-imt}\Psi_n.$$

Since

$$E\Psi'_n = e^{-imt}(E + m)\Psi_n,$$

the reduced system becomes

$$\begin{aligned} (E + m - \mathbf{P} \cdot \boldsymbol{\sigma})\Psi_2 - m\Psi_1 &= 0, \\ (E + m + \mathbf{P} \cdot \boldsymbol{\sigma})\Psi_1 - m\Psi_2 &= 0. \end{aligned}$$

Solve the first equation for the upper component:

$$\Psi_1 = \frac{(E + m - \mathbf{P} \cdot \boldsymbol{\sigma})\Psi_2}{m}.$$

Substituting this into the second equation and multiplying by m gives

$$0 = ((E + m + \mathbf{P} \cdot \boldsymbol{\sigma})(E + m - \mathbf{P} \cdot \boldsymbol{\sigma}) - m^2)\Psi_2.$$

Expand the operator product as

$$\begin{aligned} &(E + m + \mathbf{P} \cdot \boldsymbol{\sigma}) \\ &\quad \times (E + m - \mathbf{P} \cdot \boldsymbol{\sigma}) - m^2 \\ &= (E + m)^2 - m^2 - (\mathbf{P} \cdot \boldsymbol{\sigma})^2 \\ &\quad + [\mathbf{P} \cdot \boldsymbol{\sigma}, E]. \end{aligned}$$

Now

$$\begin{aligned} -(\mathbf{P} \cdot \boldsymbol{\sigma})^2 &= -\mathbf{P}^2 - i(\mathbf{P} \times \mathbf{P}) \cdot \boldsymbol{\sigma}, \\ [P_m, P_n] &= [-i\partial_m - qA_m, -i\partial_n - qA_n] \\ &= iq(\partial_m A_n - \partial_n A_m), \\ \mathbf{P} \times \mathbf{P} &:= ([P_2, P_3], -[P_1, P_3], [P_1, P_2]) \\ &= iq\boldsymbol{\nabla} \times \mathbf{A} \\ &= iq\mathbf{B}, \end{aligned}$$

so

$$-i(\mathbf{P} \times \mathbf{P}) \cdot \boldsymbol{\sigma} = q \mathbf{B} \cdot \boldsymbol{\sigma}.$$

Also,

$$\begin{aligned} [\mathbf{P} \cdot \boldsymbol{\sigma}, E + m] &= [\mathbf{P}, E] \cdot \boldsymbol{\sigma} \\ &= iq(\nabla V + \partial_t \mathbf{A}) \cdot \boldsymbol{\sigma} \\ &= -iq \mathbf{E} \cdot \boldsymbol{\sigma}, \end{aligned}$$

where

$$\mathbf{E} := -\partial_t \mathbf{A} - \nabla V, \quad \mathbf{B} := \nabla \times \mathbf{A}.$$

Defining

$$F := (\mathbf{E} + i\mathbf{B}) \cdot \boldsymbol{\sigma},$$

one has

$$-iqF = q \mathbf{B} \cdot \boldsymbol{\sigma} - iq \mathbf{E} \cdot \boldsymbol{\sigma}.$$

Therefore the exact lower-component equation is

$$\begin{aligned} 0 &= ((E + m)^2 - m^2 - \mathbf{P}^2 + q \mathbf{B} \cdot \boldsymbol{\sigma} + [\mathbf{P} \cdot \boldsymbol{\sigma}, E]) \Psi_2 \\ &= (E^2 + 2mE - \mathbf{P}^2 + q \mathbf{B} \cdot \boldsymbol{\sigma} - iq \mathbf{E} \cdot \boldsymbol{\sigma}) \Psi_2 \\ &= (E^2 + 2mE - \mathbf{P}^2 - iqF) \Psi_2. \end{aligned}$$

This can be iterated recursively as

$$\begin{aligned} E &= \frac{\mathbf{P}^2 - E^2 - q \mathbf{B} \cdot \boldsymbol{\sigma} - [\mathbf{P} \cdot \boldsymbol{\sigma}, E]}{2m} \\ &= \frac{\mathbf{P}^2 - q \mathbf{B} \cdot \boldsymbol{\sigma}}{2m} + O(1/m^2). \end{aligned}$$

At low energy, the dominant terms satisfy

$$0 \approx (2mE - \mathbf{P}^2 + q \mathbf{B} \cdot \boldsymbol{\sigma}) \Psi_2.$$

Solving for $i\partial_t \Psi_2$ yields the Pauli equation

$$i\partial_t \Psi_2 = \left(\frac{(-i\nabla - q\mathbf{A})^2}{2m} + qV \right) \Psi_2 - \frac{q}{2m} (\mathbf{B} \cdot \boldsymbol{\sigma}) \Psi_2.$$

The nonrelativistic spin coupling is therefore not an extra postulate: it is the leading field correction that survives when the massive Dirac system is expanded around its rest-energy oscillation [9, 18].

6.5.13 Spin Up and Spin Down

Let the magnetic-field direction define the spin axis,

$$\mathbf{u} := \frac{\mathbf{B}}{\|\mathbf{B}\|},$$

and use the projection operators

$$\begin{aligned} \Pi(\pm) &= \frac{1}{2}(1 \pm \mathbf{u} \cdot \boldsymbol{\sigma}), \\ \Pi(\pm)^2 &= \Pi(\pm), \\ \Pi(+)\Pi(-) &= 0, \\ \Pi(+) + \Pi(-) &= 1, \\ (\mathbf{u} \cdot \boldsymbol{\sigma})\Pi(\pm) &= \pm \Pi(\pm), \\ \Pi(\pm)^R &= \Pi(\pm), \\ \Pi(\pm)^* &= \Pi(\mp). \end{aligned}$$

Accordingly,

$$\mathbf{u} \cdot \boldsymbol{\sigma} = (+1)\Pi(+) + (-1)\Pi(-).$$

Decompose the Pauli state by

$$\begin{aligned} \Psi(\pm) &:= \Pi(\pm)\Psi, \\ (\mathbf{u} \cdot \boldsymbol{\sigma})\Psi(\pm) &= \pm \Psi(\pm), \\ \Psi &= \Psi(+) + \Psi(-), \end{aligned}$$

so $\Psi(+)$ and $\Psi(-)$ are the spin-up and spin-down parts along the field axis.

Write the Pauli equation as

$$\begin{aligned} \mathbf{P} &:= -i\nabla - q\mathbf{A}, \\ i\partial_t \Psi &= \left(\frac{\mathbf{P}^2}{2m} + qV \right) \Psi - \frac{q}{2m} \|\mathbf{B}\| (\mathbf{u} \cdot \boldsymbol{\sigma}) \Psi. \end{aligned}$$

Equivalently,

$$i\partial_t \Psi = H\Psi,$$

with

$$H := \frac{\mathbf{P}^2}{2m} + qV - \frac{q}{2m} \|\mathbf{B}\| \mathbf{u} \cdot \boldsymbol{\sigma}.$$

Applying $\Pi(\pm)$ to the time evolution gives

$$\begin{aligned} i\partial_t \Psi(\pm) &= i(\partial_t \Pi(\pm))\Psi + \Pi(\pm)(i\partial_t \Psi) \\ &= i(\partial_t \Pi(\pm))\Psi + \Pi(\pm)(H\Psi). \end{aligned}$$

Add zero in the form $H\Pi(\pm)\Psi - H\Pi(\pm)\Psi$:

$$\begin{aligned} \Pi(\pm)(H\Psi) &= \Pi(\pm)(H\Psi) + H\Pi(\pm)\Psi - H\Pi(\pm)\Psi \\ &= H\Psi(\pm) + [\Pi(\pm), H]\Psi. \end{aligned}$$

Therefore

$$\begin{aligned} i\partial_t \Psi(\pm) &= H\Psi(\pm) + (i\partial_t \Pi(\pm) + [\Pi(\pm), H])\Psi \\ &=: H\Psi(\pm) + S(\pm). \end{aligned}$$

In explicit form,

$$\begin{aligned} i\partial_t \Psi(\pm) &= \left(\frac{\mathbf{P}^2}{2m} + qV \right) \Psi(\pm) \\ &\quad - \frac{q}{2m} \|\mathbf{B}\| (\mathbf{u} \cdot \boldsymbol{\sigma}) \Psi(\pm) + S(\pm), \end{aligned}$$

with coupling term

$$S(\pm) := \left(i\partial_t \Pi(\pm) + \frac{[\mathbf{P}^2, \Pi(\pm)]}{2m} \right) \Psi.$$

If $\mathbf{B}$ is uniform and static, then $\Pi(\pm)$ is constant and

$$S(+) = S(-) = 0,$$

so the two spin sectors decouple. In that case

$$(-\mathbf{B} \cdot \boldsymbol{\sigma})\Psi(\pm) = \mp \|\mathbf{B}\| \Psi(\pm),$$

and the Zeeman effect appears as

$$i\partial_t \Psi(\pm) = \left(\frac{\mathbf{P}^2 \mp q\|\mathbf{B}\|}{2m} + qV \right) \Psi(\pm).$$

The field direction therefore selects a preferred axis, and the two projector parts evolve with opposite Zeeman shifts.

7 Conclusions

The algebraic rules fixed in Section 1 generate interval geometry, covariant transformations, electromagnetism, and relativistic wave equations without changing the underlying language. The later physical equations follow from the earlier algebraic setup rather than appearing as separate formalisms placed side by side.

The main structural point is the complex paravector split introduced in Eqs. (15), (18), and (19). Once a quantity is resolved into real scalar, imaginary scalar, real vector, and imaginary vector parts, the same algebraic product can be read component by component across several parts of physics. In the space-time sections this split yields the interval and proper-velocity formulas. In electromagnetism it packages V with $\mathbf{A}$, and $\mathbf{E}$ with $\mathbf{B}$, into one differential framework. In the wave-equation sections it reappears in the paired field couplings of Eq. (99), where the two component fields carry the conjugate pair $+iqF^*$ and $+iqF$. The benefit is not only compact notation. The same bookkeeping keeps the scalar/vector content visible while also showing which formulas come in natural complementary pairs. Within the flat-space theory developed here, it also avoids explicit tensor notation through the main line of argument.

Electromagnetism is the clearest example. Equation (81) produces the sourced equations from the real scalar and real vector parts, while the complementary imaginary scalar and imaginary vector parts give the magnetic-divergence and Faraday relations in Eqs. (82) and (83). Tables 19 and 20 show that this is not an isolated coincidence: once the full real/imaginary scalar/vector decomposition is kept, the derivative algebra naturally generates partner formulas in which electric and magnetic roles are exchanged. The same pattern appears in the force product. Equation (87) keeps the real part because it gives the observed power and Lorentz-force terms, while its imaginary companion is

$$\text{img}(FU) = \gamma(\mathbf{v} \cdot \mathbf{B} + (\mathbf{B} + \mathbf{E} \times \mathbf{v}) \cdot \boldsymbol{\sigma}),$$

which is the complementary electric-magnetic partner of the real-force expression, with the cross-product orientation adjusted by the algebraic product order.

Not every complementary component carries an accepted physical interpretation. Some are standard and necessary, such as the homogeneous Maxwell equations. Others are better read as algebraically natural partner relations that may encode dual descriptions, hidden symmetries, or possible extensions that have not been fully developed. That distinction matters. The complex scalar/vector paravector framework is rich enough to place established relativistic physics and these additional companion relations inside one common algebraic system. The remaining task is to determine which of these companion formulas are alternate forms of known results and which, if any, carry genuinely new physical content.

The extra companion formulas are therefore worth retaining in view, but they should be interpreted with discipline. Some are already standard parts of accepted theory, while others remain algebraically suggestive rather than physically established.

7.1 Toward General Relativity

The natural gravitational extension is already suggested by Section 1.11. There the basis was allowed to depend on a transform element $T(x)$, giving the local sigma basis

$$\sigma'_n(x) := T(x)^{-1} \sigma_n T(x), \quad d\sigma'_n = [\sigma'_n, T^{-1}dT].$$

To move toward general relativity, one would promote that moving basis from a calculational device to the actual local spacetime frame. The interval would then still be built from the same scalar paravector metric as before, but now with position-dependent basis elements:

$$dX = dt + d\mathbf{r} \cdot \boldsymbol{\sigma}'(x), \quad ds^2 = \langle\langle dX \rangle\rangle^2.$$

So the metric is still encoded by the same paravector product used throughout this paper; the new ingredient is that the sigma basis is no longer fixed globally.

The corresponding connection is the differential transform element already introduced in Eq. (53):

$$d\Theta := T^{-1}dT.$$

The transformed-basis calculation in Section 1.11 shows that only the parts of $d\Theta$ that fail to commute with the basis move the frame. In the decomposition

$$\Omega = T^{-1}(dT/dt) = \Omega_s + \frac{1}{2}(k + i\omega) \cdot \boldsymbol{\sigma}',$$

the real-vector part k is the infinitesimal boost data and the imaginary-vector part ω is the infinitesimal rotation data. A curved extension would therefore treat these local boost and rotation pieces as the gravitational connection written in the document's own sigma language, rather than introducing separate Christoffel symbols from the start.

For any chosen coordinate direction d_a , define the directional connection element

$$\Omega_a := T^{-1}(d_a T).$$

Then, for a transported paravector $Z' = T^{-1}ZT$, the same transport identity already derived for $X' = T^{-1}XT$ suggests the local derivative law

$$D_a Z' := T^{-1}(d_a Z)T + [Z', \Omega_a].$$

Curvature is then the failure of these local derivatives to commute:

$$[D_a, D_b]Z' = [Z', F_{ab}],$$

with curvature element

$$F_{ab} := d_a \Omega_b - d_b \Omega_a + [\Omega_a, \Omega_b].$$

This is the same algebraic pattern that already appeared in the moving-frame commutator formulas, now interpreted as genuine spacetime curvature. In other words, curvature would be the obstruction to patching the local transforms $T(x)$ into one global flat frame.

The Einstein equation would then be most naturally written not as one bare paravector product, but as an equality between linear maps on real paravectors $A \in \mathbb{R} \oplus \mathbb{R}^3$:

$$G(A) := \text{Ric}(A) - \frac{1}{2} A R = 8\pi G_N \mathcal{T}_m(A).$$

Here $\text{Ric}(A)$ is the Ricci contraction of the curvature built from the local basis $\{1, \sigma'(x)\}$, R is the associated scalar curvature, and $\mathcal{T}_m(A)$ is the matter stress-energy map. This says the same thing as the usual Einstein equation, but it packages the contractions through the scalar paravector metric, commutators, and linear maps instead of explicit tensor indices [14, 15].

So the answer remains qualified yes, but in a form closer to the present manuscript: general relativity can be written largely without tensor indexing by extending the already-developed similarity transforms, moving-basis formulas, commutators, and scalar paravector metric to local frames. What cannot be avoided is the need for linear maps. Curvature and stress-energy are not captured by one ordinary product of a single paravector; they require the local transform field $T(x)$, the connection elements Ω_a , and the induced curvature and source maps.

7.2 Remaining Work

The gravitational extension sketched above is not developed in the present paper. Every calculation proved here still assumes a fixed flat-space sigma basis, so curvature, covariant transport, and Einstein dynamics are only prospective at this stage. To turn that sketch into a full development, the interval, geodesic law, Maxwell system, and Dirac system would all have to be rewritten in local frame form, and the source discussion around Eq. (89) would have to be replaced by the full stress-energy map required by the Einstein equation. In short, the next gravitational step is not a small correction to the present notation; it is a local-frame version of the whole flat-space construction.

Several other developments are also still missing. The present document does not yet develop non-Abelian gauge structure, so the algebra has not been tested here against the internal symmetries of weak and strong interactions. It does not yet treat many-particle or entangled quantum states, where one needs more than a single paravector or a single 2×2 auxiliary system. It also does not yet give a full account of constitutive laws for anisotropic media, dissipative systems, or continuum mechanics, where higher-rank linear response objects usually appear. On the mathematical side, operator domains, boundary terms, and regularity hypotheses still need to be stated more systematically whenever differential operators are factored or commuted.

Nothing in the present calculations shows a direct incompatibility between STA and the standard relativistic theories treated here. The limitations are instead ones of scope and expressiveness in the current presentation. A single paravector layer does not by itself encode arbitrary higher-rank multilinear structure, bundle data, or global topological patching. Those features may still be representable in STA, but they require additional constructions beyond what has been developed in this document.

7.3 Advantages and Limits of Avoiding Tensors

The main advantage of the present approach is clarity of geometric meaning. The same algebraic product carries scalar contraction, oriented cross-product structure, projector calculus, and transform laws, so many long index derivations collapse to a few lines without losing component information. That compression is not merely cosmetic. It is what makes it possible to place kinematics, electromagnetism, and relativistic wave equations inside one algebraic framework while still recovering the scalar and vector parts explicitly at each step.

The cost is that tensor content has not truly disappeared; it has been repackaged. In flat-space problems that repackaging is often a gain because it removes index bookkeeping and keeps the geometry visible. In curved-space problems the story is more mixed. Once one introduces local frames, connections, curvature, and source maps, the theory again has genuinely multilinear structure even if it is written in STA form. The practical drawback is therefore twofold: first, some problems are not naturally expressed by paravectors alone; second, most of the surrounding literature, software, and standard formulations remain tensor-based, so translation back and forth is unavoidable.

For that reason the tensor-free character of this formulation is best treated as a strong feature, not as a universal replacement principle. In the regime developed here, the algebra gives a compact and coherent reformulation with real conceptual advantages. Beyond that regime, especially in general relativity and in theories built on richer internal or multilinear structure, the more useful question is how much tensor content can be absorbed into the algebra without losing precision, generality, or contact with standard formulations.

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8 Scope Notes and Cautions

The list below marks formulas or claims that still require interpretation, correction, or stronger hypotheses than the surrounding exposition alone makes explicit.

1. **The spin-axis commutator estimate in Section 1.9 should not be confused with the full Robertson or Robertson–Schrödinger state-variance inequality.**
2. **Any diagonalized matrix form of ds^2 must retain explicit Lorentz-signature information; the Euclidean shorthand $ds^2 = dx' \cdot dx'$ is not a Lorentzian identity.**
3. **In quaternion intermediate expansions, the linear term carries the factor a , as in Eq. (56).**
4. **The line $T = \frac{1}{2}FF^R$ should be read as a frame-conditioned energy-density/Poynting paravector, not the full electromagnetic stress-energy tensor.**
5. **Equation (105) is an on-shell projector statement for normalized free states, not the general Dirac equation.**
6. **An earlier sign variant of Eq. (99) with $-iqF^*$ in the upper component is inconsistent with the operator ordering in $W(i\partial - q\Phi^*)$; the corrected sign is $+iqF^*$.**
7. **Several operator equalities stated in differential form implicitly assume smooth test functions and benign boundary behavior; those hypotheses should be stated whenever commutators of differential operators are used rigorously.**

9 Verification and Formula Status

This section compares the principal statements with standard literature and with the symbolic checks listed immediately afterward, and it distinguishes established reformulations from the smaller set of statements that still require explicit qualification.

STA statement	Literature correspondence	Status	Notes
Eq. (31)	Pauli/GA product split [12, 15]	✓	Standard scalar+bivector decomposition.
Eq. (64)	Minkowski interval in paravector form [12, 13]	✓	Equivalent to $dt^2 - d\mathbf{r}^2$.
Eq. (81)	Compact Maxwell equation in STA [2, 12, 13, 16]	✓	Sign depends on the chosen $\mathbf{F}$ and metric conventions.
Eq. (87)	Lorentz force law in STA packaging [12, 16]	✓	Combines power and force components into one paravector equation.
Eq. (90)	Klein–Gordon equation [7, 8]	✓	Standard free relativistic scalar wave equation.
Eq. (93)	Minimal-coupling KG equation [16, 17]	✓	Operator-domain assumptions should be stated when commuting derivatives and fields.
Eq. (98)	Dirac equation in a 2×2 STA auxiliary representation [10, 12, 15]	✓	Representation-dependent, but equivalent in content once the map to the ordinary Dirac component form is fixed.
Eq. (99)	Squared Dirac system after STA factorization [10, 12, 15]	✓	The upper component carries $+iqF^*$ and the lower component carries $+iqF$ with the operator ordering used here.
Eq. (102)	Leading-order Pauli reduction [9, 18]	✓ with caveat	The displayed form is low-energy and omits higher-order electric-field terms.
Eq. (89)	Electromagnetic energy-momentum density	Needs qualification	Useful frame-dependent object, but not the full stress-energy tensor alone.
Spin-axis commutator estimate	Robertson–Schrödinger uncertainty relation [11]	Qualified	Section 1.9 records only the operator-level noncommutativity, not the full state-variance inequality.
Matrix diagonalization remark for ds^2	Diagonalization of indefinite quadratic form	Qualified	Diagonalization reorganizes the quadratic form but does not remove the Lorentzian signature signs.
Eq. (105)	Free on-shell projector condition	Qualified	It is an on-shell projector relation for normalized free states, not the general Dirac dynamics.
Eq. (56)	Quaternion/STA bridge [1, 3]	✓	The corrected linear term carries the factor a , as displayed.

The SymPy verification report below lists the formulas checked directly in code. Its green check marks indicate verified identities, and the comment lines in the $\text{\LaTeX}$ source point to `tools/verify_sta_results.py`.

Python Verification of Main Algebraic and Operator Identities

The identities below were checked symbolically in Python using SymPy and the standard Pauli-matrix realization of the sigma basis. Algebraic identities were reduced directly in matrix form, while differential identities were evaluated on generic test functions $\psi(t, x, y, z)$ with symbolic potentials and field components. The verification script is `tools/verify_sta_results.py`.

Check	Status	Detail
Eqs. (1)–(2)	✓	All nine basis products satisfy $\sigma_n \sigma_m = \delta_{nm} + \epsilon_{nmk} i \sigma_k$ in the Pauli realization.
Eq. (31)	✓	The product $(\mathbf{A} \cdot \boldsymbol{\sigma})(\mathbf{B} \cdot \boldsymbol{\sigma})$ reduces to $\mathbf{A} \cdot \mathbf{B} + i(\mathbf{A} \times \mathbf{B}) \cdot \boldsymbol{\sigma}$ for symbolic components A_n, B_n .
Eqs. (32)–(37)	✓	The complex paravector product, the Minkowski-type form $ZZ^{*\mathbf{R}} = S^2 - \mathbf{V}^2$, and the Euclidean scalar part $\text{scalar}(ZZ^{\mathbf{R}}) = S ^2 + \ \mathbf{V}\ ^2$ all reduce correctly.
Eqs. (45), (65), (66)	✓	The idempotents $\Pi(\pm)$ satisfy the projector identities, $UU^{*\mathbf{R}} = 1$, and $E^2 - \mathbf{P}^2 = m^2$ follows from $mU = E + \mathbf{P} \cdot \boldsymbol{\sigma}$.
Eqs. (79), (80), (81)	✓	The potential identity gives $S = \partial_t V + \nabla \cdot \mathbf{A}$, $\mathbf{E} = -\partial_t \mathbf{A} - \nabla V$, and $\mathbf{B} = \nabla \times \mathbf{A}$, while ∂F splits into Gauss, Faraday, and Ampère–Maxwell component equations with the manuscript’s signs.
Eqs. (86) and (87)	✓	The gauge law $\Phi' = \Phi + \partial^* \lambda$ gives $V' = V + \partial_t \lambda$ and $\mathbf{A}' = \mathbf{A} - \nabla \lambda$, and the product FU splits into the expected power term $\gamma \mathbf{v} \cdot \mathbf{E}$ and force term $\gamma(\mathbf{E} + \mathbf{v} \times \mathbf{B})$.
Eqs. (94), (95), and (96)	✓	Acting on a generic test function reproduces the expanded Klein–Gordon operator, the commutators $[E, P_m] = iqE_m$ and $[P_m, P_n] = iq(\partial_m A_n - \partial_n A_m)$, and the factorization $(E + \boldsymbol{\sigma} \cdot \mathbf{P})(E - \boldsymbol{\sigma} \cdot \mathbf{P}) = E^2 - \mathbf{P}^2 - iqF$.
Eqs. (99), (101), and (104)	✓	Eliminating one Dirac component yields the two squared equations; with the manuscript’s operator ordering, the upper component carries $-iqF^*$ after everything is moved to the left-hand side, while the lower component carries $+iqF$. The exact Pauli identity $(\boldsymbol{\sigma} \cdot \mathbf{P})^2 = \mathbf{P}^2 - q \mathbf{B} \cdot \boldsymbol{\sigma}$ holds on a test function; and for uniform $\mathbf{B} = B_0 \mathbf{u}$ the projectors satisfy $(\mathbf{B} \cdot \boldsymbol{\sigma})\Pi(\pm) = \pm B_0 \Pi(\pm)$.
Eq. (56)	✓	The quaternion correspondence satisfies $QQ^{\mathbf{R}} = a^2 + \mathbf{b}^2$ and $Q^2 = a^2 - \mathbf{b}^2 - 2ia \mathbf{b} \cdot \boldsymbol{\sigma}$; the omitted factor of a in the older intermediate line is indeed an error.

These symbolic checks complement the literature comparison in Section 9. They separate purely algebraic and operator-theoretic identities from statements whose interpretation still depends on physical conventions.